

Pi AutoSim User Guide

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Contents

About this User Guide	8
Safety and Compliance.....	9
Disclaimer	11
Health and Safety Information	11
Electrical safety warning	12
Maximum load warning.....	12
FCC compliance	13
CE compliance.....	14
Quick start	15
Introduction to Pi AutoSim.....	17
User Interface PC	19
I/O System	19
Installation	23
Installing the User Interface software.....	23
Troubleshooting.....	28
Setting up Networking for Pi AutoSim	28
Tutorial	31
Getting to know AutoSim	31
Starting the Pi AutoSim I/O System.....	31
Connecting to the I/O System.....	35
Loading a model	36
Creating a control panel.....	39
Moving, resizing and aligning instruments.....	42
Switching out of Design mode	42
Running and stopping the model	43
Exiting Pi AutoSim	44
The Pi AutoSim On-line Help system.....	44

Software.....	49
Connecting to the I/O System	51
Connecting an ECM to the I/O System	53
Managing I/O Systems	54
Using the Navigator	54
Managing workspaces	57
Working with Models	58
Editing model and variable properties	60
Using Open Loop Models	68
The MFX branch	68
Hardware branches.....	68
Statistics Branch	82
User Variables Branch	83
Serial Communications Branch	83
Cosmetic branches and reference variables.....	83
Working with Panels	85
Creating and manipulating panels	87
Adding, Deleting and Manipulating Instruments.....	89
Editing instrument properties	92
Common instrument properties	93
Instruments and Controls	97
IOS Scripts.....	109
Using the Editor	112
Working with IOS Scripts summary	115
IOS Script commands.....	117
Working with GUI Macros	122
Using the GUI Macro Editor	123
GUI Macro Commands	127
Using the Data Logger.....	130
Logging data and managing log files	131
Viewing and playing back logged data	136
Session Logs.....	138

Serial Communications	140
The COMMS Navigator branch.....	140
Message variables	144
Trigger variables.....	148
Setting up and managing message logging.....	151
Setting up the Real Time Viewer.....	154
Using the Message Log and Real Time Viewers	156
 Hardware.....	 163
I/O System.....	165
Front panel features	166
Hardware installation	170
 Power supplies	 173
External battery power supply	173
Internal power supply.....	175
Front panel power connections.....	176
Front panel power controls.....	181
 I/O cards	 183
Master card	184
Card configurations.....	184
I/O channel assignment	186
Component location and description	188
 Discrete I/O card.....	 190
Example application.....	190
Circuit description	190
The resistor daughterboard	195
Card configurations.....	198
I/O channel assignment	200
Component location	201
 Analog Input card.....	 205
Example application.....	205
Circuit description	205
Card configurations.....	206
I/O channel assignment	207
Component location	208
Daughterboards	209
Circuit description	210

Analog Output card	211
Example application.....	211
Circuit description	211
I/O channel assignment	214
Component location	215
Daughterboard options	216
Analog Output card specifications	230
Pulse Width Modulation Input card	233
Example application.....	233
Circuit description	233
Card configurations.....	234
I/O channel assignment	235
Component location	236
PWM Input card specifications	238
Pulse Width Modulation Output card	240
Example application.....	240
Circuit description	240
Card configurations.....	241
I/O channel assignment	242
Component location and description	243
PWM Output card specifications.....	245
Programmable Waveform Generator card	247
Circuit description	248
Card configurations.....	252
I/O channel assignment	254
Component location and description	255
Card specifications	257
Creating pattern files.....	258
Event Timer card	260
Example application.....	260
Circuit description	260
Card configurations.....	261
I/O channel assignment	264
Component location	265

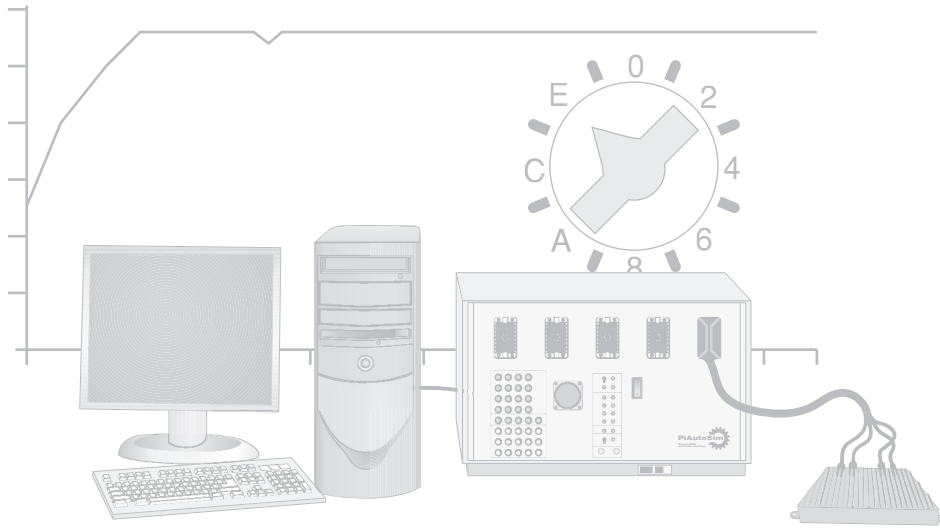
Index ----- 269

Contact information----- 274

About this User Guide

This user guide is divided into the following main sections:

- Quick Start describes the system and explains how to install the User Interface software on your PC. It also contains a tutorial, which quickly familiarizes you with the Pi AutoSim software.
- Software explains how to connect your ECM to the I/O System, and gives detailed information about using the Pi AutoSim software, including details on logging data, recording and playing GUI Macros, running IOS Script files and using serial communications.
- Hardware describes the Pi AutoSim I/O System and cards.



Safety and Compliance

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Electrical safety warning



Front panel

Do not connect any pin or socket (of any connector) to more than 50volts AC or 120 volts DC.



Back panel

Do not connect any of the 4mm sockets to more than 50volts AC or 120 volts DC.



PC panel

Do not connect any pin of any connector to more than 50volts AC or 120 volts DC.



Back panels

Hazardous voltages exist inside the simulator. Only qualified persons should work inside this equipment.

Note: This equipment must be earthed (grounded).

Maximum load warning

The maximum load that you can place on top of the I/O system is 25kg; it will support most 17-inch CRT monitors.

FCC compliance

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference, in which case the user is required to correct the interference at his own expense.

However, there is no guarantee that interference will not occur in a particular installation. You can determine whether this equipment is causing interference by turning this equipment off. If the interference stops, it was probably caused by the equipment or one of its peripheral devices. If this equipment causes interference to radio or television reception, try to correct the interference by using one or more of the following measures:

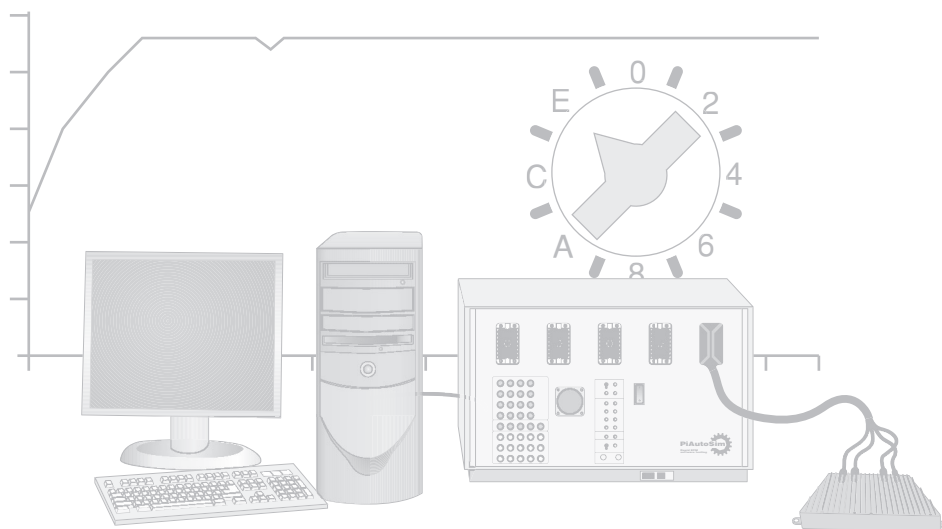
- Reorient or relocate the receiving antenna until the interference stops.
- Move the equipment to one side or the other of the television or radio.
- Connect the equipment into an outlet on a circuit different from that to which the television or radio is connected.
- Consult the dealer or an experienced radio/TV technician for help.

Warning: Modification of this equipment without Pi Technology's authorization may result in this equipment no longer complying with FCC requirements for Class A digital devices. In that event, your right to use the equipment may be limited by FCC regulations, and you may be required to correct any interference to radio or television communications at your own expense.

CE compliance

Declaration Of Conformity CE	
Manufacturer's Name	Pi Technology
Manufacturer's Address	Milton Hall Ely Road Milton Cambridge CB4 6WZ United Kingdom
declares that the product:	
Product Name	Pi AutoSim
Model Numbers	Pi AutoSim
Product Options	All
conforms to the following Product Specifications:	
Safety	EN 61010-1:1993
EMC	CISPR 22:1997/EN 55022:1998 – Class A
	FCC Rules (1995), Part 15, Subpart B, Class A
	EN 50082-1:1992
Supplementary Information	The product complies with the requirements of the Low Voltage Directive 73/23/ECC and the EMC Directive 89/336/EEC amended by 93/68/EEC and carries the CE Mark accordingly.





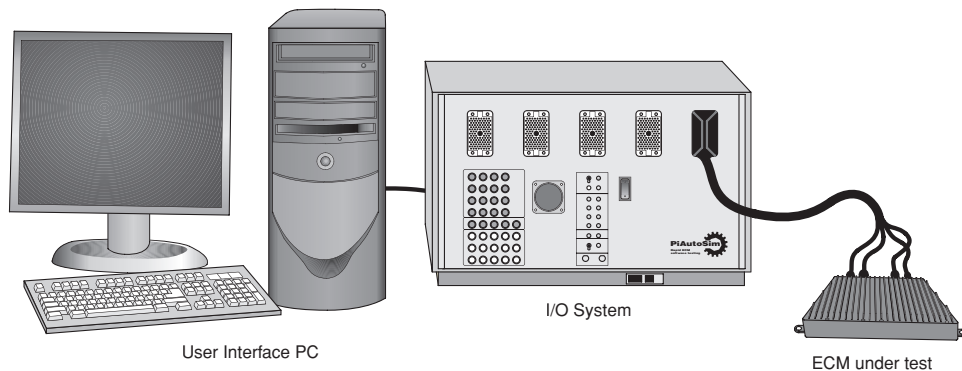
Quick start

Introduction to Pi AutoSim

Pi AutoSim enables you to test your electronic control module (ECM) in a real-world environment by simulating the component it will control, such as an engine, transmission, chassis or complete vehicle. Pi AutoSim consists of two components:

- The I/O System runs a real-time model of the environment under control and communicates with the ECM via its I/O cards.
- The User Interface runs on a desktop PC, communicates with the I/O System over a network connection, and manipulates the model and the ECM to run interactive testing or execute automated tests.

The picture below shows a typical AutoSim setup, with an ECM plugged into one of the front panel connectors of the I/O System.



A typical AutoSim setup

Pi AutoSim generates the appropriate signals to send to the ECM and responds to the signals received in return, thus making the ECM under test act as if it is connected to a real automotive environment. The User Interface gives access to all of the model variables – ECM inputs and outputs – and the simulated data. Virtual instruments and controls are used to display and control this data.

The virtual instruments and controls are pictorial representations of real instruments and controls such as a thermometer or a slider control. You simply choose how you want to

represent each model variable, and use the controls to change settings such as throttle demand or engine load.

You can test an ECM using two different types of model:

- An open loop model can be used to stimulate the ECM's inputs and measure the responses via the ECM's outputs. You can control and display stimulants and responses by means of the virtual controls and instruments.
- A closed loop model can provide real-time simulation of the automotive environment that the ECM will control. It measures the ECM's outputs and inputs, and calculates from them the appropriate values to feed back to the ECM's inputs.

You can test an ECM manually, or automate the process using Pi AutoSim's scripts and macros. Manual testing involves stimulating the ECM inputs and monitoring the outputs using virtual instruments and controls. Scripts allow you to automate sequences of I/O System operations, and give a high level of repeatability. Macros add another level of control, allowing you to manipulate entire scripts and models automatically.

The I/O cards in the I/O System provide the interface with the ECM by handling real automotive voltage levels. Your Pi AutoSim system can simulate a specific automotive environment by the configuration of the different I/O cards available. When your requirements change, you can change the I/O card configuration (perhaps to simulate a different environment or to provide extra controls).

You can install the following cards:

- Discrete card with 32 channels (16 input, 16 output)
- PWM Input card with 16 channels
- PWM Output card with 8 channels
- Analog Input card with 32 channels
- Analog Output card with 16 channels
- Event Capture card with 32 channels to monitor spark and injection signals
- Programmable Waveform Generator card with 8 channels for cam, crank and wheelspeed signals.

User Interface PC

The Pi AutoSim user interface runs on a typical desktop PC. Installation of the user interface software is described in the next chapter.

The minimum configuration necessary to run the Pi AutoSim user interface software is given below:

- IBM-compatible PC with Pentium processor, keyboard and mouse.
- 128MB RAM
- 20MB hard disk space for the software, and up to 20MB hard disk space for log files, script files and multiple configurations
- Network card with 10-baseT interface, and TCP/IP protocol support
- SVGA color monitor, supporting 256 colors and 1024x768 resolution
- Windows® 98SE/NT/2000/XP operating systems.

Recommended items

It is useful, but not essential, to have a more sophisticated setup, including:

- 17-inch SVGA color monitor, supporting 16-bit color resolution
- A printer.

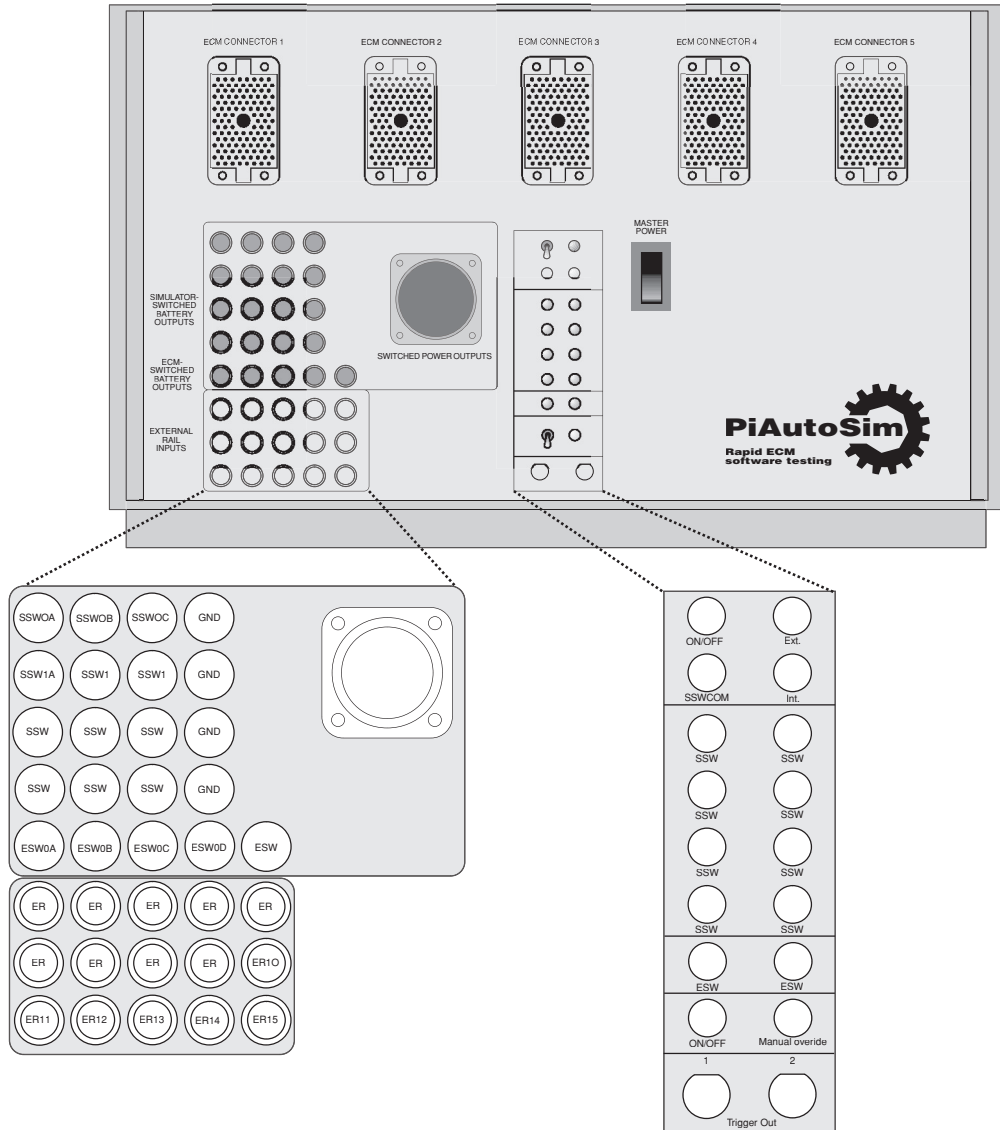
Note: Connections from the processing system are standard PC connections and are not protected against accidental damage due to incorrectly applied voltages.

I/O System

This section briefly describes the front and back panels of the AutoSim I/O System: a much more detailed description can be found in the Hardware section.

Note: Before you connect the ECM to the I/O system, it is essential that you read the rest of the information in this chapter and Connecting to the I/O system in the Pi AutoSim Software manual.

Front panel



Note: The panel layout may vary slightly depending on the version of Pi AutoSim.

ECM connectors

The five front panel connectors connect the ECM under test to the I/O System. Information about the configuration of the front panel is found later in this document.

Master power switch and LED indicators

The master power switch controls the power supplied to Pi AutoSim. When the master power switch is off, no external power can be applied. There are three LEDs:

- The external power LED indicates that external power is being applied to Pi AutoSim.
- The internal power LED indicates that power is being applied to the I/O cards so that they can be identified and set up by the software.
- The common switched supply LED indicates that the common switched supply is switched on.

Circuit breaker

The master switch also operates as a 20A circuit breaker for the unit, and protects all power output circuits simultaneously. Additionally, no power is output when the battery input voltage is either above 30V or below 0V.

Note: If you power any part of the circuit from a power supply other than those routed through the simulator, the simulator cannot protect either itself or the system under test from overload, over voltage, or negative voltages.

External rail inputs

There is one external rail input for each I/O card in the system – these inputs enable you to select the voltage applied to each individual card.

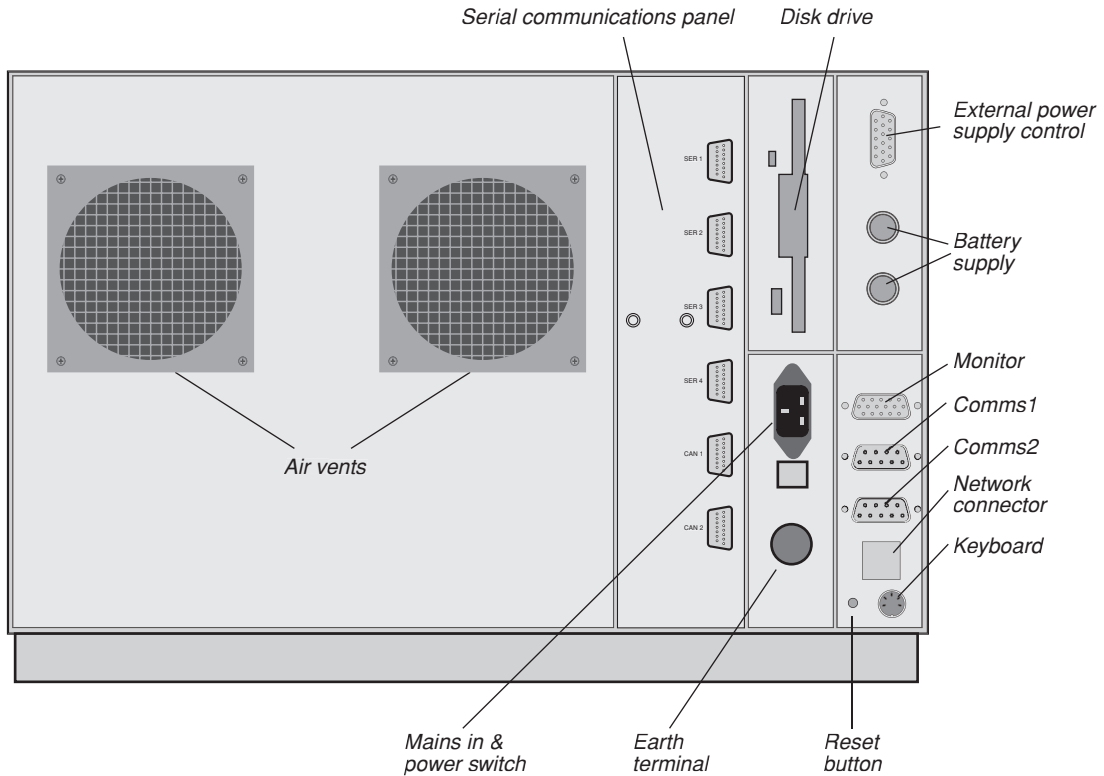
Manual override

The manual override option enables you to maintain power to certain outputs, even if the software has switched them off or is no longer running. Pushing the switch momentarily up enables the manual override feature; pushing it momentarily down disables the feature. The manual override LED is illuminated when the feature is enabled.

Trigger Out

There are two triggers, which are under software control and enable you to trigger external devices (e.g. an oscilloscope).

Rear panel



Note: The panel layout may vary slightly depending on the version of Pi AutoSim.

Note: Connections from the processing system are standard PC connections and are not protected against accidental damage due to incorrectly applied voltages.

Floppy disk drive

The disk drive is not needed in normal operation of the system but may be used during system setup or maintenance.

Installation

This chapter explains how to install the Pi AutoSim User Interface software, and gives some troubleshooting hints.

After you have installed the software, you need to start the external I/O system, and connect to it via the network. You can then familiarize yourself with the system's features by working through the example in the Tutorial section.

Installing the User Interface software

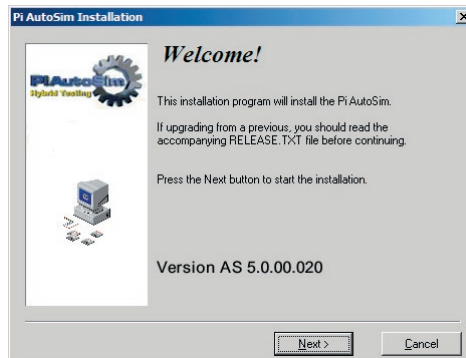
This section describes how to install the Pi AutoSim User Interface software onto your computer. The AutoSim User Interface software is supplied on a CD-ROM.

It is important to follow the installation instructions carefully.

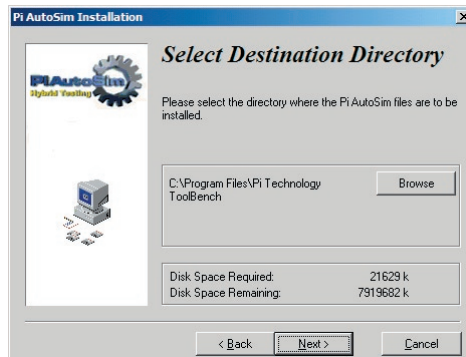
To install the Pi AutoSim software:

1. Switch on the computer and start Windows®.
2. If there is an earlier installation of AutoSim on the computer, you should remove it before installing the new version. (From the Windows Task Bar, choose Start ► Settings ► Control Panel to display the Control Panel; double-click on Add/Remove Programs to display the Add/Remove Programs Properties dialog; select Pi Technology AutoSim in the list; and click on Add/Remove. This will remove the program and associated settings, but it will not remove any of your user files such as models, panels, scripts and so on. Make sure you disconnect any I/O Systems from your PC.)
3. Insert the CD into your CD drive.
4. Choose Run from the Start menu, and type
d:\SetupAutoSim
into the text box, where d is the drive letter of your CD drive. Click OK.

5. A Welcome dialog is displayed:

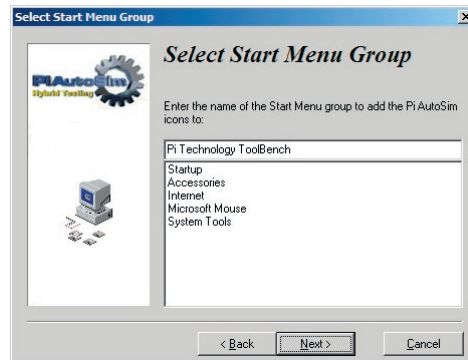


6. Click on Next to display the Select Destination Directory dialog:

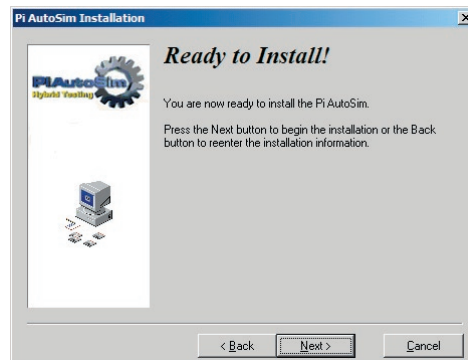


You can use this dialog to choose a directory in which to install the software. However, the default directory name follows Windows® recommended practice, and you are advised not to change the name.

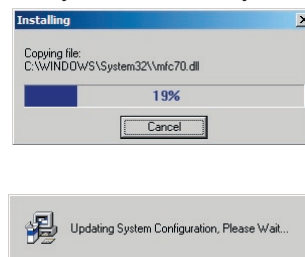
7. Change the name for the Start Menu Group, or use the default. Click on Next.



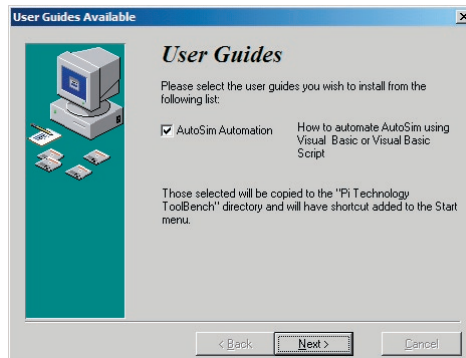
8. The software is ready to install. Click on Next to continue the installation.



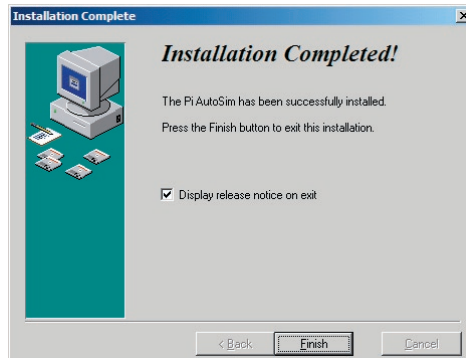
9. The software is installed on your PC and its system configuration is updated:



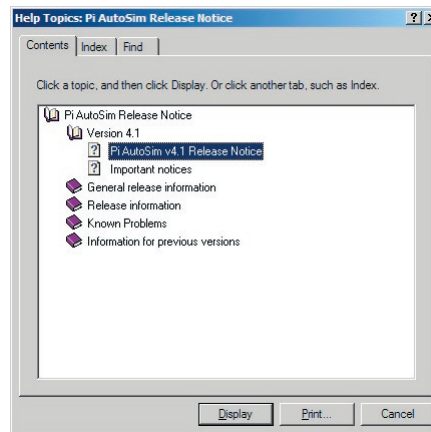
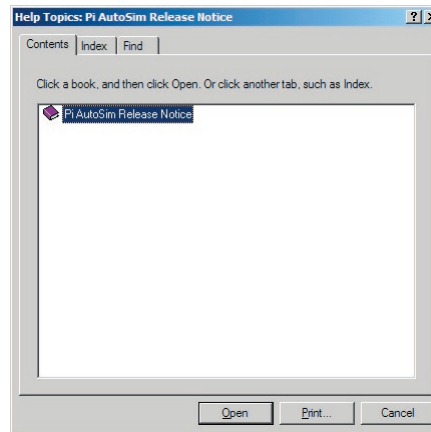
-
10. Select the User Guides you would like copied to your PC. Click on Next:



11. The installation is complete. Click on Finish:



12. The Help dialog is displayed and you can select a help topic to display.



Troubleshooting

It is unlikely you will experience any trouble in installing and setting up the Pi AutoSim system. However, this section describes some potential problems that may occur during installation or use, and the actions you should take if they do.

System file problems

During installation, AutoSim will try and update some of your Windows® system files. If you receive an error message telling you some of these files are out of date, you will need to install the latest version of the system files. If required, consult your Systems Administrator for further assistance.

Image clipping on high-resolution screens

If you are using a high-resolution screen and any controls on a panel are displayed with some parts of the image clipped (e.g. there are only partial graduations on a Slider's scale), check the Windows® screen settings by right-clicking on the Desktop and choosing Properties from the pop-up menu displayed. In the dialog box displayed, click on the Settings tab to display the options selected – in particular, make sure that Small Fonts is selected, not Large Fonts.

Setting up Networking for Pi AutoSim

Pi AutoSim uses a Local Area Network (LAN) for communications between the I/O System (IOS) and the User Interface (GUI) PC. During the installation process, and if you relocate a GUI PC, you need to ensure that the appropriate networking components are installed and correctly configured.

Overview

Pi AutoSim uses TCP/IP networking to communicate between the I/O System and the GUI PC. TCP/IP is a widely used and supported communication protocol, which is 'vendor-independent'.

As part of the TCP/IP setup, you need to assign a unique address to each of the computers on the network, which will include your Pi AutoSim chassis, the GUI PC, and any other

computers that are using TCP/IP on the same network. This address is known as the IP address of the computer, and is described in the following sections.

Starting the Pi AutoSim chassis

Switch the Pi AutoSim on using the switch located next to the mains connector on the rear panel and wait for the chassis to boot up. This may take several minutes.

Note: You should not have connected an external power supply or module to the AutoSim chassis at this stage.

Setting an IP Address

The IP Address on the GUI PC can be displayed or changed as follows:

1. From the Windows® Task Bar select Start ► Settings ► Control Panel.
2. Double-click on the Network icon.
3. On the Configuration page, select the TCP/IP entry, and click Properties.
4. Select the IP Address tab.

Please see documentation installed with the Pi AutoSim User Interface software for instructions on setting the IP Address on the Pi AutoSim chassis.

When setting an IP Address you should refer to your Systems Administrator to ensure that you use the appropriate method for your network, and ensure that you do not use an IP address that conflicts with another user on the same network.

If you need to manually set an IP address, you should also set an appropriate Subnet Mask, which determines the range of addresses that are permitted on your network. For the I/O System and the GUI PC to communicate, they must have the same Subnet Mask and the same network address within the overall IP Address.

As an example, if your network address is 193.131.70.X and you have a subnet mask of 255.255.255.0, you may allocate IP addresses in the range 193.131.70.1 to 193.131.70.255.

Note: AutoSim is deliberately configured such that the GUI PC and the I/O System must be on the same subnet. The GUI PC will not locate the I/O System if it is configured to operate on a different subnet.

Connecting the I/O System to the GUI PC

There are two main methods for interconnecting the I/O System and the GUI PC: a hub-based connection and a point-to-point connection.

The hub-based connection requires an active 10-BaseT hub that allows you to connect several computers together. This scheme would typically be applied in larger installations where there are many PCs and a structured networking system installed. It has the advantage of integrating the AutoSim chassis with the main network, and allows different GUI PCs to connect to an AutoSim chassis. You may choose to implement a simpler scheme with a small hub providing four or eight connections within a laboratory or on a test bench. In either case, each system requires a single UTP (RJ45) cable connection to the hub.

The point-to-point scheme can be used where you have only one GUI PC and one I/O System. Instead of using an active hub, you can instead use a special 'point-to-point' or crossover cable to link the two machines and provide the necessary crossover between the Rx and Tx signals that is usually provided by the hub. These cables are often sold for connecting two PCs together for games playing.

In either case, you must still ensure that the IP addresses are correctly set to configure the systems onto the same logical subnet.

Troubleshooting the Network Connection

If you cannot establish communications between the I/O System and the GUI PC, carry out the following tests:

Check first that you have TCP/IP installed on your User Interface PC. Next, ensure that the IP addresses are set correctly on both the PC and the I/O System:

1. Check that the Subnet Masks and the corresponding network address within the IP addresses match on both the GUI PC and the I/O System.
2. Use the DOS 'PING' program to check communications between the two systems. From a DOS session on the GUI PC, enter PING and the target IP Address to check that the IP communications are established.

If you cannot establish IP communications that are verified by PING, check your cabling and, if necessary, refer the problem to your System Administrators so that they can check any internal network routing that might be required.

Tutorial

Getting to know AutoSim

This tutorial section introduces the basic principles and techniques needed to work with Pi AutoSim. It uses a test model and an I/O System.

Pi AutoSim lets you create your own virtual control panels, which can be used to test an ECM as a flexible alternative to a physical control panel. Throughout this tutorial, comparisons are made between the old hard-wiring method and Pi AutoSim's 'virtual' instrumentation.

This section of the Tutorial tells you how to:

- Start the I/O System and the AutoSim software
- Use the Navigator to manage models, controls and instruments
- Connect to the I/O System
- Load a model
- Enable the variables you want to control and monitor
- Set up a simple control panel
- Connect model variables to the control panel
- Run the test model.

Once you've got a feel for setting up and using Pi AutoSim's panels and instruments, the rest of this Tutorial gives you detailed information on each aspect of Pi AutoSim.

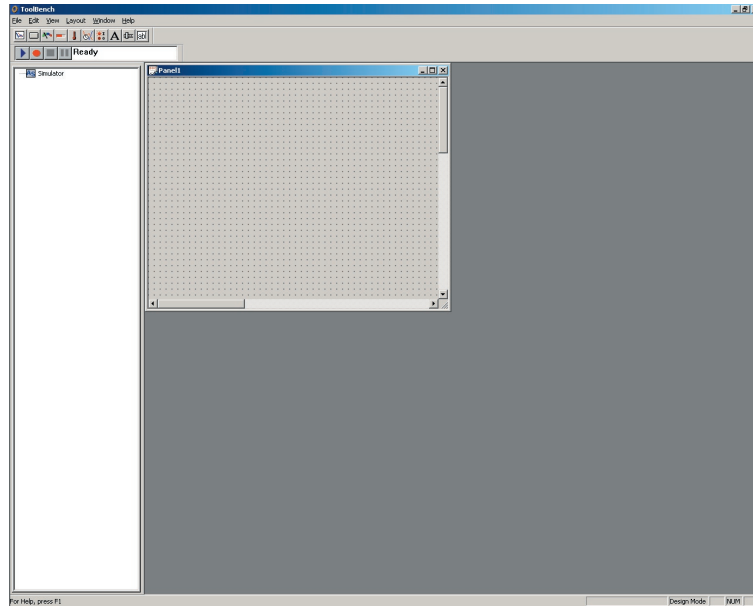
Starting the Pi AutoSim I/O System

Before you start the User Interface software, you need to start the I/O System and have it connected to your PC with the network cable. Please see the 'Setting up Networking for Pi AutoSim' section in the Software Manual.

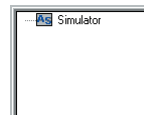
To start the AutoSim software:

1. Click on the Windows® Start button.
2. Choose Start ► Programs ► Pi Technology ToolBench ► Pi AutoSim.

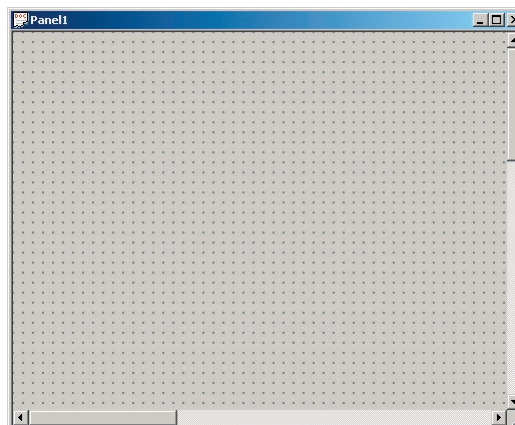
The AutoSim application window appears:



- The left-hand pane of the window contains the Navigator, which you use to manage models, controls and instruments.



- On the right is an empty control panel (Panel1), ready for you to add instruments and controls.



- Along the top are two 'docked' toolbars (the General and Controls toolbars – see below).

The Toolbar options

AutoSim provides the following toolbars:

- General toolbar



This allows you to switch between Design and Run modes for a control panel, and gives you access to standard Windows® file commands (New, Save, etc.).

- Controls toolbar



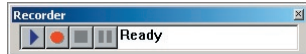
This allows you to add various 'virtual' instruments and controls to your control panel.

- Alignment toolbar



This allows you to align, center and space control panel instruments.

- Recorder toolbar



This allows you to record and play a sequence of actions in an automated test (you can close this toolbar, as we won't be using it in the tutorial).

Managing toolbars

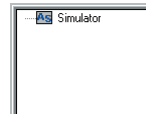
By default, the General and Alignment toolbars are docked at the top of the AutoSim window, while the Controls and Recorder toolbars are floating. You can perform the following operations on any toolbar:

- Dock a floating toolbar by dragging it into the toolbar area at the top of the window, or by double-clicking on its title bar.
- Float a toolbar by dragging it out of the toolbar area, or double-clicking on its border.
- Hide or display a toolbar using the appropriate View menu option. You can also hide a floating toolbar by clicking its Close icon.

The Pi AutoSim Navigator

Pi AutoSim's Navigator is used to navigate and manage models, IOS scripts and logs, and to assign model variables to an AutoSim instrument panel. It looks and behaves like a standard Windows® Explorer window.

By default, the Navigator is displayed as a docked toolbar on the left-hand side of the Pi AutoSim window:



If the Navigator is not visible, choose View ► Navigator from the main menu. The section Using the Navigator in the Software chapter tells you how to undock the Navigator (i.e. turn it into a floating window that you can reposition).

The Navigator may contain two or more branches. You need to work with the Simulator branch, which provides the connection to local or remote I/O Systems, as described in the next section.

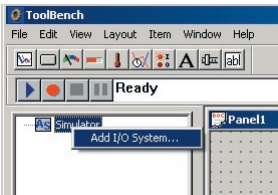
Connecting to the I/O System

Before adding any instruments to the panel, you need to connect to the I/O System. This enables you to load and run models, and makes model variables available for you to display and control.

When you are using Pi AutoSim to test an ECM, the I/O System is a separate server, which houses all the I/O cards and signal conditioning hardware needed to convert between ECM and PC signals.

To add an I/O System:

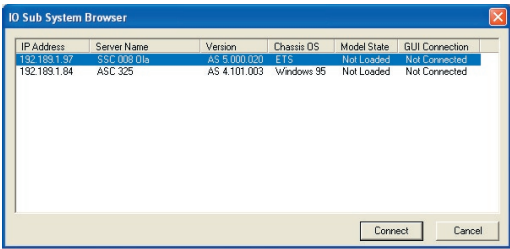
- 1. Right-click on the Simulator branch in the Navigator to display a context menu:



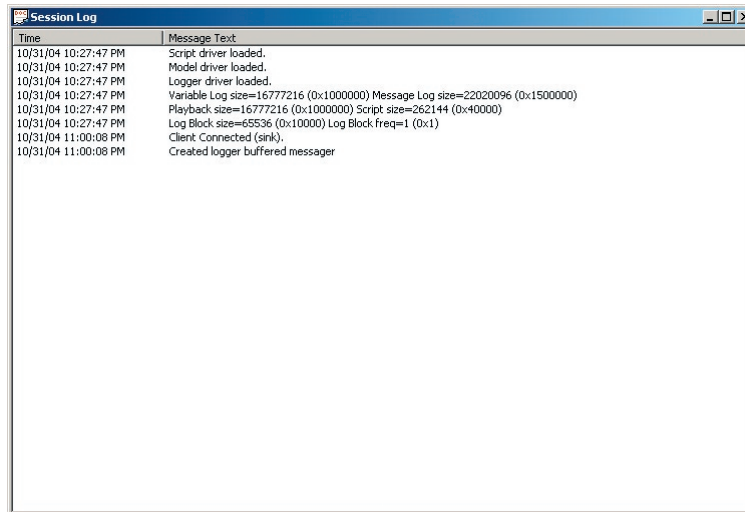
- 2. Choose Add I/O System to display the I/O System Browser dialog.

Note: If you choose Add I/O System and there are no I/O Systems visible in the I/O System Browser dialog, there may be a problem with your TCP/IP settings. Please see the section Troubleshooting the Network Connection under Setting up Networking for Pi AutoSim.

- 3. In the I/O System Browser dialog, select the entry for your I/O System and click Connect:



The Session Log window appears:



This displays messages generated by the I/O System and its models and scripts, and any other active components in the system, such as Calibration Devices – see Session Logs near the end of the Software chapter for details. When the Session Log window is selected, the main menu changes to show menus for working with it – click on the Panel window to display the Panel menus again. Normally, you can work with the Session Log window minimized.

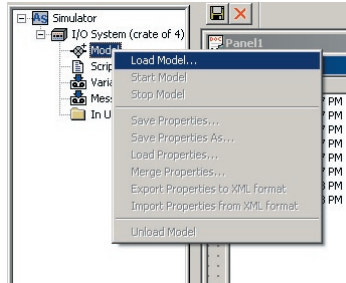
Note: When using the Navigator to manage Pi AutoSim entities, right-clicking on a particular item will often display a context-sensitive menu. The same options are usually available in the main application menu: left-click on the item in the Navigator to select it and then choose Item ► (option) from the main menu if the Panel window is selected.

Loading a model

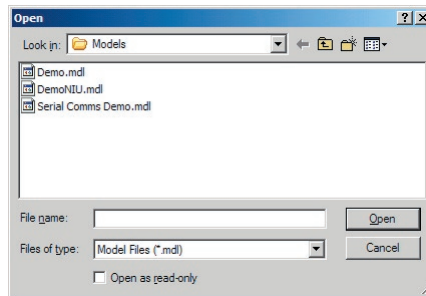
After you have connected to an I/O System, you need to load a model. The model defines all the ECM's inputs and outputs, and can also provide a dynamic simulation of, say, a vehicle powertrain. It provides all the output signals to the ECM, and responds to any input signals from the ECM. It is possible to create your own models, but to follow this tutorial, you should load the test model supplied with AutoSim.

To load a test model:

1. In the Navigator, expand the Simulator branch by clicking the plus button next to its name, or by double-clicking its name.
2. Expand the I/O System branch in the same way.
3. Right-click on the Model branch and choose Load Model.



This displays the Open dialog:



4. If the Open dialog isn't already looking at the Pi Technology ToolBench Models folder, go to:

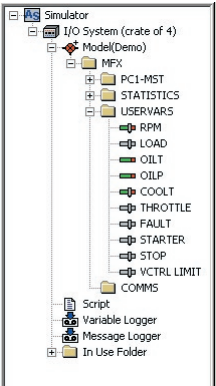
C:\Pi AutoSim Project Files

5. Select the model Demo.mdl.
6. Click Open.

To view the model variables:

- 1. Click the plus button  next to the Model branch to expand it.
- 2. Expand the MFX branch in the same way.
- 3. Expand the USERVARS branch.

The various signals available using this model are displayed in the Navigator (e.g. RPM and LOAD):



The icons next to the variables show whether the variable is:

	read only
	write, or
	read/write (combining the color of the read icon and the slider of the write icon)

Note: There are two sets of variables visible in the navigator: hardware variables, grouped by I/O card, and user variables, which have been specially created in the model. For the purpose of the tutorial, the hardware variables should be ignored and only the user variables (RPM, LOAD, etc.) used. Normally you would configure hardware variables (i.e. rename them, etc.) and would use these in conjunction with any user variables you may have specified in your model.

All the I/O signals are now available, ready to be ‘wired up’ to appropriate instruments and controls. (AutoSim refers to signals as ‘model variables’.)

Creating a control panel

The next step is to create a control panel. You do this by dragging model variables from the Navigator onto the control panel. This automatically configures the appropriate instrument. You can also use AutoSim's Controls Toolbar to add 'disconnected' controls to a panel and assign model variables to them.

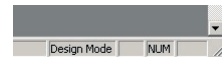
In this section a control panel is created containing the following controls:

- Start and stop push-buttons
- A slider to alter the throttle setting
- An analog meter to display engine RPM
- A slider to alter the engine load
- A thermometer to display the oil temperature.

Note: See Troubleshooting near the end of the Installation section earlier in this chapter if you are using a high-resolution screen and some parts of controls are clipped (for example, there are only partial graduations on a scale).

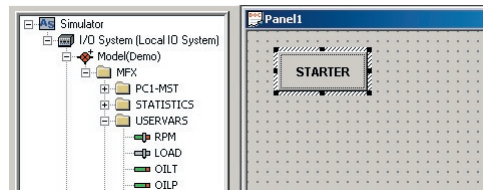
To create a control panel:

1. Resize the Panel1 window, to give yourself room to add the controls. You can either drag the panel's resize icon, or click on the Maximize icon.
2. Make sure you're in Design mode – the words 'Design Mode' should appear in the Status bar along the bottom of the AutoSim window:

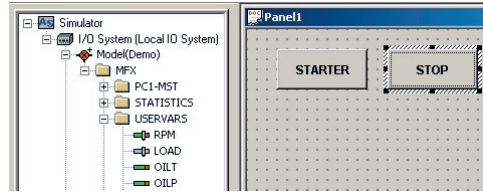


If you're not in Design mode, click the Switch Mode button  in the General toolbar or choose File ► Design Mode.

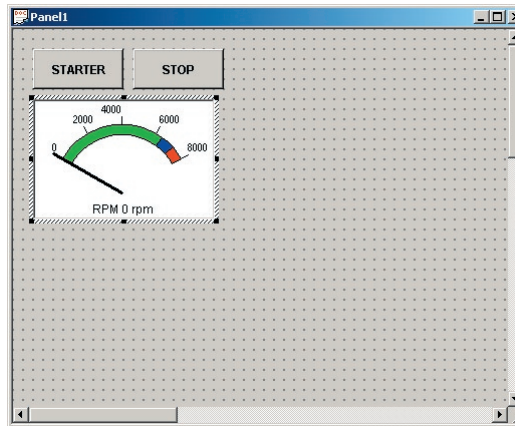
3. Drag the STARTER item from the Controls branch of the Navigator and drop it onto the panel. A STARTER button appears:



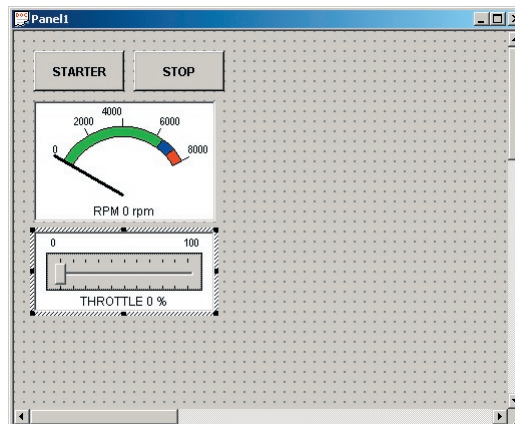
4. Drag the STOP item from the Navigator and drop it onto the panel:



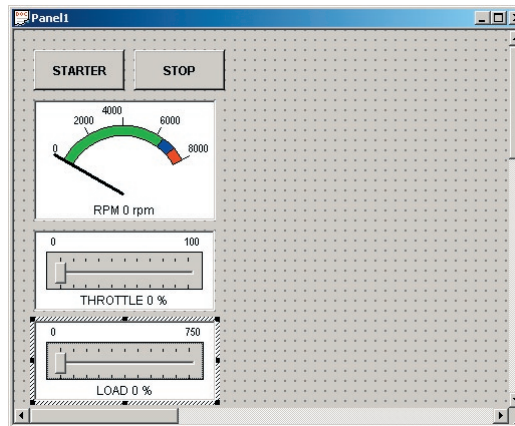
5. Drag and drop the RPM item from the Navigator. An analog meter appears:



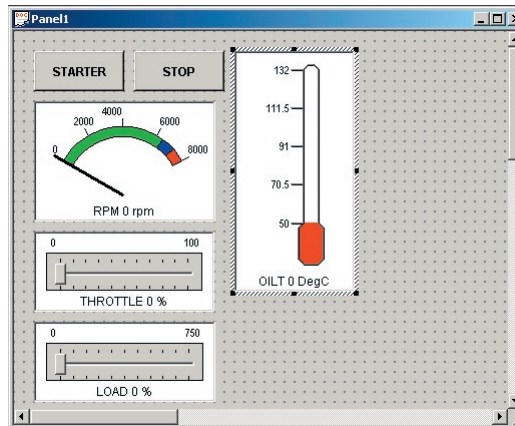
6. Drag and drop the THROTTLE item from the Navigator. A slider control appears:



7. Drag and drop the LOAD item. A second slider control appears:



8. Drag and drop the OILT item. A thermometer appears:



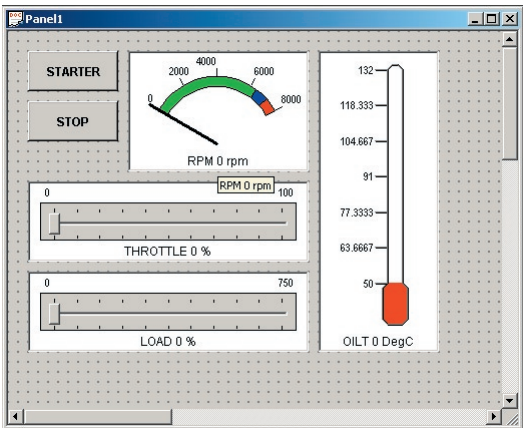
You now have a series of instruments and controls in the panel, which are automatically connected to the correct model variables.

Moving, resizing and aligning instruments

You can rearrange the control panel by moving, resizing and aligning the various instruments and controls.

To ...	Do this ...
Move an instrument	Left-click on it and drag its border when the pointer changes to a double arrow.
Resize a selected instrument	Drag a control point on an edge or a corner.
Align two or more instruments	Press [Shift] or [Ctrl] to select them or, use the alignment toolbar.

For example, you can move and resize the controls so that the panel looks similar to this:



Switching out of Design mode

Having finished configuring the Control panel, click on the Switch Mode button in the General toolbar to switch into Run mode or choose File ► Design Mode again.

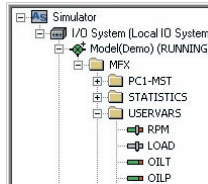
Running and stopping the model

Now the instruments and controls are connected up, you need to start the model running, and then start up the model 'engine'.

To run and stop the model:

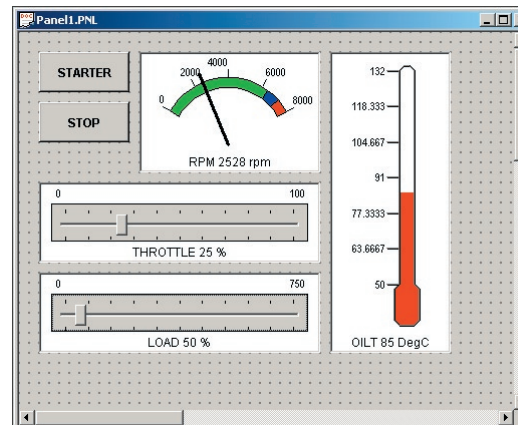
1. In the Navigator, right-click on the Model branch and choose Start Model from the context menu.

The Model branch in the Navigator shows a RUNNING status:



The dynamic model is being run by the I/O System, so it can respond to the controls on the panel. You now need to start and run the engine.

2. Click the STARTER button in the panel.
The simulated engine will start running, and you'll see the RPM change on the analog meter.
3. Drag the slider control to alter the THROTTLE and LOAD settings and watch RPM and OILT change accordingly on the analog meter and thermometer:



Note: If you set an excessive load for the engine, the model may stall the engine. In order to restart the engine you need to reset the 'Fault' variable of the model. This can be done by placing the Fault variable on a text edit control and changing its value from 1 to 0. You will then be able to restart the engine.

4. Click on the STOP button to stop the engine running.

This very simple demonstration shows how a panel can be set up to interact with an engine model. In a full system, the ECM would respond to the THROTTLE via one of its inputs, and the engine speed would be calculated by the model from measurements of injections and sparks. However, the principles involved in setting up the panel and running the model are the same.

Exiting Pi AutoSim

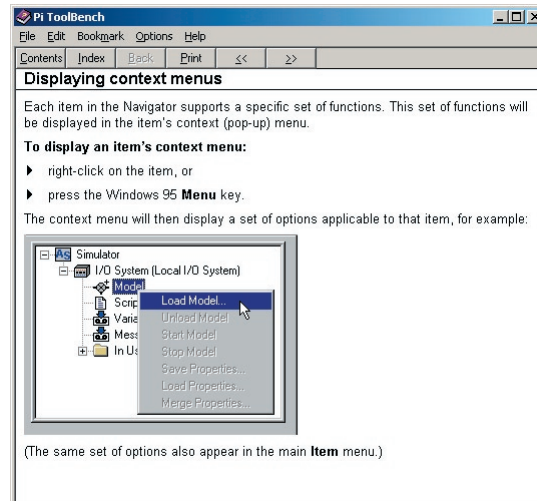
When exiting Pi AutoSim, you will need to stop and unload the active model.

To exit Pi AutoSim:

1. In the Navigator, right-click on the Model item and choose Stop Model from the context menu.
2. Click on the Close icon in the top right-hand corner of the AutoSim window.
3. You'll be asked if you want to save the panel configuration that you were using. Cancel these operations and exit Pi AutoSim.

The Pi AutoSim On-line Help system

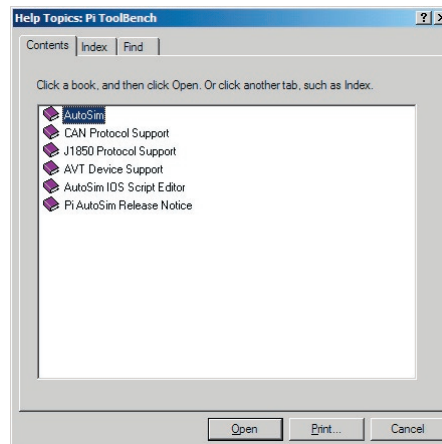
Pi AutoSim includes an on-line Help system for using the AutoSim Software: you can access Help by choosing Help Contents from the Help menu. Help is displayed in the standard Windows® Help Viewer. Below is a typical Help topic, explaining how to display a context menu:



Many Help topics feature hyperlinks (underlined green text) to other related topics. Click on a hyperlink to jump to the named topic, return to the previous topic by clicking the [Back](#) button. You can page through the Help topics using the browse buttons [>>](#) and [<<](#).

Finding a Help Topic

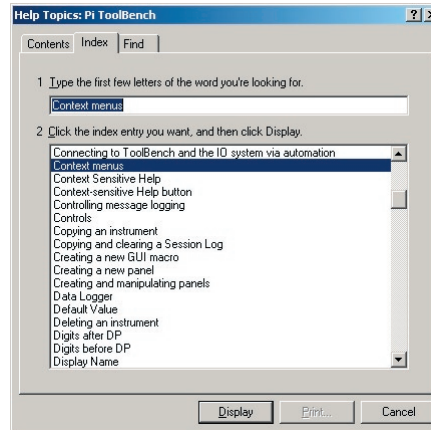
Along with context-sensitive Help, you can also search for a specific topic by accessing Help from the Help Topics menu item. Click the Contents tab to see a list of topics:



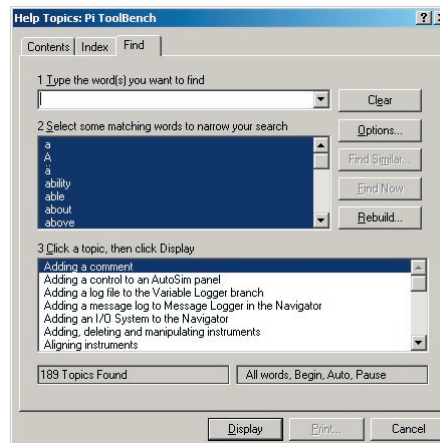
Topics are organized by category, mirroring the organization of the AutoSim User Guide.

Double-click on a book to see what topics are in that category. To see a topic, double-click on it. You can close a book by double-clicking on it.

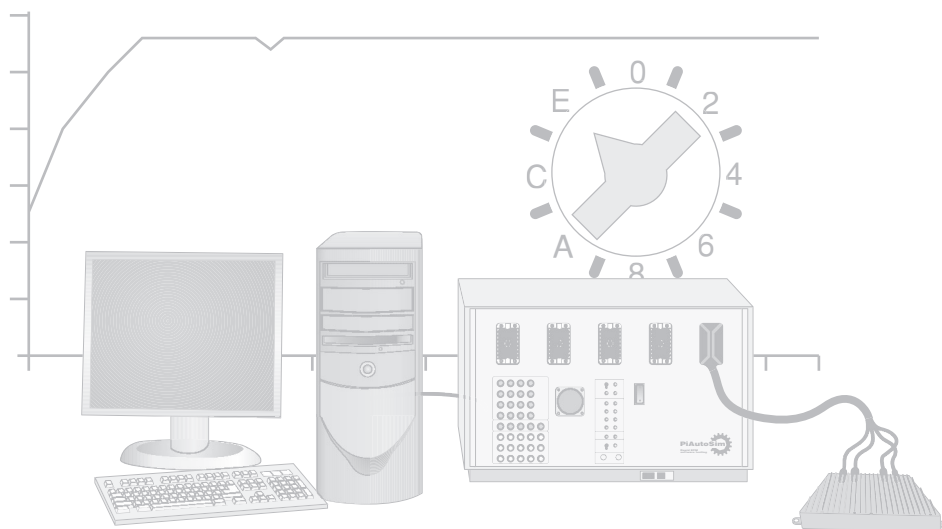
The Index tab displays an alphabetical list of Help topics; select a topic by double-clicking on it, or click a topic and click Display.



The Find tab allows you to specify topics according to words or phrases they contain.



Type the word or phrase you want to find in the top entry field. You can enter a few characters and use the word list below to see if any words match what you type. If you want to specify two or more words, separate them with a space. Click on the word you are looking for in the second field: the topics containing this word are listed in the third field. Double-click on the topic to view it, or click a topic then click Display.



Software

Connecting to the I/O System

Once you have installed the AutoSim user interface software, you need to connect to the I/O System in order to access the I/O card inputs and outputs and manipulate the ECM under test. The I/O System is connected to the user interface PC by its network connection. The steps you need to take in order to connect the I/O System are similar to those you may have carried out in 'Getting Started' to connect to the local I/O System and are described below.

Note: to ensure optimal performance the version of the AutoSim user interface software and the version of the AutoSim chassis software should be the same

To locate the I/O System:

1. Run the Pi AutoSim user interface by choosing
Start ► Programs ► Pi Technology ToolBench ► Pi AutoSim.
2. Right-click on the Simulator branch in the Navigator and choose Add I/O System from the pop-up menu. The system scans the network for any active I/O Systems and presents you with a list.

Note: See Troubleshooting and Setting up Networking for Pi AutoSim near the end of the Installation section of the Getting Started chapter if the I/O Systems are not listed.

3. Select an I/O System and click Connect.
4. Expand the Simulator branch (by clicking on the plus button) to make the I/O System branch visible.

To load a model:

Once you have connected to the I/O System, you need to load the model that defines the inputs and outputs of the I/O System.

1. In the Navigator, expand the I/O System branch.
2. Right-click on the Model branch and choose Load Model from the context menu.
3. In the Open dialog, navigate to **C:\PiAutoSim Project Files** (or the folder holding the model if it is not in the default directory) and select the model you want to load. Click Open after highlighting the model name.
4. Expand the Model branch to view the I/O System inputs and outputs. Once you have loaded the model, You are able to view all the simulator input and output variables.

Note: Before connecting your ECM to the I/O System, it is essential to configure the model and variable property pages, and in some cases the physical I/O card settings, in order that your model and I/O System are correctly configured for the ECM under test.

Working with Models describes how to configure the I/O System and model, although a brief overview is provided here.

The majority of the I/O System configuration is under software control. There are eight different types of I/O card (including the Master card). Each card and channel has slightly different properties, which can be configured from the appropriate property page. Each channel is represented as a model variable and comes with a default configuration, although you will still need to confirm that the default properties are correct for your ECM, and alter the configuration if they are not.

The properties that you will need to configure include:

- Voltage and bias levels for discrete I/O channels
- Default, minimum and maximum values for variables
- Units for variables.

As some aspects of the system are not under software control, you will occasionally need to carry out some hardware configuration by altering the settings on an I/O card itself. In such cases, please refer to the Hardware chapter for detailed descriptions of each type of card.

It is also useful at this stage to rename the model variables so that they accurately reflect the configuration of the ECM. For example, you might rename a variable 'Aircond on/off' rather than use a default name like S01-DIO:CHAN00 (i.e. channel 0 on the discrete I/O card in slot 1 of the I/O System).

Once you have configured the I/O System and model correctly for your ECM, you are ready to connect your ECM to your I/O System and begin ECM testing.

Connecting an ECM to the I/O System

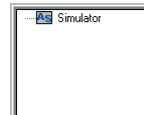
The ECM is connected to the front panel connectors via a custom-made loom. If the loom has been provided by Pi Technology, you simply need to follow the instructions supplied with this loom in order to connect your ECM to Pi AutoSim.

If you are making your own custom loom, you will need to refer to the pin-out diagrams provided with your I/O System. These describe the I/O cards included in your system and where the signals are located in the front panel connectors.

Once you have connected the ECM to AutoSim, you are ready to test your ECM, either manually or using automated tests.

Managing I/O Systems

This section describes how to use AutoSim's Navigator to manipulate I/O Systems. When AutoSim components are installed on a machine, the Navigator automatically loads and displays the Simulator Data Manager, which appears as a branch labeled Simulator:

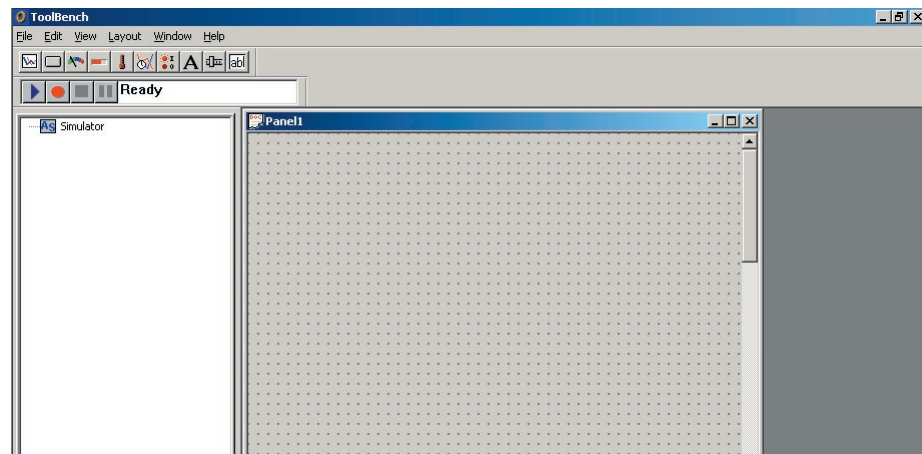


You can then add Simulator I/O Systems to the Navigator. They appear as a set of branches representing the I/O System, its model, script and logger modules, and the variables associated with the model.

From here, you can inspect model variables and use them for panel configuration.

Using the Navigator

The Navigator is a dockable panel that appears on the left-hand side of the Pi AutoSim window:



The Navigator automatically displays any active Data Managers in the system (typically I/O System Drivers and Calibration Servers). Each active Data Manager is presented as a top-level branch in the Navigator, and has its own menu options in the context menu and the main Item menu.

The Navigator represents your view of the system (your workspace), which can be saved and restored between sessions. When a Navigator is restored, it will automatically attempt to connect to any devices that were previously loaded, so minimizing setup time. You can only display a single Navigator at any time.

To display the Navigator:

1. Choose View ► Navigator from the main menu.
2. While in its docked state, you can resize the Navigator by dragging the right-hand edge of the window.

To dock and undock the Navigator panel:

1. To float (undock) the Navigator panel, drag one of its borders outside the AutoSim window boundaries, away from the 'docking' area on the left hand side of the window.
2. This turns it into a floating panel, which you can drag around the desktop.
3. To re-dock the panel, double-click on its title bar.

Note: If you drag the Navigator whilst holding down the Ctrl key you can prevent it docking on the left-hand edge of the AutoSim window.

To expand and contract a branch of the Navigator:

- Double-click on its name.
- Or
- Select the branch and click the plus button or the minus button.

To display an item's context menu:

Each item in the Navigator supports a specific set of functions. This set of functions is displayed in the item's context (pop-up) menu.

- Right-click on the item.
- Or
- Press the Windows Menu key.

The context menu then displays the set of options applicable to that item. (The same set of options appears in the main Item menu.)

Drag and drop operation

You can drag and drop items from the Navigator onto instruments in a control panel. This is a fast and effective way of connecting instruments to model variables, and of manipulating items in general.

To add an I/O System to the Navigator:

1. Select the Simulator item in the Navigator.
2. Choose Item ► Add I/O System from the main menu.
(Alternatively, right-click on the Simulator item and choose Add I/O System from the context menu.)

The system will scan the network and present you with a list of active I/O Systems.

Note: See Troubleshooting and Setting up Networking for Pi AutoSim near the end of the Installation section of the Getting Started chapter if the I/O Systems are not listed.

3. Select an I/O System and click Connect.

The system will verify communications to the I/O System and add it to the Navigator (the Session Log window will also be displayed automatically). If communications cannot be verified, an error message is displayed.

If the I/O System branch isn't visible, expand the Simulator branch.

To remove an I/O System from the Navigator:

1. Select the I/O System item in the Navigator.
2. Choose Remove I/O System from the context menu.

You are prompted to save the workspaces before the I/O System and all underlying items are removed from the Navigator.

Note: Removing an I/O System from the Navigator does not delete any items from the system or stop the I/O System running; it merely removes them from the Navigator.

Managing workspaces

A workspace is a snapshot of a Navigator view that can be saved and restored between sessions. By saving a workspace, a model and model properties, you can keep a complete working environment, consisting of the I/O system link, the model and the properties, plus the log and script files referenced in the Navigator.

Note: When a workspace is saved, the system will save references to the various items and files shown in the Navigator, not copies of the items. This ensures that a workspace does not take its own local copy of a model file, or a panel, but references a shared copy stored on the User Interface PC.

To save the current Navigator workspace:

1. Choose File ► Save Workspace from the Main menu.
2. Enter a name and click Save. The file will automatically be given a .wrk extension.

The system will save the contents of the current Navigator, including references to the active model and scripts, so that they can be restored at a later date. In this way, you can create different workspaces for different operating configurations, perhaps with different I/O Systems, models or scripts.

To restore the contents of a saved Navigator workspace:

1. Choose File ► Open workspace from the main menu.
2. Select the required workspace file by browsing the file system (look for files of type .WRK).
3. Click Open.

The system closes the current Navigator workspace and all files associated with it. You are prompted to save the current workspace, and the model properties if you have not already specified a model property file. The system then loads the contents of a new Navigator workspace, connecting to I/O Systems and loading new model and script files as necessary.

Note: Where a reference to a file or item in the Navigator cannot be restored, an error message is displayed and you may then review the specific errors.

Working with Models

Each I/O System runs a single model reflecting the particular hardware configuration in use, determined by the physical I/O cards fitted to the rack and the particular ECM connected. It presents the current input and output values as a set of model variables, which can act as data sources to the various instruments, controls and scripts used in the system.

At their simplest, models may just transfer values from the I/O hardware to the user interface with minimal scaling or other translation. However, models typically implement some dynamic behavior of the system being simulated and perform real-time calculations based upon values read from both the ECM under test and the user interface.

Open loop models

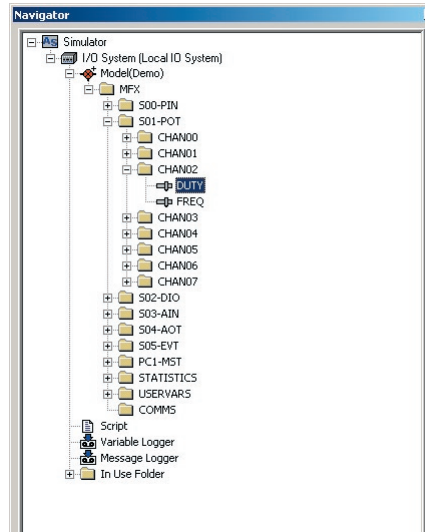
Pi AutoSim comes supplied with a basic open loop model, in which all the available hardware channels are presented as a set of raw, un-configured variables.

The Navigator organizes variables according to the following hierarchy:

1. Card
2. Daughterboard
3. Channel
4. Variable.

Each I/O System supports a single active (running) model, which is displayed in the Model branch of the Navigator for that I/O System.

For example, the picture below shows a model called 'commstest'. You can see a variable called 'Duty' on CHAN02 (Channel 2) on the POT (Pulse Width Modulation Output) card:



Initially, all variables are set up with default names, linear scaling and no filtering, and are marked as 'in use'. You can configure model variables, specifying names, scaling and filtering values appropriate to the signals being measured or simulated.

Once you have set up variables, they are marked as 'in use' and are added to the In-Use folder in the Navigator – this contains a list of variables scanned by the model (to minimize system processing overheads, unused variables are not scanned).

Closed loop models

You can develop your own closed loop models using the AutoSim C++ Model Development Kit or the AutoSim Simulink Model Development Kit and the appropriate Microsoft or Matlab development environment. Once you have created your closed loop model, you may then distribute it and use it on your AutoSim system.

To load a model on an I/O System:

1. Right-click on the Model branch.
2. Choose Load Model from the context menu.
3. Browse the file system and select a model file (open and closed loop model files both have a .MDL extension).

When you load a model:

- It becomes the active model.
- A status message is sent to the Session Log.
- The name of the currently active model is displayed in the I/O System Model branch.

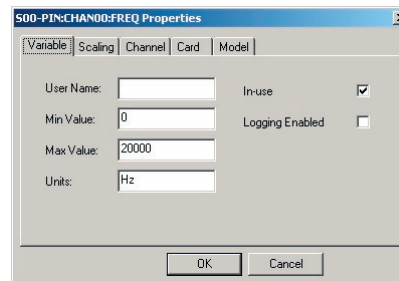
You are now ready to start the model, or configure it by editing the model variable properties.

Editing model and variable properties

Each item in the open loop model has a set of properties associated with it, which reflect how it is set up. In many cases these properties can be changed to alter the way in which the channel or variable operates.

You can display the properties by right-clicking on an item in the Navigator and choosing Properties from the context menu. The dialog displayed contains a number of tabbed sections, relating to items at and above the level of the item you have selected.

For example, if you had selected a model variable at the lowest level of the tree (a PWM Input Frequency variable for example), you would see all the available tabs:



If you had selected an I/O card, you would only see the properties for that card.

The Variable tab is the same for all variables, though some properties may be hidden if they are not relevant to a particular type of variable (for example, the Trigger Enabled property is only shown for Trigger variables). The Scaling tab is also the same for all variables, though it will only be displayed if scaling has been defined for the variable.

Channel and card properties vary according to the type of card in use.

Note: You can also display the properties page for a model variable by right-clicking on an instrument in the panel and choosing ...Control Properties ► Data Source Properties from the context menu (where ... is the type of control).

This section lists the software-editable properties common to most variables. The following section gives a brief overview of the properties pages for I/O cards, daughter boards and channels – for more details on the hardware-specific properties of variables, channels, daughter boards and I/O cards, see the information on individual I/O cards in the Hardware chapter.

Note: See Serial Communications near the end of this Software chapter for a description of Receive Message and Transmit Message variables.

Model variable properties

This section lists the basic set of model variable properties. Note that often only a subset of properties is available, depending on the model and type of variable (read only, write only, read/write or trigger).

There are two properties tabs associated with model variables:

- Variable
- Scaling.

Variable properties tab

This allows you to supply a user name for the variable, set up various defaults and enable logging, I/O script and model access:

The screenshot shows a dialog box titled "S01-POT:CHAN00:FREQ Properties" with a close button (X) in the top right corner. The dialog has five tabs: "Variable", "Scaling", "Channel", "Card", and "Model". The "Variable" tab is selected. It contains the following fields and checkboxes:

- User Name: [text box]
- Min Value: [text box with "0"]
- Max Value: [text box with "20000"]
- Units: [text box with "Hz"]
- Default: [text box with "0"]
- In-use: ☒
- Logging Enabled: ☐
- Script Access Enabled: ☒

At the bottom of the dialog are "OK" and "Cancel" buttons.

- User Name – This allows you to apply a custom name to the variable, which is more meaningful than the default name.
- Min and Max Value – Allow you to specify the default limits.
- Units – The units in which the variable is displayed.
- Default – The default value applied to the variable when the model is started.
- In-use – When set, the variable is scanned by the model. By default, this flag is unset for all variables (no variables are scanned, thus minimizing processing overheads). When you edit a variable's properties, this flag is set automatically (i.e. it is assumed that when you edit a variable, you want it to be scanned).
- Logging Enabled – When set, logging is enabled for the variable.
- Script Access Enabled – When set, the I/O Script can update the variable. When unset, the script cannot update the variable – it is under manual control from the User Interface. By default, this flag is set. This option is only present for write only and read/write variables.
- Model Access Enabled – When set, the model can update the variable. When unset, the variable is under manual control from the User Interface. For example, this is typically used for RPM. A closed-loop model usually calculates RPM, but you may want to overwrite it to set a fixed engine speed for a particular test. This option is only present for read/write variables.
- Trigger Enabled – This check box is only available for Trigger variables. These variables play a particular role in serial communications, but can be used more generally. If the trigger is enabled and the value of the variable is changed to some non-zero value, some action defined in the model is carried out. Typically, the

action would be the transmission of a message, but the model author could use a trigger to initiate many other types of action. If the Trigger variable is not enabled (check box cleared), changing the value of the Trigger variable will not initiate the action defined for it.

Note: Because the Trigger does not fire when the value is set to zero, if you link the variable to a non-latching button, it will only fire once when you press the button (when the value is changed from zero, but not when it changes back to zero). Note also, however, that buttons are latching by default, so you will need to change a trigger button to non-latching after you have created it.

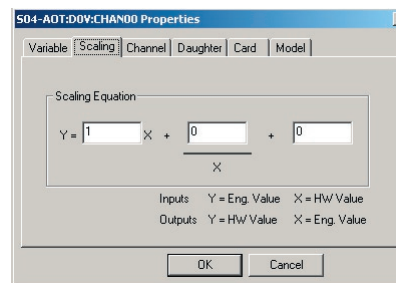
Scaling properties tab

Many variables support a Scaling tab, which allows you to set the scaling for that variable.

Note: The Scaling tab may not be present for a variable that supports scaling until you have set up scaling. To do this, choose Add Scaler from the context menu for the variable to display the Add Scaler dialog. This is identical to the Change Scaler Type dialog (see below), and is used to select the type of scaling. Once you have chosen the type of scaling to use, the Scaling tab is displayed in the Properties dialog for the variable so that you can set values for the scaling.

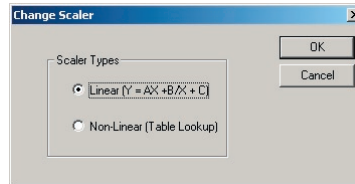
Two types of scaling are supported: linear scaling and non-linear scaling via a lookup table.

The default method is linear scaling, as shown here:

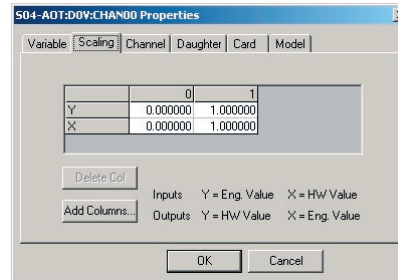


You can also apply non-linear scaling by changing the variable's scaling method:

1. Choose Change Scaler option on the context menu for the variable. You'll see the following dialog:



2. Select Non-Linear (Table Lookup) and click OK.
3. Display the Scaling tab in the Properties page again, and you'll see:



4. You can then enter a table of scaled engineering values against hardware values. Click Add Columns to add additional columns.

The table can contain between two and forty columns, and the model will apply scaling by means of linear interpolation between appropriate points in the lookup table. Where a value falls outside the range specified in the table, the model will extrapolate from the last point specified. The Engineering values are the values shown on controls and used in scripts, while the hardware values are the raw output values from the channel expressed in volts, or hertz, for example.

Displaying model variables

To display model variables in the Navigator, you should expand the variable groups in the I/O System Model branch (variables are grouped by I/O card). The Navigator shows which variables are present in the model, but not their current values.

- Viewing real-time model values

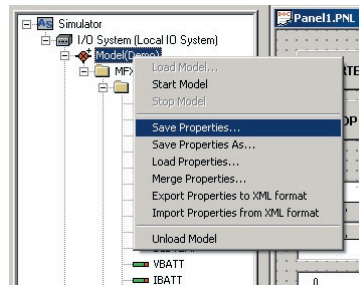
You can view the value of a variable in real time by dragging the variable from the Navigator and dropping it on a Panel. This creates a new instrument displaying the variable's real-time value. You must ensure that the variable is marked as in-use as described below.

- Viewing used/unused variables

The main model branch shows all variables, sorted by I/O card. Variables that are configured as 'in use' are also shown in the In-Use folder, which is part of the I/O System branch. (To configure a variable as in use, right-click on it in the Navigator and choose Properties from its context menu.)

To save the configuration of a model:

1. Choose Save Properties from the model's context menu:



2. Enter a name with a .MPR extension and click Save.

This will save the properties of each model variable in a .mpr file.

To load a previously-saved configuration:

1. Choose Load Properties from the model's context menu.
2. Use the Open dialog to locate the required folder and select the .MPR file to load.

If any of the properties cannot be matched, an error message is displayed. You may then choose to review a list of errors encountered.

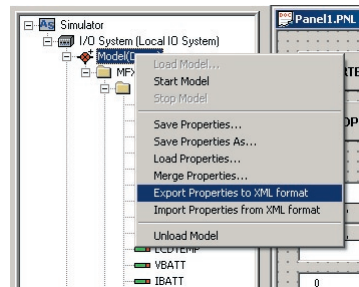
Importing and exporting of XML properties

Toolbench (3.2 and later) supports importing and exporting of XML properties. This allows the user to view the property files 'offline'. Exporting of model properties to XML format can only be performed after the model and '.mpr' properties have been loaded onto the IOS. Importing XML model properties can only be performed after a model has been loaded onto the IOS.

Exporting XML Properties

Follow these steps to export AutoSim model properties to the XML format:

1. Right-click on the Model branch of the Navigator.
2. Select the 'Export properties to XML format' menu item from the model context menu.



3. Select the file path to export the properties to.
4. Select the 'Save' button on the 'Save As' file dialog to export the model properties

The session log displays the progress of the export.

Importing XML Properties

Follow these steps to Import model properties into AutoSim model.

1. Right-click on the Model branch of the Navigator.
2. Select the 'Import properties from XML format' menu item from the model context menu.
3. Select the file path to import the XML properties from.
4. Select the 'Open' button on the 'Open' file dialog to import the model properties

The session log displays the progress of the import.

Please refer to the online user guide, 'XML model props User Guide.doc', located in the install directory, for more details.

To start a model on an I/O System:

1. Right-click on the Model branch in the Navigator.
2. Choose Start Model from the context menu.

To stop the active model:

1. Right-click on the I/O System in the Navigator, or the Model branch.
2. Choose Stop Model from the context menu.

Note: Stopping a model will stop script and logger activity on the I/O System, since they only work when a model is running.

Using Open Loop Models

The sample model provided with AutoSim, and the closed loop models that can be developed with the AutoSim Model Development Kits, all support a common open loop model. This model provides a direct interface to the hardware cards and the underlying channels and variables. This section describes the hardware structure and the individual variables that you can use to control the output and measure the input. Since closed loop models are essentially built on the infrastructure provided by an open loop model, much of the information in this section applies equally to closed loop models.

Overall structure

When you load a model, the GUI transfers the model to the I/O System. This then scans the installed hardware and returns a Navigator configuration that reflects that hardware, and any closed loop variables that the model also defines.

The MFX branch

The MFX branch is the top level model branch, which contains all of the hardware and user variable branches. The only property available for the MFX branch is the model property page, which allows you to adjust the model frame rate. This is the model cycle rate and may be set in 100 μ s steps from 500 μ s to 25.5ms. The new frame rate is applied when you next start the model. MFX is an acronym for **M**odel **F**ramework **eX**tension.

Beneath the MFX branch are individual branches for each of the installed cards, and branches for system statistics, user-defined variables and serial communications. These branches are described below.

Hardware branches

Each of the cards in the I/O System is represented by a branch in the Navigator. The top-level branch for each card tells you two things: the type of card, and the slot in which it is located. The slot numbers are in a range of 0 to 15, whilst the card types are represented by three-letter acronyms, which are given in the following table.

Acronym	Description
MST	Master card (Power Control)
AOT	Analog Output card
AIN	Analog Input card
DIO	Discrete I/O card
PIN	PWM Input card
POT	PWM Output card
PWG	Programmable Waveform Generator card
EVT	Event Timer card

Thus a branch with the label 'SO1-AOT' indicates that there is an Analog Output card (AOT) in slot 1 (SO1) of the rack. Each of the card types has its own configuration beneath the main card branch, and these are described in the following sections.

Master card Branch (MST)

The Master card is present in every system and is always reported as the first entry in the MFX branch (PCI-MST). The structure of the Master card is as follows:

VariableName	Range	Resolution	Description
TRIGGER 1...2	on/off		Scope Trigger output
SSWCOM	on/off		Internal switched supply rail control
SSW0...7	on/off		External switched supply rail control
VBATT	0–66V	16.1mV	External PSU, 'battery' voltage measurement
TEMP	0–50°C	0.1°C	Internal board temperature
VCTRL	0–10V	0.01V	External PSU, voltage control
ICTRL	0–10V	0.01A	External PSU, current control
	0–21A	5.1mA	External PSU, 'battery' current measurement
D0V:CHAN00...04	0–25V	0.01V	Analog inputs
D1V:CHAN05...07	0–25V	0.01V	Analog inputs
LOOMID	0–25V	0.01V	VLoom identifier input

TRIGGER1...2

This controls the digital trigger output on the I/O system front panel. The signal can be controlled from the user interface, or from within a model. Within a model, it may be used as momentary output to detect specific conditions, or to trigger an oscilloscope or logic analyzer.

SSWCOM

This is the main Sensor Supply rail within the AutoSim chassis, which controls the output from all of the other AutoSim I/O cards. When SSWCOM is off, all outputs are clamped to zero, simulating the lack of a sensor power supply on the simulated 'vehicle'. With SSWCOM enabled, the supply rails on the other I/O cards are enabled, and the outputs become live. The SSWCOM signal is switched from the External Battery PSU input to the I/O Chassis.

SSW0 ... SSW7

These are switched battery rails, which are derived from the External Battery PSU input to the chassis and are provided on the front panel of AutoSim. They are typically used to provide the main battery supply to the controller under test, and to provide switched power signals simulating different positions of an ignition switch.

VBATT

This is an internal measurement of the External PSU voltage being supplied to the I/O Chassis.

TEMP

This is an internal temperature measurement taken from the Master card and is used by the system as over-temperature protection.

VCTRL / ICTRL

These signals can be used to control the External PSU via two dedicated 0-10V analog outputs. The VCTRL line is used to control the voltage level, while ICTRL is typically used to set the PSU's current limit.

Note: It is very important to note that the output voltage specified for VCTRL is the control voltage to the External PSU, not the actual supply level that the PSU will generate. Therefore, when you set 5V on the VCTRL output you will be requesting 50% of the PSU's output, which may actually be 15V if it is a 30V PSU. Of course if you set a higher control voltage you may generate a supply rail that is sufficient to destroy your controller. It is therefore very important that you set appropriate scaling and limits for the VCTRL output. You should typically set the range limits to the acceptable range for the controller, not the full range of the PSU, which will usually be greater.

When setting scaling for the VCTRL output, you should use the following formula:

Scale Factor (A) = PSU Max Control Voltage / PSU Max Output Voltage

Thus, for a power supply with a 0 – 30V range and a 0 – 10V control range, the scale factor is calculated as:

Scale Factor (A) = 10 / 30 = 0.333

D0V:CHAN00 ... D1V:CHAN07

The D0V and D1V branches represent the two analog input daughter boards on the Master card, with each daughter board having four input channels. A standard Master card will be fitted with voltage input cards (hence the V suffix), but these may be replaced with current input cards, which will be shown by a C suffix. Each branch expands to display the individual channels, which represent the input voltage (or current) being measured.

Code	Description	Channels	Output Range
V	Voltage input	4	0–25V
C	Current input	2	–20mA to +20mA

LOOMID

This is the value read from the Loom ID inputs, which can be used to form an identification code for each cable harness. The Loom ID inputs are pulled high by the AutoSim, but you can create a unique Loom ID number for each cable harness. Unused inputs will be read as a logical 1, while grounded inputs are reported as logical 0. If no pins are grounded, the LOOMID will show a value of 255.

Analog Output card (AOT)

The Analog Output card supports up to four daughter boards, each of which provide up to four output channels, with a particular type of signal conditioning. Expanding the AOT branch shows the four daughter board branches with names D0X to D3X, where X represents the daughter board type, as shown below.

Code	Description	Channels	Output Range
V	Voltage output	4	0–20V
C	Current output	2	0–1.25A
R	Thermistor output	4	40Ω–100kΩ
T	Thermocouple output	4	0–49.75mV
L	Lambda sensor output	2	–20mA to +20mA
P	LVDT sensor output	2	See Hardware section
D	Differential voltage output	2	See Hardware section

Within each daughter board branch, the channels are shown in the form CHAN00, with numbers applied according to the daughter board's position on the Analog Output card. Each channel represents the output signal to be generated, and the default scaling and ranges are shown in the units applicable to that daughter board type. Thus a current output card will support two channels, each with the range 0 to 1.25A. These can be rescaled to provide engineering values by means of the Variable and Scaling property pages for each channel variable branch.

Applying Hardware Limits

The Voltage and Current cards also support a Channel property page, which allows you to set an absolute limit on the voltage or current generated by that channel. This is very useful in preventing damage to the controller under test by setting a hardware limit for the voltage output on a per channel basis. AutoSim will ensure that the output voltage never exceeds this maximum value irrespective of incorrect scaling, erroneous user input, script operation or model operation.

Analog Input card (AIN)

The Analog Input card supports up to eight daughter boards, each of which can provide up to four channels, with either voltage or current measurement capability. Expanding the AIN branch shows the eight daughter board branches with names D0X to D7X, where X represents the daughter board type:

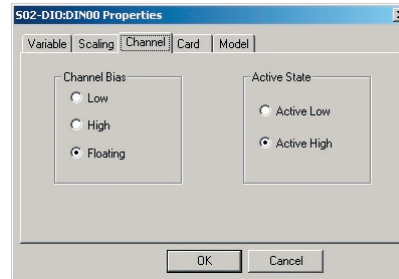
Code	Description	Channels	Output Range
V	Voltage input	4	0–25V
C	Current input	2	–20mA to +20mA

As with the Analog Output card, within each daughter board branch, the channels are shown in the form CHAN00, with numbers applied according to the daughter board's position on the main card. Each channel represents the input signal being measured, and the default scaling and ranges are shown in the units applicable to that daughter board type. Thus a voltage input card will support four channels, each with the range 0 to 25V. These can be rescaled to provide engineering values by means of the Variable and Scaling property pages for each channel variable branch.

Discrete Input/Output card (DIO)

The Discrete Input / Output card provides 16 channels of discrete input and 16 channels of discrete output. Each channel is represented by a single variable, named in sequence from DIN00 or DOT00 according to the channel type.

Each channel supports the standard Variable and Scaling property pages, but they also support a Channel page, which supports different properties for input and output channels.



Channel Bias (Digital Inputs)

This setting determines whether the input channel is configured to pull up to the high rail, down to ground or left floating. The channel bias must be set to work in conjunction with the drive circuit of the signal being measured to ensure that the driven and floating states can be distinguished. Thus, if the Controller's output circuit can only drive low (short to ground), the channel bias should be set to pull high so that the un-driven state can be detected.

Active State (Digital Inputs)

This determines the logical level for the channel. It does not affect the physical characteristics of the channel, but does affect the reported state of the channel variable.

Active State (Digital Outputs)

This determines whether the output is driven high (to the high rail) or low (to ground) when active.

Pulse Width Modulation Input card (PIN)

The PWM Input card supports up to 16 channels of PWM input measurement. When you expand the PIN branch you will see the 16 channels listed as CHAN00 to CHAN15. You can then further expand each channel to view the following variables associated with each channel:

Variable Name	Range	Resolution	Description
S99-PIN:CHAN01:Freq	0–20kHz	0.05%–1%	PWM input frequency
S99-PIN:CHAN01:Duty	0–100%	0.05%–1%	PWM input duty cycle
S99-PIN:CHAN01:State	on/off		PWM input state
S99-PIN:CHAN01:OnTime	0–10s	1μs	PWM input on-time

Therefore, for each input channel you can view four characteristics of the input signal; the frequency, duty cycle, state and on-time. The frequency and duty cycle are the main measurements taken for the input signal, but the current state and time spent in the on (or active) state may also be measured. Each of these variables has its own range and scaling, which can be set by the standard property pages.

There are some properties, however, that are related to the input channel rather than the individual variables. These are given in the following table:

Variable Name	Range	Resolution	Description
Active State	Hi/Lo		Sets active Hi (1) or Lo (0) state
Low Freq. Limit	0–20kHz	1Hz	Sets a minimum input frequency
Channel Bias	Hi / Lo / None		Sets channel input bias

Active State

This determines the logic level of the channel. It does not affect the physical characteristics of the channel, but does affect the calculations for duty cycle, state and on-time.

Low Frequency Limit

This sets a minimum frequency below which AutoSim determines that the input signal has become steady state, and sets the frequency to 0Hz, and the duty cycle to either 0% or 100% depending upon the current input state. If you do not use this option, AutoSim will continue to measure a low frequency signal on the input channel until it can confirm that it

is below the minimum measurable frequency of 0.12Hz. Since the latter can take up to 8 seconds to detect, it is often desirable to set a low frequency cut-off, which can then give loss of signal detection within a few model frames.

Channel Bias

This setting determines whether the input channel is configured to pull up to the high rail, down to ground or left floating. The channel bias must be set to work in conjunction with the drive circuit of the signal being measured to ensure that the driven and floating states can be distinguished.

Pulse Width Modulated Output card (POT)

The PWM Output card supports up to 8 channels of PWM output. When you expand the POT branch you will see the 8 channels listed as CHAN00 to CHAN07. You can then further expand each channel to view the following variables associated with each channel:

Variable Name	Range	Resolution	Description
S99-PIN:CHAN01:Freq	0 – 20kHz	0.1%	PWM output frequency
S99-PIN:CHAN01:Duty	0 – 100%	0.1%	PWM output duty cycle

Thus for each channel you can set the two main characteristics of the output signal: the frequency and duty cycle. Each of these variables has its own range and scaling, which can be set by the standard property pages. Note that the default value for the duty cycle is 50%.

Each variable also presents a Channel property page, which contains the Active State property, which is common to both variables:

Variable Name	Range	Resolution	Description
Active State	Hi/Lo		Sets active Hi (1) or Lo (0) state

Active State

This determines whether the output is driven high (to the high rail) or low (to ground) when active. Changing the active state will thus affect the polarity of the output waveform.

Programmable Waveform Generator card (PWG)

The Programmable Waveform Generator card supports 8 output channels, which can be configured to replay data patterns that are loaded from a pattern file during model initialization. The 8 channels are represented by the 8 channel branches CHAN00 to CHAN07. Within each channel, five variables are provided to give overall control of the output waveform:

Variable Name	Range	Resolution	Description
S99-PWG:01:Rpm	0 – 10000	1rpm	Engine or wheel speed
S99-PWG:01:Amplitude	0 – 75V	0.1V	For analog outputs only
S99-PWG:01:Pattern	0 – 15	1	See Note 1
S99-PWG:01:Slave	On / Off		Enable slave mode
S99-PWG:01:Phase	0 – 720°	0.1°	Offset from master channel

Rpm

The Rpm variable governs the rate at which the pattern is replayed. Changing Rpm has an immediate effect upon the output signal.

Rpm is scaled on the basis of a 0.1 degree sample rate in the pattern data. If you use a different resolution, you will need to change the scaling factor for Rpm to achieve a correct output.

Amplitude

The Amplitude variable determines the magnitude of the analog signal being generated, which can be up to 75V for a full-scale signal with no load. The actual value delivered by the output channel is dependent upon both the pattern data and the load it is driving. Changing the Amplitude has an immediate effect upon the output signal, but note that for digital outputs (see hardware details) it has no effect.

Pattern

The Pattern variable selects the pattern to be generated from the 16 patterns loaded for each bank of four channels. When a change to the pattern number is made, the change takes effect when the current pattern reaches its end point.

Slave

The Slave variable allows the channel to be slaved to the frequency (Rpm) of a Master channel, so that several channels may be controlled at a common frequency. When Slave mode is enabled, the channel's replay rate is locked to its Master channel, and its phase relationship is fixed by the Phase variable. The Master channel is the first preceding channel that is not itself in Slave mode. It is therefore possible to set up several Master/Slave relationships on a single card.

Phase

The Phase variable sets the phase relationship between the output channel and its Master when the channel is in Slave mode. The value specified sets the phase lag between the Master channel and the Slave channel. Where a phase lead is required, it should be entered as a phase lag of $(720 - \text{Phase Lead})$. The phase relationship actually changes at the end of each pattern cycle for the Master channel. When several channels are slaved to a common Master, the phase relationships are all relative to the Master channel.

Loading Patterns

The pattern data for the PWG card is split into two banks, each containing up to 16 patterns of 8192 bytes each. If you are using the default scaling provided for the Rpm and Phase channels, you should use patterns with a 0.1 degree resolution. The patterns are stored in pattern files and you can load them by specifying the file names in the 'Card' property page, accessible from the model variables or from the card branch itself. The pattern files may be specified as a CSV file (Comma Separated Variable, generated by Excel) or as an AutoSim PAT file. When loading a CSV file, AutoSim will automatically create a PAT file in the same directory that stores the data in a binary format. See the PWG card in the Hardware for information on how to generate pattern files using Excel.

Event Timer card (EVT)

The Event Timer card supports 32 input channels, which can measure events in both the time and angle domain to resolutions of 250ns and 0.1 degrees. The angle measurement is achieved by linking the PWG card to the EVT card within the AutoSim chassis, so that the angle clock used on PWG Channel 0 is also used for EVT measurement. Thus as the simulated engine speed varies, the measurements are taken against the actual engine position to a 0.1 degree resolution. However, in order to use this facility, you must use Channel 0 as the master engine speed. If you wish to use an alternate channel for this purpose you must change the hardware configurations of the PWG and EVT cards.

The EVT card supports four daughter boards, each of which manages eight input channels.

Each of the 32 channels supports the following five variables:

Variable Name	Range	Resolution	Description
S99-EVT:CHAN00:State	On / Off	Current state	
S99-EVT:CHAN00:Start	0 – 720°	0.1°	Start angle
S99-EVT:CHAN00:Stop	0 – 720°	0.1°	Stop angle
S99-EVT:CHAN00:Duration	0 – 10s	1µs	Duration of 'On' time
S99-EVT:CHAN00:Count	0 – 99	0.5	Number of pulses in 720°

State

The State variable indicates the current state of the input. For fast moving or transient events this will not show a correct indication on the GUI, but may be logged or tested within a script on the I/O System.

Start

The Start variable indicates the start angle of the event, where the start of the event is the transition from the inactive to active state. Currently, all angles are measured relative to the start of the pattern being generated by the PWG card (Channel 0). If no transitions occur during a complete pattern cycle, the value is set to 0.

Stop

The Stop variable indicates the stop angle of the event, where the input changes from the active to the inactive state. As with the Start variable, angles are measured relative to the

start of the pattern being generated by the PWG card (Channel 0). If no transitions occur during a complete pattern cycle, the value is set to 0.

Duration

The Duration variable indicates the time that the input signal was in the active state: the time between the start and stop events. Duration is measured in the time domain, and is independent of the PWG pattern. If no transitions occur during a complete cycle, the value is set to 0.

Count

The Count variable indicates the number of complete pulses that have occurred within a complete cycle. Where a pulse has started but not finished (or vice versa), an odd 0.5 is shown.

Channel Properties

The EVT card supports a set of properties for each channel that affect the set of variables associated with that channel. Each channel has the following properties:

Property	Range	Resolution	Description
Active State	Hi / Lo		Sets active Hi (1) or Lo (0) state
Channel Bias	Hi / Lo / None		Controls hardware setting
TDC Value	0 – 720°	0.1°	TDC for this input
Event Number	1 – 5	1	Event to measure
Configuration	See below		Input channel configuration

Active State

This determines the logic level of the channel. This does not affect the physical characteristics of the channel, but does affect the calculation of Start, Stop and Duration values.

Channel Bias

The Channel Bias setting determines whether the input channel is configured to pull up to the high rail, down to ground or left floating. The channel bias must be set to work in conjunction with the drive circuit of the signal being measured to ensure that the driven and floating states can be distinguished.

TDC Value

This setting is not used currently, but is used to make all angle measurements relative to a TDC (Top Dead Centre) value for each channel.

Event Number

The Event number determines which event should be measured and reported. By default this should be set to 1, but it may be set to 2 or higher where multiple events are expected on a single channel.

Configuration

The Configuration property sets the operation of the conditioning logic on the daughter board for that channel. The available options are given in the following table:

Option	Input
Single	Single inputs
Dual	Dual (differential) inputs
Dual Filtered	Dual (differential) inputs with pulse filtering
Mode 3	Reserved – do not use

Special channels

The EVT card is internally configured to measure two system events: the cycle position at the start of a model frame, and the total number of cycles completed. As with all other EVT channels, these values are measured using the angle reference from the PWG card (typically Channel 0).

Channel 29: Cycle Count

The count variable displays the total number of cycles completed since the model was started. The remaining variables for this channel are not updated.

Channel 31: Cycle Position

The Start and Stop variables display the angular cycle position at the beginning of the current model frame. The remaining variables for this channel are not updated.

In engine-based closed loop models, the Cycle Position indicates the instantaneous engine position at the start of each model frame.

Statistics Branch

The statistics branch contains timing information that may be useful when you are developing or working with closed loop models. Each branch provides information on a particular part of the I/O System's operation, and provides minimum, maximum, average and last values for the parameter being measured.

The individual branches are labeled as follows:

FRAMETIMES	The elapsed time from the start of one frame to the start of the next frame. This should correspond to the frame time set in the Model property page.
READHWINPUTS	The time taken to read all hardware inputs in a single frame, excluding any scaling or manipulation of model variables.
WRITEHWOUTPUTS	The time taken to write all hardware outputs in a single frame, excluding any scaling or manipulation of model variables.
MODELTIMES	The time taken to execute the model code, including variable scaling and any closed loop model code.
SCRIPTTIMES	The time taken to execute one iteration of the script (when active).
LOGGERTIMES	The time taken to execute one iteration of the logger (when active).
PROCESSTIMES	The total time taken to execute the model, script, logger and hardware input/output.

Note: The timing information is measured by the I/O System processor and is scaled according to the processor clock speed, which is set during I/O System software installation. The figures generated are therefore subject to rounding errors and should only be used as a guide to performance. All figures are reset to zero when the model is started.

The Statistics branch also includes two other variables, which are useful in checking system operation:

FRAMECOUNT	The number of frames executed since the model was started.
MISSEDTICKS	The number of missed frames since the model was started.

If you wish, you can of course log these variables in the normal way.

User Variables Branch

The USERVARS branch contains any variables that have been defined by the closed loop model, which is under the control of the model author. Refer to the model author for details of the variables provided and the actions that they perform. Note that closed loop model variables do not support user scaling.

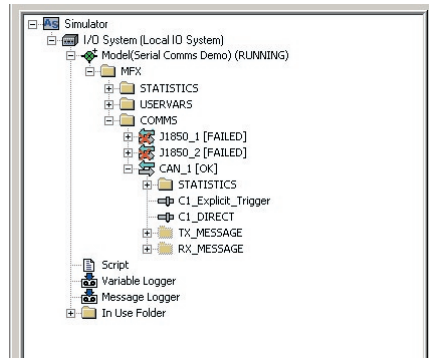
Serial Communications Branch

The COMMS Branch allows you to configure the protocols (if any) used for serial communications, and provides access to their message and trigger variables, and operating statistics. See Serial Communications near the end of this Software chapter for a full description of the Serial Communications Branch and how to use it.

Cosmetic branches and reference variables

The model author can create sub-branches within the USERVARS and COMMS branches to help organize what might be a very large number of variables so that you can find an individual variable more easily. These sub-branches are called cosmetic branches, as they do not reflect different AutoSim structures but are provided to ease access and usability.

Cosmetic branches are distinguished from the fixed branches by their 'ghost' folder icon with a gray outline. In the following example, the model author has created the cosmetic TX_MESSAGES and RX_MESSAGES cosmetic branches in the CAN_1 serial comms branch; the other branches in the picture are all fixed:



The example shows a typical use of cosmetic branches: the model author has created cosmetic branches so that the message variables for a particular serial comms card are grouped together with the receive and transmit message divided into separate groups.

This organization of the variables will help you find a specific one when you need it. Also, if variables are grouped logically so that those you will need to use together are grouped together, when you are working with a particular set, You are able to simplify the Navigator display by opening the branch containing the variables you are interested in and closing all the others. However, there could be a problem if a variable could logically be grouped in two or more branches. To get around this, the model author can define reference variables.

A reference variable is a pointer to a real variable, and is essentially indistinguishable from it – you can edit the properties of the real variable by opening the Properties sheet for the reference variable; you can link the reference variable to a meter control to monitor the value of the real variable; and you can link the reference variable to a slider control to change the value of a real variable. Any changes you make to a real variable are immediately reflected in any of its reference variables, and vice versa.

Working with Panels

This section tells you how to create, edit and save instrument panels, i.e. how to build up custom sets of Pi AutoSim instruments and controls. Please see the later section (Instruments and Controls) for details on how to configure individual instruments. You use panels to display instrumentation data, control outputs and provide editing facilities for tabular data. Instruments can be connected to model variables that are derived either from the Simulator I/O System.

You can create multiple panels, subject to system limitations such as disk space, and display different subsets of panels according to the tests you're performing.

The toolbar options

The following section displays four of the toolbars featured in AutoSim. It explains their use within the system.

- General toolbar – Allows you to switch between Design and Run modes for a control panel, turn the grid on and off, and gives you access to standard Windows file commands (New, Save, etc.):



- Controls toolbar – Allows you to add various 'virtual' instruments and controls to your control panel:



- Alignment toolbar – Allows you to align, center and space control panel instruments:



- Recorder toolbar – Allows you to record and play a sequence of actions in a macro file:

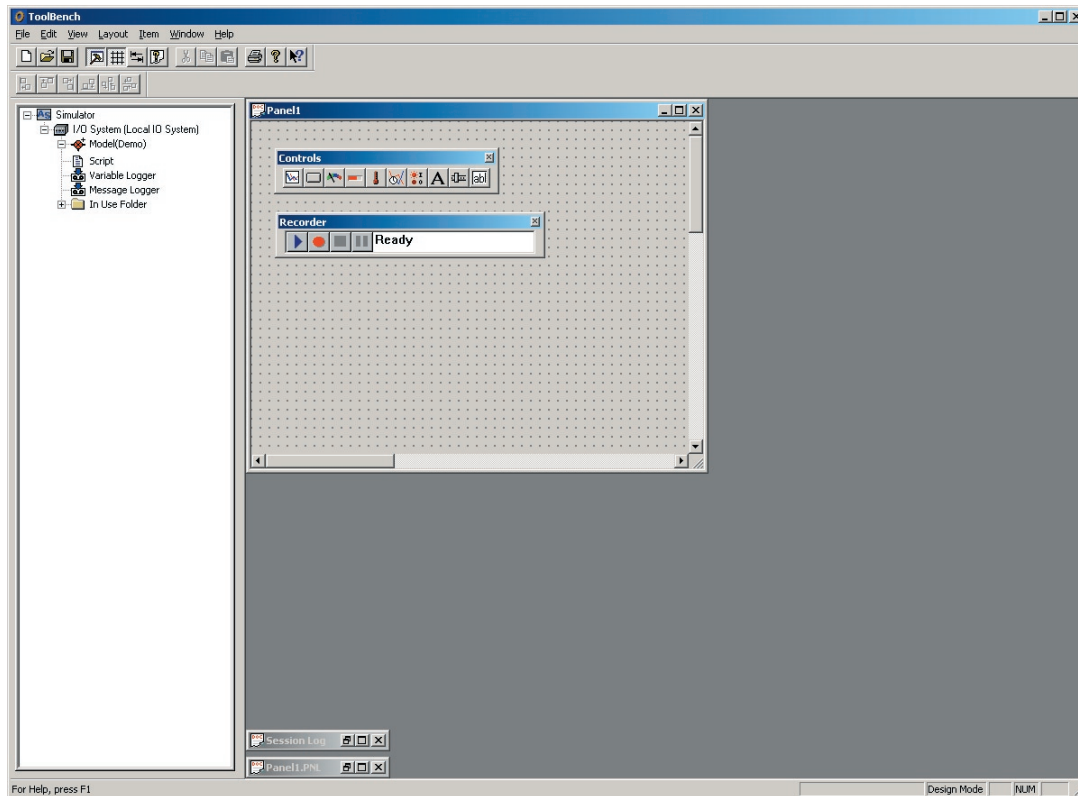


Managing toolbars

You can perform the following operations on any toolbar:

- Dock a currently undocked toolbar by dragging it into the toolbar area at the top of the window, or by double-clicking on its title bar.
- Float a toolbar by dragging it out of the toolbar area, or by double-clicking on its border.
- **Ctrl**-drag a toolbar to float it over an area of the screen where it would normally be docked.
- Hide or display a toolbar using the View menu (you can also hide a floating toolbar by clicking its Close icon).

The picture below shows the General and Alignment toolbars docked at the top of the window, while the Controls and Recorder toolbars are floating:



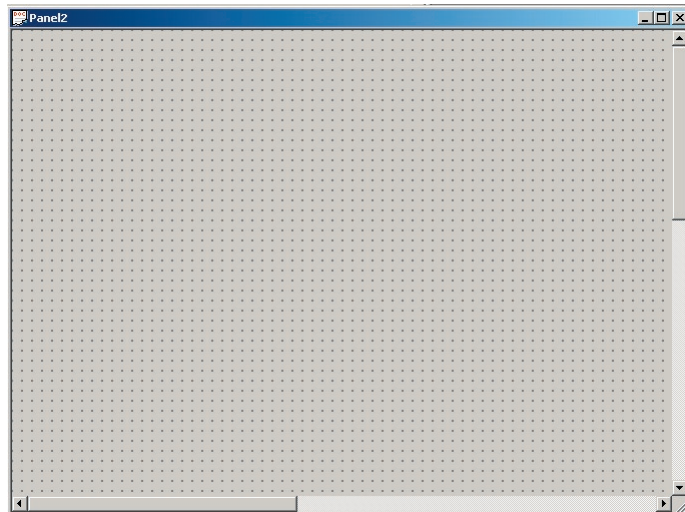
Creating and manipulating panels

By default, when you start Pi AutoSim, you'll be presented with an empty panel. You can then add instruments and controls to this blank panel, or open an existing panel (of type .PNL).

This section explains how to create and delete panels; the following section Adding, Deleting and Manipulating Instruments tells you how to add instruments and controls to a panel.

To create a new panel:

1. To open a new blank panel, choose File ► New ► Panel.
2. A blank panel will then be displayed ready for instruments to be added:



To close the active panel:

1. Choose File ► Close, or
2. Click the Close icon in the top right-hand corner of the panel window.

If any changes have been made to the panel's instruments, You are prompted to save the changes before the panel is closed.

Note: Closing a panel will not have any effect upon the I/O Systems being monitored by that panel. The active model and any active scripts will continue to run as normal.

To save an open panel to disk:

1. Click in the panel window.
2. Choose File ► Save (or Save As, if you want to specify a new filename).

Panel files are usually saved with an extension of '.PNL'. If you stick to this convention, your saved panel files will automatically appear in the Open dialog.

To open an existing (saved) panel:

If you have previously saved an instrument panel, you can open it as follows:

1. Choose File ► Open ► Panel. The standard Open dialog appears.
2. Browse the file system and select a panel file to open.

When the panel opens any instruments on the panel are loaded, and will attempt to connect to the model variables for which they were configured when the panel was originally saved. If these model variables are already available, the panel is displayed with its instruments configured and active.

If the model variables are not available (e.g. if you are running a different model), the instruments are displayed in an unconfigured state ready to accept new model variables.

To switch between panels:

If the panel you want to make active is visible, you can click in it to make it active.

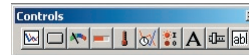
Alternatively, you can use the Windows menu to find a particular open panel, or use **Ctrl+F6** to cycle through all of the active windows.

Adding, Deleting and Manipulating Instruments

You must be in design mode to add instruments to a panel – when you are in design mode, Design Mode is checked in the File menu, the Status bar along the bottom of the AutoSim window shows Design Mode and the Design mode button in the General toolbar is shown depressed. To switch design mode on or off, choose File ► Design Mode or click the Design mode button in the General toolbar.

To add an instrument to a panel:

- To create automatically configured panel instruments, drag the model variables from the Navigator onto the panel,
- or,
- Use the Controls toolbar to add unconfigured controls to a panel:



If the Controls toolbar is not visible, choose View ► Controls Toolbar to display it.

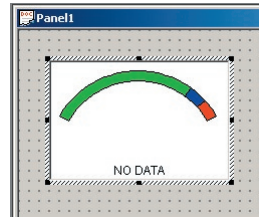
Note: You can dock or float the Controls toolbar in the same way as the Navigator – for example, double-click along its top edge to float it and double-click in the floating toolbar's title bar to dock it again.

Note: See Troubleshooting near the end of the Installation section in the Getting Started chapter if you are using a high-resolution screen and some parts of controls are clipped (for example, there are only partial graduations on a scale).

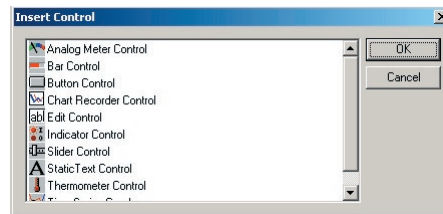
To add an analog meter:

- Click on the Analog control tool , then drag and drop it onto the panel (this creates a default analog meter),
- or,
- Click and release the tool in the toolbar, then drag out the size and shape of meter you want.

The instrument appears in the panel, ready for you to resize and position it and assign it a model variable.



You can also add an instrument by choosing Edit ► Insert Visual Object and selecting from the list of available instruments:



To delete an instrument from the panel:

1. Click on the instrument to select it and then do one of the following:
2. Choose Edit ► Delete.
3. Right-click on the instrument and choose Delete from the context menu.
4. Press **Ctrl** + **Del**.

To align two or more instruments:

1. Select the instruments by dragging a box around them, or clicking on them while holding down **Ctrl** or **Shift**.
2. Click the appropriate icon on the Alignment toolbar to align the instruments:
 - Align to a common edge (left, top, right or bottom):



- Align to a common vertical or horizontal centre line:



Note: You can achieve the same results using the Layout ► Alignment Controls menu options.

To snap instruments to a grid:

You can ensure that instruments automatically snap to a predetermined grid when you place or resize them:

1. Choose View ► Show Grid if the grid is not already displayed, or click on the grid icon  on the General toolbar.
2. Right-click on the panel and choose Snap To Grid (this toggles snapping on and off).
3. If you want to change the grid setup, right-click in the panel and choose Change Grid Spacing and change the settings in the dialog.

To move and resize instruments:

1. Select the instrument on the panel.
2. Drag one of its edges or corners, in the standard Windows fashion. The cursor will change to indicate the action you are about to perform.

You can move several instruments together by selecting them and then dragging one of them by its edge in the normal way. The selected instruments will move together, maintaining their relative spacing and alignment.

To copy and paste an instrument:

You can copy and paste an existing instrument on a panel to create new instruments with the same properties (size, shape, color, range etc.) as the existing instrument. Once you have pasted an instrument, you can modify its individual properties, or connect it to a new data source.

1. Select the instrument on the panel (you can only copy and paste one instrument at a time).
2. Choose Edit ► Copy or press **Ctrl+C** to copy or choose Edit ► Cut or press **Ctrl+X** to cut the instrument (or right-click on the instrument and choose the same commands from the context menu).
3. Choose Edit ► Paste or press **Ctrl+V** to paste the instrument (or right-click on the panel and choose the same command from the context menu).

The new instrument is pasted next to the currently selected instrument, or in the top left-hand corner of the panel if no instrument is selected.

Note: You can copy and paste instruments between different panels if required.

To change the orientation of an instrument:

1. Select the instrument.
2. Drag one of the corner resize handles.

Note: As you decrease the size of some instruments beyond a certain limit, the caption (i.e. the name, value and units) and graduations may disappear. However, the caption is displayed as a ToolTip when you move the pointer over the control (assuming that the caption is enabled via the General properties tab for the instrument).

To connect a model variable to an instrument:

1. Select the model variable in the Navigator.
2. Drag and drop it onto an instrument in the panel.

The instrument will automatically re-configure itself to accept the new model variable. The re-configuration process will include setting up the instrument to show the correct model variable name, units and value range.

Editing instrument properties

This section lists all the instruments and controls provided by Pi AutoSim, along with their configurable properties (note that you must be in Design mode to add controls to a panel). In general, you can configure an instrument's properties by:

- Left-clicking on the instrument in the panel and choosing, for example, Edit ► Analog Meter Control Object ► Properties... from the main application menu bar,
- or,
- Right-clicking on the instrument and choosing, for example, Analog Meter Control Object ► Properties... from the context menu.

Using any of the above methods will display the Properties dialog for the instrument. This usually consists of two or more tabbed panels.

Some properties (e.g. those in the General tab) are common to many instruments. They are described in the following three sections. Subsequent sections describe the properties particular to certain instruments.

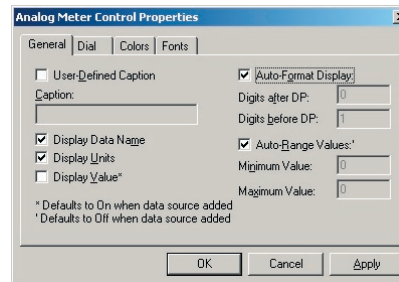
The General, Colors and Fonts properties tabs are common to the following instruments:

- Analog meter
- Slider control
- Bar graph
- Thermometer control
- Indicator control
- Edit control
- Button control.

Common instrument properties

To modify the general options:

Use the General tab to change an instrument's caption, and format the numerical display and range:



Changing an instrument's caption (name, value and units)

A caption displays an instrument's name, its current value and the units in which the value is expressed. You can change the default name, and display or hide any or all parts of a caption. The caption is also displayed as a ToolTip when the pointer moves over a control.

For example, in the caption for the Analog control shown below, RPM is the name (in this case, the default name), 753 is the value, and rpm is the unit:



- User-defined Caption

If this is unchecked, the name defaults to the name of the model variable in the Navigator (e.g. THROTTLE). Alternatively, check the option and enter some text in the Caption text box.

- Display Data Name

Display Units

Display Value

Check or uncheck these to change the caption displayed below the control, and in the ToolTip.

Formatting the display

You can specify the number of digits before and after the decimal point, or let AutoSim display the value in the most appropriate format:

- If Auto-format Display is checked (it is unchecked by default), AutoSim automatically formats the value with the correct number of digits before and after the decimal point.
- If it is unchecked, you can specify the Digits before DP (1 by default) and Digits after DP (0 by default).

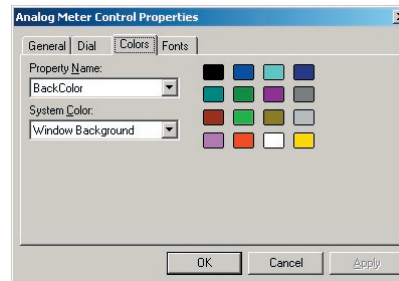
Setting the display range

If Auto-range Values is checked, AutoSim will dynamically change the range of the instrument as necessary.

If it is unchecked, you can specify the Minimum Value and Maximum Value to be displayed. Values that fall outside this range cannot then be displayed.

To modify the color options:

Use the Colors tab to change an instrument's foreground color (used to display the caption and legends), the background color, and other aspects of its appearance:



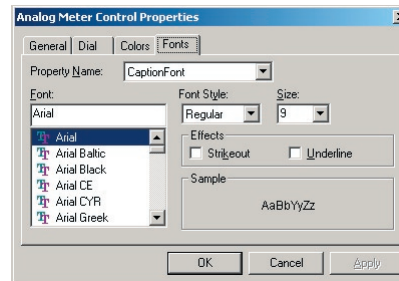
1. Choose the property you want to change from the Property Name box.
2. Choose a color from the System Color box, or click on one of the color swatches on the right of the panel.

When specifying a system color, the color changes in accordance with the current Windows color scheme.

3. Click Apply to apply the change and leave the dialog displayed.
4. Click OK when you have finished.

To modify the font options:

Use the Fonts tab to change the font used to display an instrument's caption and graduations, if appropriate:



1. Choose the property for which you want to change the font.
2. Specify the font name, style, size, and any effects (e.g. Underline). Arial, a Microsoft sans-serif font, generally gives a clear, uncluttered appearance.
3. Click Apply or OK.

Note: The color of the font is determined by the Foreground Color property, as described in the previous section.

Instruments and Controls

This section lists all the instruments and controls provided by Pi AutoSim.

You must be in design mode to add controls to a panel. When you are in design mode, Design Mode is checked in the File menu, the Status bar along the bottom of the AutoSim window shows Design Mode and the Design mode button in the General toolbar is shown depressed. To switch design mode on or off, choose File ► Design Mode or click the Design mode button  in the General toolbar.

In general, you can configure an instrument's properties by:

- Left-clicking on the instrument in the panel and choosing (for example) Edit ► Analog Meter Control Object ► Properties... from the main application menu bar,
- Right-clicking on the instrument and choosing (for example) Analog Meter Control Object ► Properties... from the context menu.

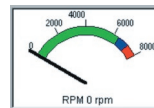
Either of the above methods will display the Property pages for the instrument.

Some instruments can only be used to display data (such as oil pressure), whereas others can be used to control Pi AutoSim outputs to a model, as well as display the current values of those outputs (e.g. load).



Analog meter

This is a display-only instrument. It displays a single data value on a dial, with a needle representing the current value:

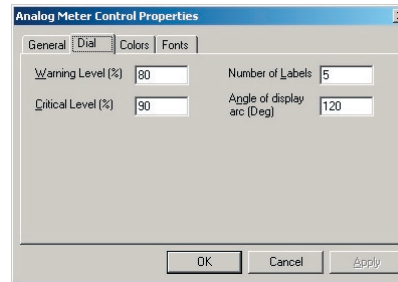


By default, there are three regions marked on the dial:

- Normal region (green) – values less than 80% of maximum
- Warning region (blue) – values greater than 80% of maximum
- Critical region (red) – values greater than 90% of maximum.

As well as the standard General, Colors and Fonts properties tabs described earlier, you

can use the Dial tab to change the following instrument-specific properties:

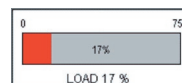


- Warning Level – Defines the start of the blue portion of the display (must be an integer)
- Critical Level – Defines the start of the red portion of the display (must be an integer, and should be greater than the Warning level, unless you want no warning level displayed)
- Number of Labels – Sets the number of graduations on the display (must be an integer between 3 and 21)
- Angle of display arc – Defines the angle of arc used for the display (must be an integer between 30 and 180).



Bar control

This is a display-only instrument. It displays a single data value as a vertical or horizontal bar graph:



By default, the bar is red on a background color taken from the current system color in the Windows desktop setup (this is gray in the default Windows color scheme). You can change the Windows default, or specify a custom background color using the instrument's properties page.

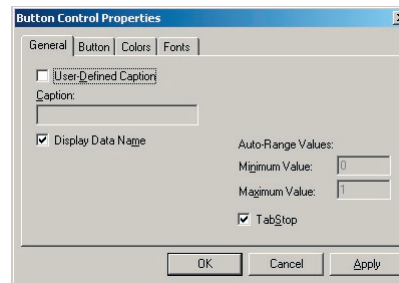


Push-button control

You can use a push-button for control and/or display. It displays a standard Windows push-button, with a legend showing the output being controlled.



You can use a push-button control to toggle the state of the simulator output (e.g. to start the simulator engine). As well as the standard General, Colors and Fonts properties tabs described earlier, you can specify the behavior of the button:



- Button Type – On/Off Latch, On/Off Non-Latch or Spin Dial.

If you set the type to Spin Dial, you also have the following options:

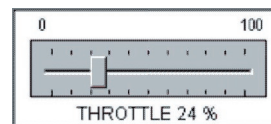
- Direction – None, Up, Down, Start, End, Centre or Reset.
- Step Size – Small or Large.

Note: The Spin Dial options can only be used with calibration data sources.



Slider control

You can use a slider for control and/or display. It updates as the value changes, but you can also use it to set a new output value. Either click the mouse to the left or right of the current bar position to increase or decrease the value, or grab the slider and drag it to the required position. You can also move the slider to the limits of its travel by selecting the slider itself and using **[Home]**, **[End]**, **[PgUp]** and **[PgDn]** or the cursor keys.



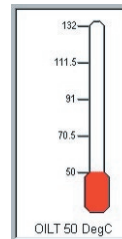
As well as the standard General, Colors and Fonts properties tabs described earlier, you can use the Slider control tab to:

- Graduations – sets the number of graduations on the display.
- Mouse Select – when enabled (the default) the user can adjust the slider using the mouse; when disabled, the user can only adjust the slider using the keyboard. Disabling mouse adjustment prevents users accidentally adjusting the slider setting when they select it with the mouse.
- Highlight – when enabled the control is highlighted when it is selected and responding to keyboard controls. In a complex panel this makes it clear which slider will respond to keyboard presses.



Thermometer

This is a display-only instrument. It displays a single data value on a vertical thermometer:



There are no warning or critical levels visible on this control. By default, the value is displayed by a red 'mercury' line on a graduated scale.

As well as the standard General, Colors and Fonts properties tabs described earlier, you can use the Thermometer tab to change the number of graduations on the display.

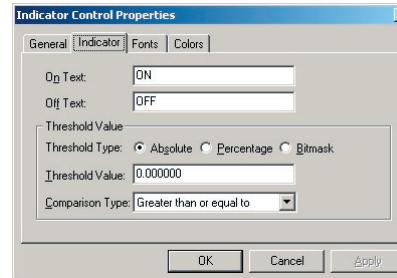


Indicator control

You can use an Indicator control to display a variable's status – on or off, or above or below its midpoint.



This control is useful as an LED indicator: you can reduce its size so that only a small colored box is visible. You are able to see the color change as the variable changes state. If you need to remind yourself what the indicator refers to, move the pointer over it to display the ToolTip, which shows the current caption. The control can display the word ON or OFF, or other text, as appropriate and change color. You can configure the text and background colors for each state:



ab| Edit control

You can use an edit control to display or edit a single data value in textual form (as a decimal number). It updates as the value changes, and you can use it to enter a new value for the variable.



To enter a new value, click inside the control, delete the existing value and enter a new one at the cursor. You can enter data in hexadecimal format by adding 'H' to the end of the number.

As well as the standard General, Colors and Fonts properties tabs described earlier, you can use the Edit tab to choose whether to display data in hexadecimal format. Hexadecimal data is shown with an 'H' suffix.

A Static Text control

You can use the Static Text control to add labels and instructions to the panel – unlike all the other controls, the Static Text control is not used to display or control variables (you cannot link Static Text controls to variables).

You can use the standard Colors and Fonts properties tabs as described earlier. To enter the text you want to appear in the control, display the General tab and type in the required text (the settings normally available on the General tab are not relevant for the Static Text control because the control is not linked to a variable).



Chart Recorder

You can use a Chart Recorder control to display a moving record of one or more data variable values. The Chart Recorder behaves in much the same way as a paper strip chart recorder, where the 'paper' scrolls from right to left showing a 'pen' trace of the value as it changes. However, one difference in the electronic version is that the data is lost once it goes off the left-hand edge of the display.

Each variable is assigned its own pen trace, and each trace can have its own vertical axis with an associated range and trace color. The chart's timebase can also be changed from the default 200 seconds.

To add a new trace to the Chart Recorder, simply drag the required variable from the Navigator onto the chart area. The new trace is added, and data is displayed immediately.

As with the other controls, you configure the appearance and behavior of the Chart Recorder using its Properties sheet. However, the Chart Recorder Control Properties sheet is identical to the Time Series Graph Properties sheet, and is used in exactly the same way. Therefore, to avoid repetition, they are described together after the description of the Time Series Graph control below.

Another feature shared by the Chart Recorder and Time Series Graph controls is the ability to display them in either 'Overlaid' or 'Tiled' modes. The use and behavior of these modes is identical for both controls, so again these modes are described below for both controls after the description of their Properties sheets.

A final feature shared by both the Chart Recorder and Time Series Graph controls is the use of Pan and Zoom mode to zoom in and out of the chart displays and to move around the zoomed displays. Pan and Zoom mode is described below for both controls after the description of Tiled and Overlaid modes later in this section.



Time Series Graph

You can use a Time Series Graph control to view one or more variable values recorded in a data log – see Using the Data Logger later in this chapter for how to record variable values in a log.

To add a new variable log to a Time Series Graph control:

1. Open the required log if it is not already open in the Navigator – see Using the Data Logger later in this chapter for details.
2. If necessary, expand the Navigator display so that the required variable is visible in the log.
3. Drag the variable from the log in the Navigator onto the graph area.

The new trace is added to the graph display immediately.

As with the other controls, you configure the appearance and behavior of a Time Series Graph control using its Properties sheet. However, the Time Series Graph Properties sheet is identical to the Chart Recorder Properties sheet, and is used in exactly the same way. Therefore, to avoid repetition, they are described together in the following section.

Another feature shared by the Time Series Graph and Chart Recorder controls is the ability to display them in either ‘Overlaid’ or ‘Tiled’ modes. The use and behavior of these modes is identical for both controls, so again these modes are described below for both controls after the description of their Properties sheets.

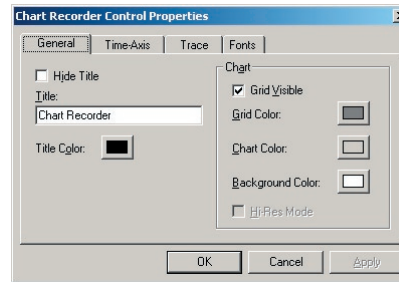
A final feature shared by both the Time Series Graph and Chart Recorder controls is the use of Pan and Zoom mode to zoom in and out of the chart displays and to move around the zoomed displays. Pan and Zoom mode is described below for both controls after the description of Tiled and Overlaid modes later in this section.

Configuring Chart Recorders and Time Series Graphs

You configure the appearance and behavior of Chart Recorders and Time Series Graphs using their Properties sheet. The Properties sheets are identical for both controls and are used in exactly the same way. This section describes how to use them.

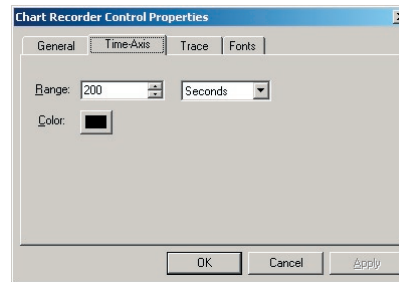
You can use the standard Fonts properties tab, described earlier, to set the font separately for the heading, x-axis scale and y-axis scale in the Chart Recorder or Time Series Graph.

The **General** tab allows you to set general properties of the control:



- Hide Title – Chooses whether to display the title text on the chart/graph.
- Title – allows you to enter or edit text to be used for the control's title.
- Title Color – Sets the color of the title text.
- Grid Visible – Chooses whether to display a scale grid on the chart/graph.
- Grid Color – Sets the color of the grid.
- Chart Color – Sets the color of the chart/graph background.
- Background Color – Sets the color of the control's background (the area outside the chart/graph itself – for example, the scales are displayed over the background).
- Hi Res Mode – This is not currently supported.

The **Time-Axis** tab allows you to set the time axis for the Chart Recorder or Time Series Graph:

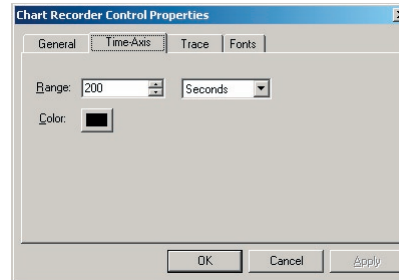


- Range – Sets the length of the time-axis: the left-hand box sets the number and the right-hand box sets the units. See Pan and Zoom mode below for how to zoom in to view a smaller section of the time axis in greater detail – you cannot adjust the Range setting for any of the traces when you are in Pan and Zoom mode.

Note: For a Chart Recorder, changing the Range resets the recorder when you click OK or Apply, so any existing recordings are removed.

- Color – Sets the color of the scale and scale text.

The **Trace** tab allows you to set the properties of the trace(s) displayed by the Chart Recorder or Time Series Graph:



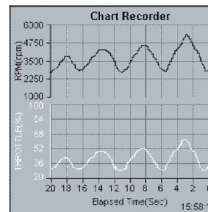
- Trace – Selects the trace to which the other settings on the tab apply; in Overlaid mode it also sets the trace that appears on top (see Tiled and Overlaid modes below for more details).
- Delete – Deletes the selected trace from the control.
- Color – Sets the color of the selected trace, its scale and its scale text.
- Show Units – Chooses whether to include the units in the scale text for the selected trace.
- Unit – Shows the unit for the selected trace: you cannot change the unit.
- #Graduations – Sets the number of labeled tick marks (and grid lines) along the y-axis for the selected trace: enter 0 to let AutoSim choose a suitable value.
- Position – Chooses where to display the scale for the selected trace. This setting has no effect in Tiled mode – the scales are always on the left-hand side in Tiled mode. If you are recording or displaying two quantities, you can display the scale for one on the left-hand side and the scale for the other on the right-hand side so that both are visible at the same time. If the scales for two or more traces are set to be displayed on the same side, the scale for the most recently selected trace is displayed.
- Digits after DP – Sets the number of digits to be displayed after the decimal point in the scale for the selected trace.
- Digits before DP – Sets the minimum number of digits to be displayed before the decimal point in the scale for the selected trace.

-
- Minimum – Sets the lowest value on the axis for the selected trace. See Pan and Zoom mode below for how to zoom in to view a smaller section of the y-axis in greater detail – you cannot adjust the Minimum setting for any of the traces when you are in Pan and Zoom mode.
 - Maximum – Sets the highest value on the axis for the selected trace. See Pan and Zoom mode below for how to zoom in to view a smaller section of the y-axis in greater detail – you cannot adjust the Maximum setting for any of the traces when you are in Pan and Zoom mode.

Tiled and Overlaid modes

When you are recording or displaying more than one variable, you can display a Chart Recorder or Time Series Graph in either Tiled or Overlaid mode.

- Tiled mode:



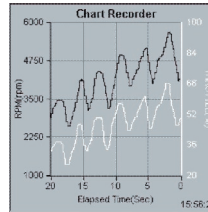
In Tiled mode the traces are displayed on separate charts arranged one above the other with a common time axis. The scale for each trace is displayed on its left-hand side, whatever the Position settings in the Trace tab.

To change from Overlaid to Tiled mode:

From the control's context menu, choose ...Control Object ► Tiled Mode (where ... is Chart Recorder or Time Series Graph).

Note: You cannot change between Overlaid and Tiled modes if you are in Pan and Zoom mode – see below.

■ Overlaid mode:



In Overlaid mode the traces are all displayed on the same chart with a common time axis, but can be distinguished from each other by setting different colors for them (a new color is set by default when you add a new variable for chart recording).

The Position settings in the Trace tab determine where the scale for each trace is displayed. However, if the scales for two or more traces are set to be displayed on the same side, the scale for the most recently selected trace in the Trace tab is displayed.

To change from Tiled to Overlaid mode:

- From the control's context menu, choose ... Control Object ► Overlaid Mode (where ... is Chart Recorder or Time Series Graph).

Note: You cannot change between Tiled and Overlaid modes if you are in Pan and Zoom mode – see below.

Pan and Zoom mode

Pan and Zoom mode allows you to view areas of a Chart Recorder record or Time Series Graph in greater detail.

Note: In Tiled mode you can only adjust the time axis; in Overlaid mode, you can also adjust the y-axes of all the traces simultaneously.

To switch Pan and Zoom mode on or off:

- From the control's context menu, choose ... Control Object ► Pan and Zoom Mode (where ... is Chart Recorder or Time Series Graph).

When a Chart Recorder or Time Series Graph is in Pan and Zoom mode, you can zoom in to produce a magnified view by dragging out a rectangle over the area of the display that you want to view in more detail. You can also zoom in using the Pan/Zoom tool, which also allows you to zoom out and to pan across the magnified display.

When Pan and Zoom mode is on, the Pan/Zoom tool is displayed:

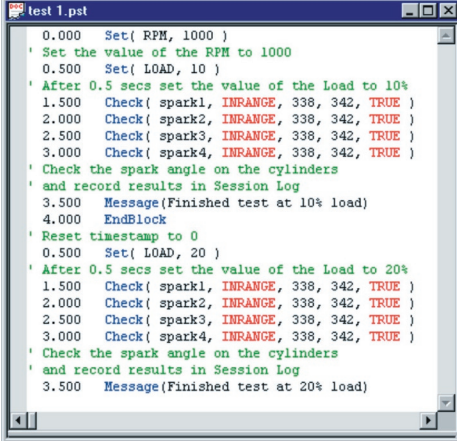


	Zooms into the chart/graph display to view the central part of the display in greater detail – alternatively, press [+] . In Overview mode, the display is magnified in both the time and y-axes; in Tiled mode, only the time axis is magnified.
	Zooms out from the chart/graph display to view the central part of the display in less detail – alternatively, press [-] .
	Resets the chart/graph display to show the full recording/logged data – alternatively, press [Shift] + [1] in the numeric keypad.
	The Pan buttons pan the display over a zoomed chart/graph in the direction shown by the arrow – alternatively, press the corresponding keyboard arrow cursor.

IOS Scripts

Pi AutoSim supports two types of automated test: IOS Scripts and GUI Macros. While both types of test can be generated using AutoSim's integrated editors and are run using the Recorder toolbar, they are different and are intended for slightly different tasks. This section concentrates on IOS Scripts. For more details on GUI macros, see the following section entitled Working with GUI Macros.

I/O Systems support interpreted scripts, which run at the frame rate of the active model. You use IOS Scripts to control output values and measure controller responses in real time:



```
test 1.pst
0.000 Set( RPM, 1000 )
' Set the value of the RPM to 1000
0.500 Set( LOAD, 10 )
' After 0.5 secs set the value of the Load to 10%
1.500 Check( spark1, INRANGE, 338, 342, TRUE )
2.000 Check( spark2, INRANGE, 338, 342, TRUE )
2.500 Check( spark3, INRANGE, 338, 342, TRUE )
3.000 Check( spark4, INRANGE, 338, 342, TRUE )
' Check the spark angle on the cylinders
' and record results in Session Log
3.500 Message(Finished test at 10% load)
4.000 EndBlock
' Reset timestamp to 0
0.500 Set( LOAD, 20 )
' After 0.5 secs set the value of the Load to 20%
1.500 Check( spark1, INRANGE, 338, 342, TRUE )
2.000 Check( spark2, INRANGE, 338, 342, TRUE )
2.500 Check( spark3, INRANGE, 338, 342, TRUE )
3.000 Check( spark4, INRANGE, 338, 342, TRUE )
' Check the spark angle on the cylinders
' and record results in Session Log
3.500 Message(Finished test at 20% load)
```

Using IOS Scripts, you can set values, generate ramps and create repeating waveforms (such as sine, square and sawtooth). You can also check input values to see whether they are within acceptable limits, with all actions typically taking place at millisecond resolution.

Each I/O System is capable of running a single IOS Script. The active script is displayed in the Script branch of the Navigator for that I/O System. You can add script files to this branch, or delete them. Scripts are actually run on the I/O System. The files required to run the script are automatically transferred to the I/O System via the network when you load a script. The following sections describe how to create, manage and run automated tests.

Note: The I/O System also supports compiled scripts that can be written in C or C++ for optimum performance and flexibility. This requires the use of a proprietary Microsoft compiler and is covered in a separate document for advanced users.

Pi AutoSim provides the following tools for working with IOS Scripts.

- Script Recorder toolbar – for recording and playing back automated tests
- Editor – for creating and editing scripts.

You'll often use a combination of both tools. For example, you might use the Script Recorder toolbar controls to record a basic test sequence, and then use the Editor to fine-tune it (e.g. alter timings). Once you've created and saved a test you can play it back.

Script Recorder toolbar

You can use the Script Recorder toolbar to record a sequence of events in an AutoSim session and save them as an IOS Script – you don't need to know command names, as the test file is created for you automatically. You can then use the Script Recorder to play back the IOS Script, possibly after editing, or to play back other IOS Scripts created in the Editor.

You can also use the Script Recorder toolbar to play back GUI macros – in fact, while you can run IOS Scripts in other ways, this is the only way to play back GUI macros.

To display the Script Recorder, choose View ► Script Recorder:



The Recorder has the following buttons:

- Play – Plays a macro or script (by default, plays the most recent)
- Record – Starts recording an IOS Script
- Stop – Stops the current IOS Script recording or GUI Macro playback session (you cannot use it to stop an IOS Script playback)
- Pause – Pauses the current recording/play back session (this control is not currently supported).

To record a new test file with the Recorder toolbar:

The Script Recorder allows you to create a new IOS Script without needing to know any commands. Here's an example of how to record a series of user actions in an IOS Script file:

1. Click the Record button 

A dialog is displayed so that you can select a folder and enter a name for the IOS Script file.

2. Enter a name for the test file.
3. Click Save.

Note: You can select an existing file here. If you do, you'll be prompted to confirm that you want to overwrite that file with the commands you're about to record.

The text in the Script Recorder toolbar changes to read Recording...00:00:01 (the timer increments automatically).

4. Perform the required sequence of actions on your panel instruments.

For each action, the corresponding script command is added to the file, with the appropriate timestamp.

5. When you have finished, click the Stop button 

This action stops the current recording session and automatically saves the file with the name you specified at step 2.

It also 'rewinds' the file so that it will play or record from the start of the file on the next Play or Record command.

To play back a test file with the Script Recorder toolbar:

1. Click the Play button: . The Playback Script/Macro dialog is displayed.
2. Use the dialog to select the required file (you can open GUI Macro and compiled IOS Script files as well as IOS Script files).
3. Click Open to play the file.

While the IOS Script is running, the Script branch in the Navigator shows the name of the script followed by the text (RUNNING).

If you have a suitable control panel open, you can see variable values changing as each command is executed.

Note: You cannot stop the playback of a script using the Script Recorder stop button. To stop a running script, right-click on the Script branch in the Navigator and choose Stop Script.

Once you've played the automated test, you might decide to edit it by changing or adding commands. See 'Using the Editor' below.

Using the Editor

You can use the Script Editor to create and edit automated script files. The Script Editor is a plain text editor with syntax coloring – when you enter a command name or other structure that the editor recognizes, such as a comment, it displays it in a distinctive color. This helps you check that you have entered the text correctly and makes it easier to read and understand it later.

See IOS Script commands later in this section for a definition of the Script commands and other language components you can include in an IOS Script.

To create a new IOS Script file using the Script Editor:

- Choose File ► New ► IOS Script.

or,

- Right-click on the Script branch in the Navigator and choose Add New.

This procedure automatically adds an entry for the new file to the Script branch.

A blank Editor window appears ready for you to enter and edit test commands.

To open an existing test file for further editing:

1. Choose File ► Open ► IOS Script.
2. Specify the file in the resulting dialog.
3. You are now ready to enter and edit test commands.

You can also open an existing IOS Script as follows:

- If the IOS Script appears in the Script branch of the Navigator, right-click on it and choose Edit Script.
- If an IOS Script file already exists on your filing system, but does not currently appear in the Navigator, right-click on the Script branch and choose Add Existing. This adds an IOS Script to the Script branch and opens an Editor window for the file.

The Script Editor toolbar and context menu

When a Script Editor window is open and selected, the Script Editor toolbar is displayed:



The tools in the toolbar allow you to cut, copy and paste text in the Script Editor, and to check its syntax (in other words, check that the Script is able to run).

The cut, copy and paste operations can also be carried out by selecting these commands from the context menu displayed by right-clicking in the Script Editor. The context menu also has commands for undoing and redoing text editing, and a command for adding or removing a bookmark (see To check the syntax of the text in the Editor later in this section for more information about bookmarks).

Copying, cutting and pasting in the Script Editor

You can copy and paste a selection of commands within a Script Editor window, or between two different windows. This is useful, for example, if you want to repeat a series of tests on a different variable, or include a test written in one script in another script.

1. Select the text you want to copy.
2. Either:
 - Right-click in the Editor and use the context menu to Copy (or Cut) and Paste the selection to the new location, or use the equivalent buttons in the Script Editor toolbar.
 - Or:
 - Drag the selection to the new location; **[Ctrl]**-drag to copy the selection to the new location

Note: When you move some code in a Script or between Scripts, there may be a problem with the ordering of the timestamps. To avoid this, you can use the EndBlock command to delimit blocks of code – the timestamps following an EndBlock command are taken relative to the EndBlock command, not the beginning of the Script.

To undo and redo actions in the Script Editor:

The Script Editor supports multiple-level undo and redo – right-click in the Script Editor and choose Undo or Redo, as required, from the context menu.

To check the syntax of the text in the Script Editor:

- Click the syntax check button in the Script Editor toolbar or press **[F7]**.

Any lines with syntax errors are marked with a bookmark (a blue circle in the margin).

You can move to the next bookmark in the editor by pressing **[F2]**.

*Note: You can also add or remove your own bookmarks by right-clicking in a line and choosing **Toggle Bookmark** from the context menu displayed.*

The results of the syntax check are also shown in the Script Editor's Status Pane. By default, the Status Pane is minimized. To view the Script Editor's Status Pane, move the pointer over the border just inside the bottom edge of the Editor window – the pointer shape will change showing that you can drag the border. Drag the border up to reveal the Status Pane. Any errors found by the syntax check are listed. You can double-click on an error message to select the line in the script causing the error.

Working with IOS Scripts

summary

To create a new IOS Script:

- Right-click on the Script branch and choose Add New,
- Or
- Choose File ► New ► IOS Script.
- Or
- Click on the Record button in the Recorder toolbar.

To add an existing saved IOS Script to the Script branch:

- Right-click on the Script branch and choose Add Existing and select the required file from the dialog displayed.

To open an existing IOS Script for editing:

- Choose File ► Open ► Script/Macro and select the required file from the dialog displayed.
- Or
- Right-click in the Script branch and choose Open.

To save an edited IOS Script file:

- Choose File ► Save or File ► Save As and enter a name for the file in the dialog displayed.

To remove an IOS Script from the Script branch:

- Right-click on the Script in the Script branch and choose Remove.

To start an IOS Script running:

- Right-click on the Script in the Script branch and choose Run Script.
- Or
- Click the Play button in the Recorder toolbar and specify the file name.

To stop a running IOS Script:

- Right-click on the Script branch and choose Stop Script.

To print a Script:

- Select the Script Editor containing the Script and choose File ► Print.

Working with multiple files

You can have more than one Editor window open at the same time. This is useful if you want to copy sections from one file into another, for example.

Overriding an IOS Script

You can override variables while a script is running by adjusting panel controls that affect those variables.

Note: The value set may be immediately overwritten unless you disable the 'Script can access' flag in the variable's property sheet, which prevents the script from altering that variable.

Running a test that doesn't match the current model

If you try to run a test in which none of the variables matches the variables in the current model, the script will not run and an error is recorded in the Session Log.

However, if only some variables do not match, the script will report an error message in the Session Log, but will continue with the script.

IOS Script commands

Interpreted script commands have the following format:

Time **Command (Parameters)**

The timestamp Time is defined in seconds, in the format 9.999, which allows you to specify events down to millisecond resolution. The Script code is executed in the order it is written, so the time specified for each line must increase as you go down the file. Generally this means that the Time numbers increase as you go down the file.

However, when an EndBlock command is executed, the timestamp is reset and the following Time numbers specify the time since the EndBlock command. In other words, timestamps are relative to the most recent EndBlock command or the start of the script.

Generally, scripts can operate at a loop time that is much faster than the device under test.

Comments

Script lines beginning with a single quote character (') are ignored by the interpreter, so can be used to add comments to the script.

Comment lines are colored green in the Script Editor.

Interpreted script commands and parameters

The following subsections describe the commands you can include in IOS Scripts.

Set **Set (Variable,Value)**

Sets the value of Variable to Value.

If the Variable is a read/write variable, it is set to manual mode so that it cannot be overridden by a closed loop model. You can return control of the Variable to the closed loop model using the Auto command.

The Set command takes effect immediately, and automatically cancels any existing Ramp or Function commands currently acting on the variable.

No range checking is performed, so it is possible to set a variable to a value that is outside its normal operating range.

Ramp

Ramp (Variable, Profile, Rate, Start, Stop)

Sets Variable to an increasing or decreasing series of values over a period of time.

The Ramp command takes effect immediately by setting Variable to Start. It then sets it to a new value on each cycle to produce the specified Profile, at the specified Rate until the Stop value is reached. When the ramp is complete, the Variable value will remain at the Stop value.

Start and Stop specify the initial and final values of the ramp as floating point numbers.

The Profile can have the following values, which also determine the interpretation of the Rate value:

Profile	Interpretation of Rate
R_ACCELERATION	units/sec ²
R_VELOCITY	units/sec

If the Variable is a read/write variable, it is set to manual mode so that it cannot be overridden by a closed loop model. You can return control of the Variable to the closed loop model using the Auto command.

Ramp automatically cancels any existing Ramp or Function commands currently acting on the variable.

No range checking is performed, so it is possible to set a variable to a value that is outside its normal operating range.

Function

Function (Variable, Pattern, Frequency, Min, Max)

Sets Variable to a continuous series of values ranging between Min and Max and following a specified Pattern at a specified Frequency.

Function takes effect immediately by setting Variable to Min and then sets it to a new value on each cycle to produce the specified Pattern, at the specified Frequency. The pattern is applied indefinitely to the Variable until the Set or Ramp command, or another Function command is applied to the Variable.

The following patterns can be used:

Pattern	Variable values set according to
F_SINE	sine wave pattern
F_SQUARE	square wave pattern
F_RAMP	ramp pattern (slope followed by step)
F_SAWTOOTH	sawtooth pattern (slope followed by slope)

Frequency specifies the frequency of the pattern in hertz. Note that the period of the signal is adjusted to be a multiple of the cycle time of the active model, so may not be exactly as specified.

Min and Max specify the initial and final values for the Variable in the pattern as floating point numbers.

If the Variable is a read/write variable, it is set to manual mode so that it cannot be overridden by a closed loop model. You can return control of the Variable to the closed loop model using the Auto command.

The Function command automatically cancels any existing Function or Ramp commands currently acting on the variable.

No range checking is performed, so it is possible to set a variable to a value that is outside of its normal operating range.

Auto Auto (Variable)

Sets a read/write Variable back to automatic operation after it has been overridden by a Set, Ramp or Function command. In automatic operation, it can be updated at each cycle by a closed loop model.

Check Check (Variable, TestType, Min, Max, LogFlag)

Checks that the value of the Variable is within the range specified by TestType, Min and Max as follows:

TestType	Check succeeds if the Variable value is
GT	greater than Min
GTEQ	greater than or equal to Min
LT	less than Min
LTEQ	less than or equal to Min
EQ	equal to Min
NEQ	not equal to Min
INRANGE	greater than or equal to Min and less than or equal to Max
NINRANGE	less than Min or greater than Max

Min and Max are floating point numbers.

LogFlag can have the value True or False. If LogFlag is True, the result of the check is recorded in the Session Log.

Use the Monitor command if you want to apply a test repeatedly.

Monitor Monitor (Variable, TestType, Min, Max, LogFlag)

The Monitor command is a continuous form of the Check command, which operates on every iteration of the model. The command monitors the value of the Variable over a period of time to check that it is within a range specified by TestType, Min and Max as follows:

TestType	Check succeeds if the Variable value is
GT	greater than Min
GTEQ	greater than or equal to Min
LT	less than Min
LTEQ	less than or equal to Min
EQ	equal to Min
NEQ	not equal to Min
INRANGE	greater than or equal to Min and less than or equal to Max
NINRANGE	less than Min or greater than Max

Min and Max are floating point numbers.

LogFlag can have the value True or False. If LogFlag is True, a message is recorded in the Session Log whenever the value of the Variable enters or leaves the specified test range.

The Monitor command will remain in effect until the MonitorStop command is executed.

MonitorStop **MonitorStop(Variable)**

The MonitorStop command terminates a Monitor command on the Variable. Summary results are recorded in the Session Log if LogFlag was set True in the Monitor command.

EndBlock **EndBlock**

Specifies the final time stamp of a block of code and resets the Script timer to zero, so that the time stamp of the next command is measured relative to the time of execution of the EndBlock command rather than from the beginning of the Script.

You can use the EndBlock command to write code blocks in which the time stamps are all relative to the start of the block. This code can then be copied into different parts of the Script, or different Scripts. Provided the code blocks are pasted immediately after an EndBlock command, they will still work correctly since the time stamps in the code blocks will still be valid.

You can have as many EndBlock commands in a script as you require.

Message **Message('message')**

Records the message in the Session Log.

You can use the message to document the log with a sign-on message showing the Script name and version when the Script starts to run, as well as to display status messages as the Script is running.

Working with GUI Macros

GUI Macros are sequences of commands that automate the high level operations of the User Interface such as loading or starting models, running scripts and performing log sessions. By building up GUI Macros you can create automation sequences that load a model, run a specific set of IOS scripts and log the results to one or more files. These sequences can then be run unattended, with the results saved to disk in the form of log files, or recorded in the Session Log as the output from the IOS scripts.

Timestamps are used, as for IOS Scripts, but to a 100ms resolution. However, timing on the GUI is not guaranteed as it depends on the multi-tasking of Windows, and whatever other applications are running at the time.

To create a new GUI macro:

- From the File menu choose New ► GUI Macro.

The GUI Macro editor opens – see Using the GUI Macro Editor later in this section for details.

Note: It is not possible to record a GUI macro.

To edit an existing GUI Macro:

1. Choose File ► Open ► GUI Macro.
2. Either enter the name of the file you are looking for or use the Browse button to locate the file.

The GUI Macro editor will open with the selected Macro loaded – see Using the GUI Macro Editor later in this section for details.

To save a GUI Macro:

- With the GUI Macro Editor selected, choose File ► Save, or Save As.

To replay a GUI macro:

The GUI Macro must be replayed from the Recorder. Do not attempt to replay macros from the Navigator Script branch, as this is used to replay real-time IOS scripts on the I/O System. GUI Macros run on the GUI PC only.

- Press the Replay button on the Recorder.

If a GUI Macro Editor window is selected, the GUI Macro in the window is played, otherwise an Open dialog is displayed so that you can select the GUI Macro (or IOS Script) file to play.

If the macro replay system encounters an error it will try to continue operation, and will resume its replay at the next command in the macro. This allows the system to make its best effort to replay the macro even if an error occurs that prevents a particular command from completing successfully.

You can press the Stop button on the Recorder to stop a GUI Macro playing.

To print a GUI Macro file:

- With the Editor selected, choose File ► Print.

Using the GUI Macro Editor

You can use the GUI Macro Editor to create and edit GUI Macros. The GUI Macro Editor displays a GUI Macro file in a split window.

The top part of the Editor window shows the contents of a GUI Macro file. If you select (highlight) a command in this part of the display, it appears in the lower part of the display, ready for editing.

You can create a new GUI Macro file from scratch, or you can open and edit an existing file. You need to know what AutoSim's GUI Macro commands do (see GUI Macro Commands at the end of this Working with GUI Macros section), but you do not need to have an in-depth knowledge of their syntax; the Editor handles this for you. Using the Editor, you can:

- Enter, change and merge commands using the context menu
- Copy and paste a selection of commands within the current file, or between two open files, by dragging and dropping
- Change a command's parameters and timestamp, and enter a comment.



Insert/Overwrite mode

The Editor supports an Insert/Overwrite mode, which allows you to automatically insert commands or overwrite existing commands. You can toggle the mode using the **Insert** key or the Insert button in the Macro Editor toolbar.

When the button is pressed, Insert mode is active.

To insert a new command into a GUI Macro file:

1. Enable insert mode.
2. Right-click on the command before or after which you want to insert the new command.
3. Choose Insert Before or Insert After from the context menu displayed.

A new default command is inserted above or below the currently selected command. By default, the timestamp is the same as that of the preceding command.

When you select a command in the top half of the editor window, its parameters are displayed in the bottom half of the window, ready for editing. To switch between the top and bottom half of the editor window, press **Ctrl** + **Tab**, or just click with the mouse.

You can also use the up and down cursor keys to move the command selection up and down the list in the top half of the Editor window.

The following subsections describe how to edit commands and their parameters.

Changing a command

To change the currently selected command:

1. Select the new command from the 'command' drop-down box. You can either use the mouse, or start typing the command name, and the best match is found.
2. Click in another field or on the selected command in the top half of the editor (or use the **Tab** key to change fields).

To copy and paste commands:

You can copy and paste a selection of commands within a GUI Macro Editor window, or between two different GUI Macro Editor windows.

1. Click to select a command.
2. If required, **Shift**-click to select a sequence of two or more commands and/or **Ctrl**-click to add to the selection.

3. Either use the context menu to Copy (or Cut) and Paste the selection within the same GUI Macro Editor window or to a different window, or drag and drop to copy it to a different GUI Macro Editor window.

The selected command(s) is inserted at the selected point and their timestamps set to the preceding command's timestamp.

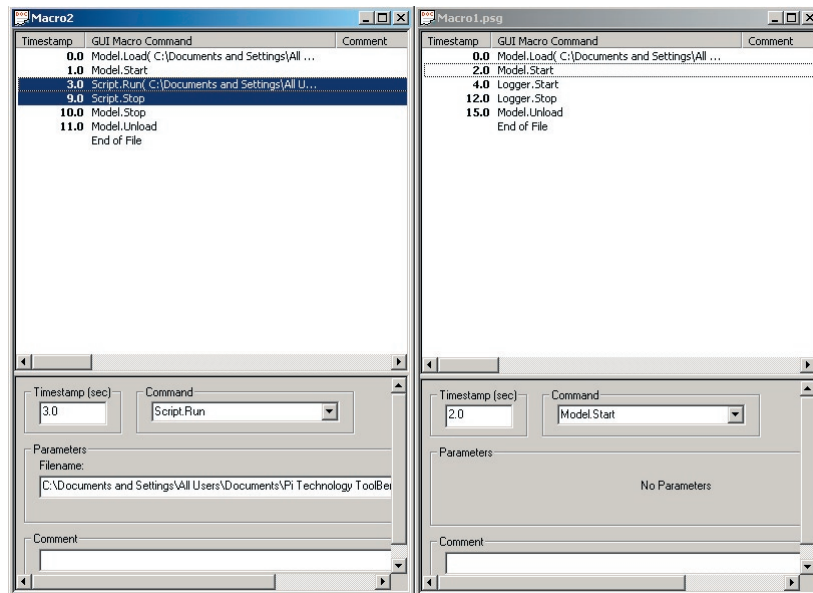
To merge commands:

Pasting (or dragging and dropping) commands places the new commands at the point in the file you select and sets their timestamps to the preceding command's timestamp.

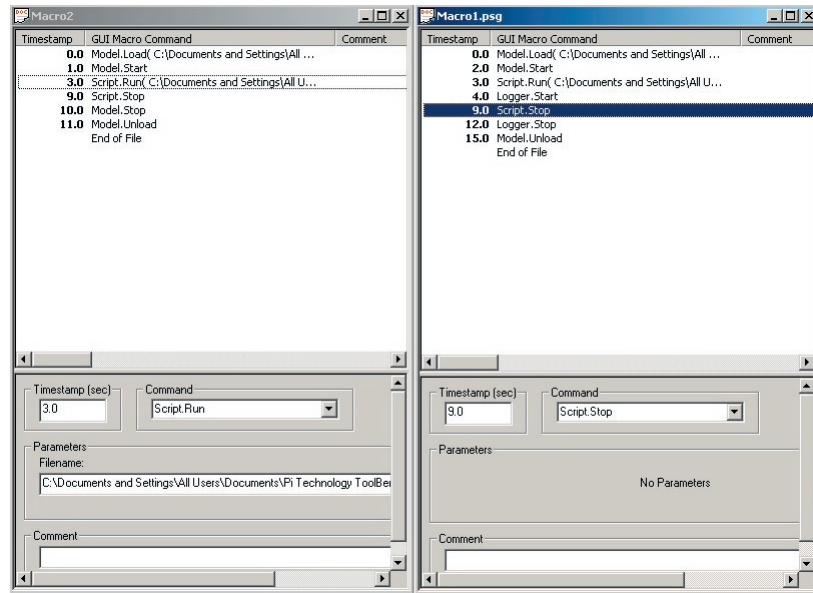
Merging commands allows you to copy commands between GUI Macro files while preserving the original timestamps – in other words, each merged command is inserted into the target file's command list according to its timestamp.

1. Open Editor windows for the source and destination files.
2. Copy a sequence of commands in the source file.
3. Right-click in the destination window and choose Merge.

For example:



The selection on the left is pasted into the file on the right

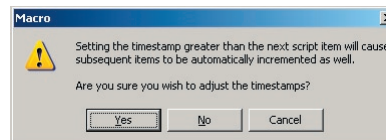


The right-hand Editor window now contains the merged commands

To change the timestamp for a command:

1. Click on the command in the top half of the editor to select it.
2. Enter a new value in the Timestamp(s) field in the bottom half of the editor.
3. Click in another field or the top half of the editor (or use the **Tab** key to change fields).

Note: If you enter a value that is greater than that for the following command, you'll be asked whether all subsequent timestamps should be incremented accordingly.



Changing a command's parameters

When a command is selected, the Parameters box changes to show the currently set parameters for the command. Many commands require no parameters. Several commands have a file as parameter, and in these cases a Browse button is displayed so that you can select the file. For other commands, one or more edit boxes are displayed for you to type in plain text parameters.

See GUI Macro Commands at the end of this Working with GUI Macros section for a list of commands and parameters.

To add a comment:

- Select a command and enter text in the Comment field in the bottom half of the GUI Macro Editor window.

The comment appears in the right-hand column of the GUI Macro Editor window.

GUI Macro Commands

The GUI Macro System supports the following commands:

Model.Load (filename)

Loads a model file on the connected I/O System, unloading an existing model if one is present. The macro command completes when the model is loaded, but before the model properties (and hence the Navigator) have been updated. You should add a delay after the command to allow time for the Navigator to update.

Model.Unload

Unloads the current model – You are asked if you want to save its properties. The command returns immediately.

Model.LoadProperties (filename)

Loads a set of model properties into the active model. The properties will typically include user names for variables, together with default values and ranges. The command returns when the properties have been loaded into the Navigator.

Model.Start

Starts the model running on the I/O System. The command returns immediately.

Model.Stop

Stops the model running on any I/O System. This will also stop any script, playback or log operations. The command returns immediately.

Script.Run (filename)

Loads and runs an IOS Script on the I/O System. This command will start the model running if it has not already been started with the Model.Start() command. The command returns as soon as the script has been started, rather than waiting for the script to complete.

Script.Stop

Stops the active IOS Script. The command returns immediately.

Logger.LoadConfig (filename)

Loads a log configuration from an existing log file. The configuration is transferred to the I/O System but logging is not started. The command returns when the configuration has been transferred to the I/O System.

Logger.Start

Starts the logger on the I/O System, starting the model automatically if required. The command returns immediately.

Logger.Stop

Stops the logger on the I/O System, but does not transfer the log file to the GUI PC. The command returns immediately.

Logger.Retrieve (filename,format)

Retrieves the log file from the I/O System and saves it on the GUI PC. The filename specifies the name and location of the file format, which may be AutoSim binary format (IOL) or Comma Separated Variable format (CSV). The command returns when the file has been transferred and the save is complete.

Logger.StartPlayback (filename)

Loads and starts the playback of a log file. The file is loaded and transferred to the I/O System and playback is started automatically. The command returns immediately.

Logger.StopPlayback

This command stops the log file playback on the I/O System. The command returns immediately.

User.Connect (ObjectName)

This connects to a third party automation 'object' that is already running on the GUI PC. This command implements the Visual Basic 'Get Object' command to allow subsequent User.Commands to control the object.

User.Command (Command Text)

Invokes the command with any specified parameters using the object defined by the User.Connect command. The 'CommandText' field is free-format, allowing the entry of any commands or parameter fields.

MessageLog.LoadConfig (filename)

Loads a message log configuration file. The configuration is transferred to the I/O System but logging is not started. The command returns when the configuration has been transferred to the I/O System.

MessageLog.Start

Starts the message logger on the I/O System, starting the model automatically if required. The command returns immediately.

MessageLog.Stop

Stops the message logger on the I/O System, but does not transfer the log file to the GUI PC. The command returns immediately.

MessageLog.Retrieve (filename)

Retrieves the message log file from the I/O System and saves it on the GUI PC. The filename specifies the name and location of the file. The command returns when the file has been transferred and the save is complete.

Using the Data Logger

The data logger allows you to log data from a running model, at the model's frame rate or some multiple of the frame rate. You can then analyze the data by viewing it in a Time Series graph on a control panel, or you can replay the log. Before starting a log session, you can define the variables that you want to log, and set the frame rate at which they are logged.

Logged data is captured and stored in a log file on the I/O System. Logged data is then transferred automatically to the User Interface PC, ready for display or replay when you stop a log session. The system asks you to confirm the transfer and assign a name to the log file.

Logging is controlled from the Variable Logger branch of the Navigator. You can add and remove log files, start and stop a data log or playback session, and change a log file's configuration (i.e. specify which model variables are logged and change the frame rate). The system allows you to play back a previously recorded data log session while logging is active, but not while a script is running on the I/O System.

Note: A model must already be running before you can use the data logger, otherwise you will not be able to start a log session.

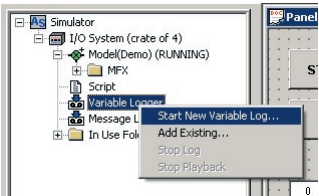
Before you start logging

Editing model and variable properties in the Working with Models section earlier in this Software chapter showed you how to enable and disable logging for individual variables (by default, logging is not enabled for a variable). However, when you start data logging you will get another opportunity to choose which variables to enable and disable for logging without having to open the Properties dialog for each variable.

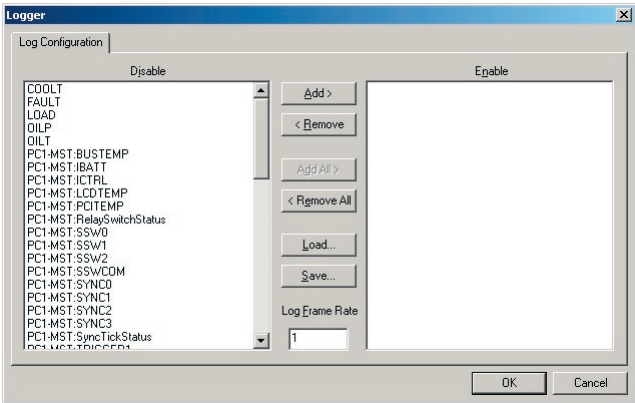
Logging data and managing log files

To start data logging with a new or saved configuration:

1. Right-click on Variable Logger in the Navigator:



2. Choose Start New Log to display the Logger dialog:



The dialog lists all the variables in the model.

The left-hand box lists the variables that have been disabled for logging; the right-hand box lists the logging-enabled variables.

-
3. To enable a variable, select it in the left-hand list box and click Add >, or just double-click on it.

You can select several variables at the same time in the usual way by **[Shift]**-clicking to extend a selection and **[Ctrl]**-clicking to add to a selection. You can then enable all the selected variables at the same time by clicking Add >.

Note: Since data logging can affect performance, Add All > is permanently disabled, to ensure that you have to make a separate decision for each variable that you want to enable for data logging.

4. To disable a variable, select it in the right-hand list box and click < Remove, or just double-click on it.

You can select several variables at the same time in the usual way by **[Shift]**-clicking to extend a selection and **[Ctrl]**-clicking to add to a selection. You can then disable all the selected variables at the same time by clicking < Remove.

5. To disable all the variables, click < Remove All.
6. To save the data logging settings, press Save and enter a name for the data logging settings file.
7. To load a previously saved set of data logging settings, press Load and select the required file.
8. Enter a number to set the Log Frame Rate – for example, enter 3 if you want the values of all the enabled variables logged on every third model iteration.

Note: The Log Frame Rate is not saved in the data logging settings file.

9. Click OK.

The Logger starts automatically.

- The Navigator shows a 'Logging' status for the current log file:

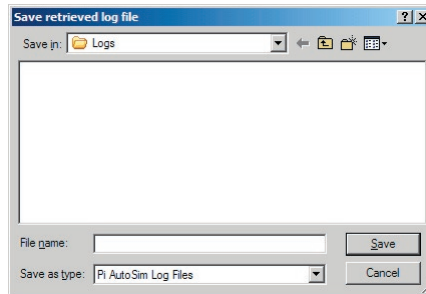


- The Session Log will record the time the Logger started.

To stop the current Logger session:

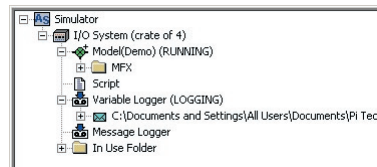
1. Right-click on Variable Logger in the Navigator and choose Stop Log from the context menu.

You are asked if you want to save the logged data to a file on disk:



2. Enter a name and location for the log file when prompted and click Save to save the log data.

The log file appears as an individual Log file branch in the Navigator, under the Variable Logger branch:



The Session Log records the time you stopped logging.

To add a log file to the Variable Logger branch:

1. Right-click on the Variable Logger branch and choose Add Log.
2. Browse the file system on the User Interface PC and select the file you want to add.

Note: The default format for saving log files is as binary data with a .IOL extension, but they can also be saved and reloaded as a comma separated values text file with a .CSV extension – see To save a log file with a different name or format later in this section.

3. Click OK.
4. The file appears as a new branch in the Variable Logger branch.

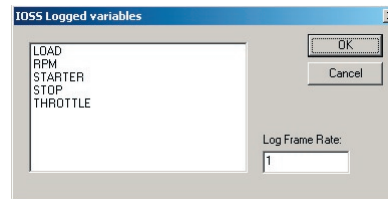
To remove a log file from the Navigator:

1. Right-click on the individual log file branch.
2. Choose Remove Log.
3. If the log file contains any unsaved data, you'll be prompted to save it.

To start data logging using the same configuration as an existing log:

Note: You cannot add to or remove variables in the variable list for existing log files. However, you can change the names of the variables to match other variables in the model. For example, you could change the variable names for a log file created for a model with different names to the names of the equivalent variables in the current model – see To rename a variable in a log file.

1. Right-click on a log file in the Variable Logger branch and choose Start log.
2. The Logger configuration is displayed:



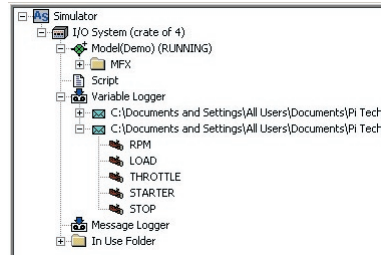
3. Change the log rate, if required.
4. Click OK to start logging data.

To save a log file with a different name or format:

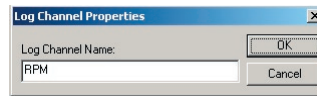
1. Right-click on the log file under the Variable Logger branch.
2. Choose Save As from the context menu.
3. Use the Save as type drop-down list box in the Save As dialog to specify whether to save the log file in:
 - ☐ AutoSim format, with a .IOL extension (this is the default)or,
 - ☐ ASCII text (CSV) format.
4. Select a folder and enter a name for the file, then click Save to save it.

To rename a variable in a log file:

1. Expand the branch for the individual log file, to display the logged variables:



2. Right-click on the variable and choose Properties.
3. Enter the new variable name in the Log Channel Name field and click OK.



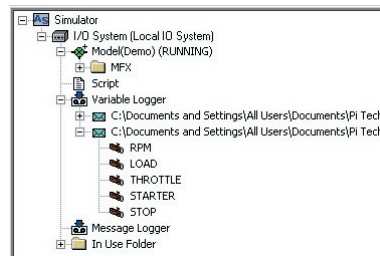
Note: If you use Save As to save the log file, it is saved with the new variable names. However, if you remove the log from the Variable Logger branch, the file is removed and any changes to variable names made since you last saved it are lost.

Viewing and playing back logged data

Once you have created a log file, you can display a graph showing how the value of the logged variables changes in time or play the log file back in a model.

To display logged data on an AutoSim control panel:

1. Expand the required log file branch to show the available logged variables.



2. Create a Time Series Graph control on the panel: 
3. Drag your chosen variable from the Navigator onto the Time Series Graph control and it is plotted automatically.

See Time Series Graph in the Instruments and Controls section earlier in this Software chapter for further details of using the Time Series Graph control.

Playing back log data

You can play back any log file under the Variable Logger branch in the Navigator – see 'To add a log file to the Variable Logger branch' earlier in this section if the file you want to use is not already loaded.

You can change the names of the variables in a log file (see 'To rename a variable in a log file' above) so that, for instance, you can play back a script recorded on a similar model with different variable names. You cannot change the number of variables logged, the frame rate or the data recorded.

To start and stop log data playback:

1. Right-click on an individual log file branch and choose Start Playback.

The Navigator shows a 'Playing' status for the active log file and a message is sent automatically to the Session Log.

2. To stop a playback session, right-click on an individual log file branch, or on the Variable Logger branch, and choose Stop Playback.

Note: If you try to play back a log file that does not match the current model, it will ignore the variables that do not match those in the current model.

Note: You can play back a file while logging is active, but not while an IOS Script is running. You cannot run an IOS script while you are playing back a log file.

Session Logs

A Session Log displays messages generated by the active components in the system, such as the Simulator I/O Systems (and their user models and scripts) or Calibration Devices. Each component generates its own status messages indicating key events (model started or stopped, script loaded, etc.) or error messages (could not load model, etc.). Each message is date and time-stamped.

The output from IOS scripts (e.g. the results of Check commands) and GUI macros is sent to the Session Log.

To display the session log:

The Session Log window is opened automatically when you connect to an I/O System, but if you close it (by choosing File ► Close when it is selected, or click the window's close button), you can reopen it:

1. Choose File ► New ► Session Log.

This opens the default Session Log as a separate window.

2. If the Session Log is already active, use the Window menu to display it, or press **Ctrl** and **F6** until it is displayed.

The new Session Log window is empty – there is no record of events that occurred while it was closed.

The columns in the Session Log window are re-sizeable. They contain the following information:

Time	Date/Time of the message in locale sensitive format
Message	The message text

The Session Log displays the list of events that have occurred since the Session Log was started. You can scroll through the log to review old messages.

When the Session Log window is selected, the menu bar in the main AutoSim window changes to show a different set of menus – in particular, there is a Tools menu and the View menu only contains the Toolbar command, which shows or hides the Session Log toolbar:



Clearing the Session Log

You can clear the contents of the Session Log by using the Tools ► Clear Session Log option or by pressing the 'Clear' toolbar button: .

Saving the Session Log contents

You can save the contents of the Session Log to a text file by using the Tools ► Export as Text menu option or by pressing the 'Save as' button on the toolbar: .

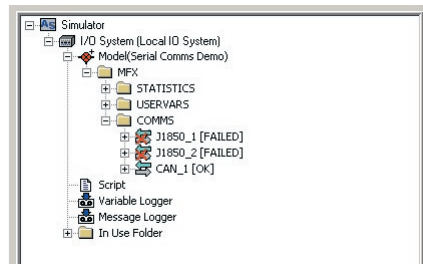
The Session Log information is saved as ASCII text format with a tab delimiter between the columns.

Serial Communications

AutoSim supports a range of different types of communications hardware and protocols, such as CAN and Class 2 J1850. This section of the manual describes features and mechanisms within AutoSim for handling local communications that are independent of the specific hardware and protocols used – see the on-line Help and separate documentation for information on specific hardware and protocols.

The COMMS Navigator branch

If a model has been defined to use serial communications, the COMMS branch in the Navigator shows serial comms icons for the protocols specified in the model (the I/O system running the model may contain other serial comms hardware, and may not contain all the hardware assumed in the model):



In this example, the model assumes two CAN connections and a J1850 connection.

The [OK] next to CAN_1, shows that the CAN_1 connection is present and enabled; the [FAILED] next to J1850_1 shows that the I/O System has not been able to detect a J1850 connection, because it is missing, disconnected or malfunctioning in some way. The icons for the serial comms protocols in the Navigator show when an item is OK but do not distinguish between the disabled and failed conditions.

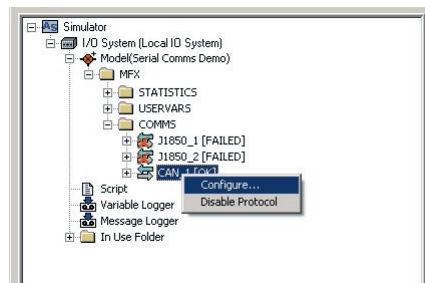
Even if the serial comms hardware used by a model is absent, has failed or has been disabled, you can still run the model – it behaves in the same way as a system connected to serial comms hardware when no messages are received, and no messages are transmitted. In particular, this means that you can run a serial comms model on an I/O System that does not have any serial comms hardware.

Configuring serial comms hardware

The default configuration for the serial comms protocols can be set on the I/O System using the IOS Comms Configuration utility on the I/O System. This means that the model can be independent of a specific configuration, so that the same model can be run on differently configured systems. It also means that you do not have to set the configuration details of the protocol configuration separately for each model.

However, if required, you can also change the configuration within the AutoSim software:

1. Right-click on the serial comms item for the hardware in the Navigator:



2. Choose Configure from the context menu to display the Protocol Configuration Properties dialog. The contents of this depend on the specific hardware selected – see the on-line Help for how to use it.

If you save the model properties (right-click on the model in the Navigator and choose Save Properties) any changes you have made to the serial comms configuration are saved with it, and can be loaded back at a later time using Load Properties.

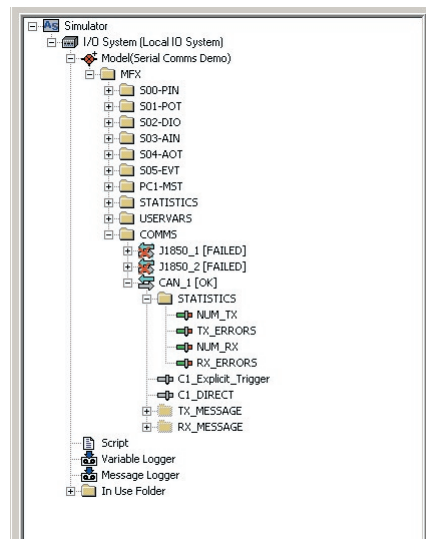
Enabling and disabling serial comms protocols

If required, you can disable a serial comms protocol used by a model even when the hardware is present and operating correctly in the I/O System:

1. Right-click on the serial comms protocol branch in the Navigator.
The context menu contains the command Disable Protocol when the protocol is enabled and the command Enable Protocol when the protocol is disabled.
2. Choose Disable Protocol from the context menu to disable the protocol in the model; choose Enable Protocol to enable it again.

Note: You can only enable or disable a protocol while the model is stopped.

Serial comms protocol branches



The serial comms protocol branch in the Navigator always contain a STATISTICS branch, but the model author will often also add:

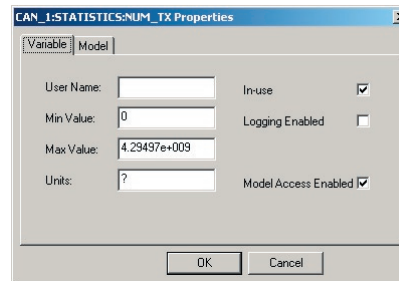
- Other variables – C1_Explicit_Trigger and C1_Direct in the example above.
- Cosmetic branches – TX_MESSAGE and RX_MESSAGE in the example above.

Serial comms statistics

The STATISTICS branch in a serial comms protocol branch in the Navigator contains variables counting the number of transmitted and received messages and the number of transmission and reception errors.

As the icons show, the comms statistics variables are read/write, unlike the model statistics variables. In particular, this means that you can reset them while the model is running using the following procedure:

1. Drag the comms statistics variable from the Navigator onto a panel to create an edit control linked to it.
2. Right-click on the variable and choose Properties from the context menu:



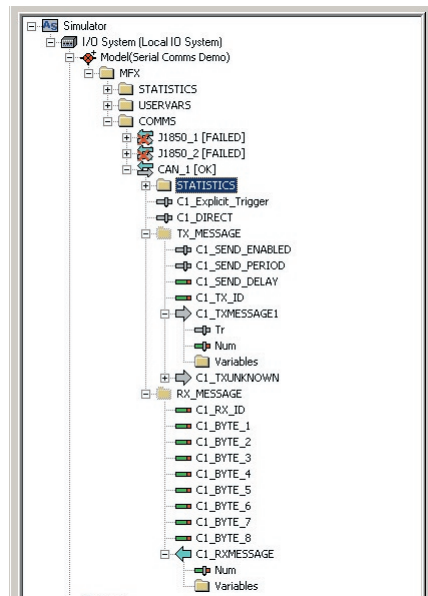
3. Uncheck Model Access Enabled and click OK.
4. Use the edit control to set the variable to 0.
5. Redisplay the Properties dialog and check Model Access Enabled again so that the model is able to increment the counter.

Message variables

AutoSim allows the model author to define message variables for individual messages and include entries for them in the Navigator. Typically, an author will place message variables within the branch for the serial comms protocol used to transmit or receive the message (either directly in the serial comms protocol branch or in a cosmetic branch contained within it), but they could be placed elsewhere in the Navigator tree, such as in a cosmetic branch in the USERVARS branch.

Note: Model authors can write code to send or receive messages without defining message variables for them. However, message variables provide a number of useful features for the model author and the AutoSim Software user – see [To control message logging and Trigger Variables](#) below for details.

When a model author defines a message variable and specifies its location in the Navigator, a number of items are automatically created within the message variable branch:



Transmit message variables always contain:

- a trigger variable with the name Tr
- a statistics variable with the name Num
- a Variables branch.

Receive message variables always contain:

- a statistics variable Num
- a Variables branch.

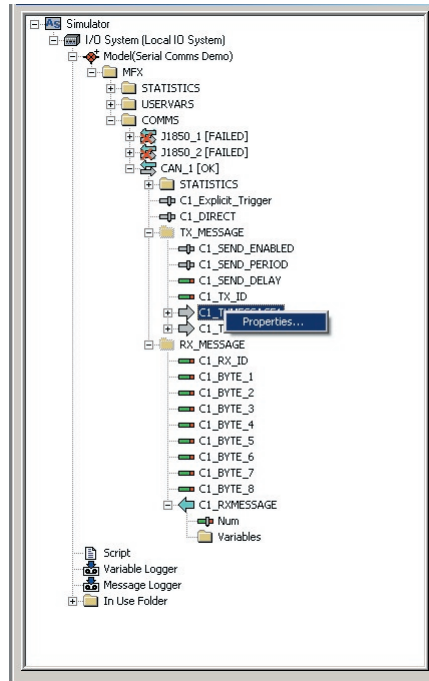
Typically, the model author will use the Variables branch for variables related to the message so that you can get to them easily without having to search through the whole USERVARS branch.

The statistics variable (Num) is a counter that is automatically incremented whenever the message is transmitted or received. The variables are read/write, so like the statistics variables in the serial comms protocol's STATISTICS branch (but unlike those in the main STATISTICS branch), you can reset them while the model is running – see Serial comms statistics earlier in this section.

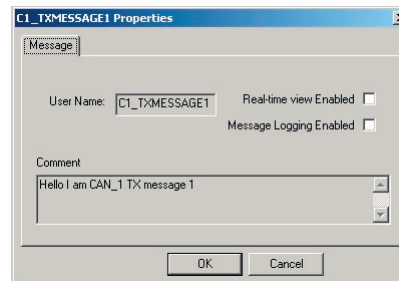
Typically, trigger variables are used to transmit the associated message, but the model developer is free to define any other action for the trigger – see Trigger Variables later in this section for details.

To control message logging:

1. Right-click on the message variable:



2. Choose Properties to display a Properties dialog:



Note: The dialog is identical for transmit and receive messages.

Note: The User Name and Comment are defined by the model author and cannot be edited by the AutoSim Software user.

3. Check Real-time view Enabled to display a record in the Real Time Viewer (see Using the Message Log and Real Time Viewers later in this section) giving details for the selected message when it is sent or received.
4. Check Message Logging Enabled to display a record in the Message Logger (see Using the Message Log and Real Time Viewers later in this section) giving details for the selected message when it is sent or received.

You can also enable and disable message logging in the Real Time Viewer and Message Logger for several variables at the same time – see Setting up the Real Time Viewer and Setting up and Managing Message Logging later in this section for details.

Trigger variables

When a model author defines a message variable, an associated trigger variable is automatically created for it, and appears within the message variable in the Navigator tree. However, model authors can also create their own trigger variables, which may or may not be used for serial communications.

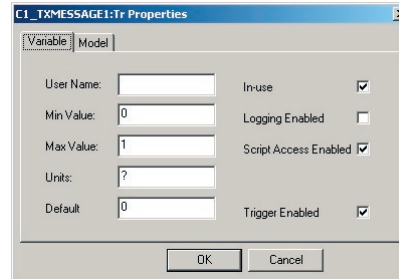
Trigger variables can be assigned values in a model, by a script or by using panel controls, in just the same way as other write variables. However, if the trigger is enabled (see the next subsection), setting the value of a trigger variable to a non-zero value also causes some action defined by the model author to be carried out at the end of the current model iteration – typically, this action would be to send the associated message, but the model author could use a trigger to carry out many other actions, or no action.

The fact that the action is carried out when the value is set to a non-zero value but not when it is set to zero means that if you link the trigger variable to a non-latching button on a panel, pressing and releasing the button will only trigger the action once.

Note: If you drag a trigger variable from the Navigator to a panel, a button control is created and linked to the variable. However, the default button is latching. If you connect a trigger variable to a non-latching button, you will need to press it twice (off then on) to trigger the action once. You can use the button's Properties dialog to make it non-latching.

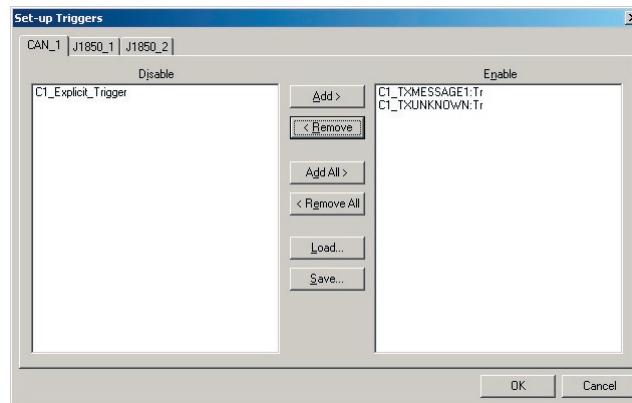
Enabling and disabling trigger variables

Trigger variables have the same properties as other write variables, which you can view and edit in the usual way. However, they also have the property of being enabled or disabled. You can enable or disable an individual trigger using the Trigger Enabled check box in its Properties dialog, which you display in the usual way by right-clicking on the trigger variable in the Navigator and choosing Properties:



However, you can also enable or disable a group of triggers together:

1. Right-click on the COMMS branch in the Navigator and choose Set-up Triggers to display the Set-up Triggers dialog:



The dialog has a tab for each of the serial comms protocol Navigator branches listing all the message variables placed within the corresponding branch in the Navigator. If the model author has placed any message variables in the Navigator outside the serial comms protocol branches, there is an Unassociated tab listing them all.

The left-hand box in each tab lists the triggers that have been disabled; the right-hand box lists the enabled triggers.

2. To enable a trigger, select it in the left-hand list box and click Add >, or double-click on it.

You can select several triggers at the same time in the usual way by (Shift)-clicking to extend a selection and (Ctrl)-clicking to add to a selection. You can then enable all the selected triggers at the same time by clicking Add >.

-
3. To enable all the triggers listed in a tab, click Add All >.
 4. To disable a trigger, select it in the right-hand list box and click < Remove, or just double-click on it.
You can select several triggers at the same time in the usual way by **[Shift]**-clicking to extend a selection and **[Ctrl]**-clicking to add to a selection. You can then disable all the selected triggers at the same time by clicking < Remove.
 5. To disable all the triggers listed in a tab, click < Remove All.
 6. To save the trigger settings for this tab, press Save and enter a name for the trigger settings file.
 7. To load a previously saved set of trigger settings for this tab, press Load and select the required file.

Note: If the settings file was saved for a different protocol, You are warned and asked if you want to load them anyway.

Note: Any triggers on the tab that are not listed in the settings file are disabled.

Setting up and managing message logging

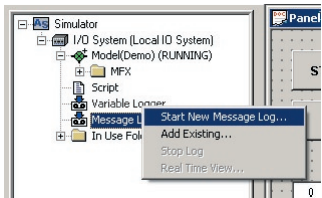
The Message Logger allows you to create a record of messages as they are transmitted and received and then view it.

Note: This section shows you how to set up and manage message logging. You view completed message logs in the Message Log Viewer. However, this is very similar to the Real Time Viewer, so they are described together in Using the Message Log and Real Time Viewers at the end of the main Serial Communications section.

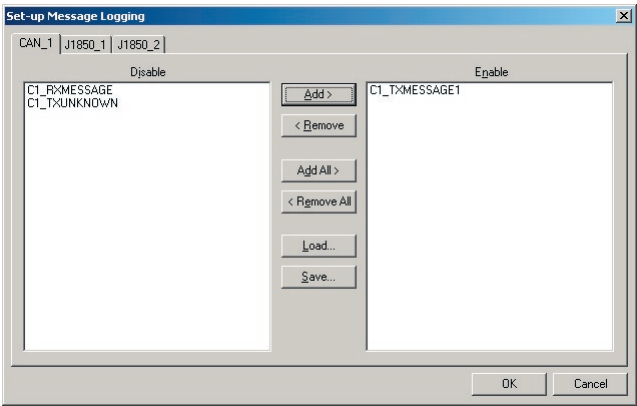
Message variables earlier in this section showed you how to enable and disable message logging for individual variables. However, when you start message logging you will get the opportunity again to choose which variables to enable and disable for logging.

To start message logging:

- 1. Right-click on Message Logger in the Navigator:



- 2. Choose Start New Message Log to display the Set-up Message Logging dialog:



The dialog has a tab for each of the serial comms protocol Navigator branches. Each tab lists all the message variables placed within the corresponding branch in the Navigator. If the model author has placed any message variables in the Navigator outside the serial comms protocol branches, there will also be an Unassociated tab listing them all.

The left-hand box in each tab lists the message variables that have been disabled for logging; the right-hand box lists the enabled message variables.

3. To enable a message variable, select it in the left-hand list box and click Add >, or just double-click on it.

You can select several variables at the same time in the usual way by (Shift)-clicking to extend a selection and (Ctrl)-clicking to add to a selection. You can then enable all the selected variables by clicking Add >.

4. To enable all the message variables listed in a tab, click Add All >.
5. To disable a message variable, select it in the right-hand list box and click < Remove, or just double-click on it.

You can select several variables at the same time in the usual way by (Shift)-clicking to extend a selection and (Ctrl)-clicking to add to a selection. You can then disable all the selected variables at the same time by clicking < Remove.

6. To disable all the message variables listed in a tab, click < Remove All.
7. To save the message logging settings for this tab, press Save and enter a name for the logging settings file.
8. To load a previously saved set of message logging settings for this tab, press Load and select the required file.

Note: If the settings file was saved for a different protocol, you are warned and asked if you want to load them anyway.

Note: Any message variables on the tab that are not listed in the settings file are disabled.

9. When you have enabled all the variables you want to log and disabled the rest, click OK to start logging. Message Logger in the Navigator shows that you are logging:



To stop message logging:

1. Right-click on Message Logger in the Navigator and choose Stop Log.
2. You are asked if you want to transfer the log to a file on your PC.
If you click No, the message log is deleted.
If you click Yes, a Save As dialog is displayed so that you can specify a name and location for the log file.

The message log is added to the Message Logger branch in the Navigator:

**To add a saved message log to Message Logger in the Navigator:**

1. Right-click on Message Logger in the Navigator and choose Add Existing to display an Open dialog.
2. Select the required file and click Open.

The selected message log is added to the Message Logger branch in the Navigator.

To remove a message log from the Message Logger in the Navigator:

1. Right-click on the log in the Message Logger in the Navigator.
2. Choose Remove Log.

To view a message log:

Either:

1. Right-click on the log in the Message Logger in the Navigator.
2. Choose Open Message Log.

Or:

1. Choose File ► Open ► Message Log to display an Open dialog.
2. Select the required file and click Open.

The selected message log is opened in the Message Log Viewer.

Note: The Message Log Viewer and Real Time Viewer are very similar and are described together in Using the Message Log and Real Time Viewers at the end of this Serial Communications section.

Setting up the Real Time Viewer

Note: The Real Time Viewer can be a useful tool for debugging models, but it can affect performance, as it may generate overruns. Where performance is an issue, you should use message logging instead.

The Real Time Viewer allows you to view a record of communications activity as it happens.

Note: This section shows you how to set up the Real Time Viewer. The Real Time Viewer and the Message Log Viewer are very similar and are described together in Using the Message Log and Real Time Viewers following this section.

Message variables earlier in this section showed you how to enable and disable real time monitoring for individual variables. However, you can also choose which variables to enable and disable for the Real Time Viewer in a single dialog, which is also displayed when you start the Real Time Viewer.

To select which message variables to monitor in the Real Time Viewer:

- Right-click on Comms in the Navigator and choose Set-up Real Time View to display the Set-up Real Time View dialog.

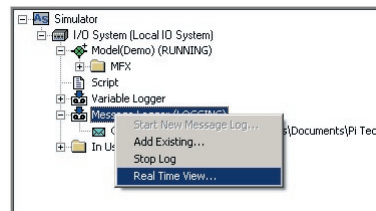
This dialog is identical in form to the Set-up Message Logging dialog and you use it in exactly the same way to enable and disable variables, and to save and load settings files – see Setting up and managing message logging earlier in this section for details.

Note: This procedure allows you to choose which variables to monitor in the Real Time Viewer, but does not display the viewer. See the following procedure for how to select the variables and start up the viewer.

Note: You can carry out this procedure while the Real Time Viewer is running, to change which variables are monitored.

To display the Real Time Viewer:

1. Right-click on Message Logger in the Navigator:



2. Choose Real Time View to display the Set-up Real Time View dialog.

This dialog is identical in form to the Set-up Message Logging dialog and you use it in exactly the same way to enable and disable variables, and to save and load settings files – see Setting up and managing message logging earlier in this section for details.

When you click OK to confirm your settings, the Real Time Viewer will open – see the following section for how to use it.

Using the Message Log and Real Time Viewers

The Real Time Viewer and the Message Log Viewer share many features, so they are described together in this section. Attention is drawn, however, to where there are differences in functionality or behavior.

Note: The maximum number of messages that can be viewed in the Viewer is 64,000. If a message log contains more messages than this, the Message Filter dialog is displayed (see To Filter the Message Log Viewer below) automatically when you open the message log so that you can make the number of messages to be displayed less than 64,000. When the Real Time Viewer contains 64,000 messages, You are prompted to clear it and given the opportunity to save the contents.

The viewer window

Type	Iterations	Time	Protocol	Name	ID	Data/Error
	0	15:43:47.845	CAN_1		00	00 00 00 00 00 00 00 00
	0	15:43:47.845	J1850_1		00	00 00 00 00 00 00 00 00
	0	15:43:47.845	J1850_2		00	00 00 00 00 00 00 00 00
	0	15:43:47.847	CAN_1		00	
	4	15:43:47.864	J1850_2			break in rx message
	5	15:43:47.875	J1850_1		00	00 00 00 00 00 00 00 00
	13	15:43:47.900	J1850_2			Warning: a tx message has not been trans...
	2647	15:43:58.435	J1850_1	J1_TXMES...	00	31 32 33 34 35 36 37 38...
	2654	15:43:58.463	J1850_1	J1_TXMES...	00	32 32 33 34 35 36 37 38...
	2656	15:43:58.469	J1850_1		00	31 32 33 34 35 36 37 38...
	2656	15:43:58.471	J1850_2	J2_RXMES...	00	31 32 33 34 35 36 37 38...
	2662	15:43:58.495	J1850_1	J1_TXMES...	00	33 32 33 34 35 36 37 38...
	2662	15:43:58.495	J1850_2	J2_RXMES...	00	32 32 33 34 35 36 37 38...
	2662	15:43:58.497	J1850_1		00	32 32 33 34 35 36 37 38...
	2670	15:43:58.527	J1850_1	J1_TXMES...	00	34 32 33 34 35 36 37 38...
	2671	15:43:58.530	J1850_1		00	32 32 33 34 35 36 37 38...

The picture shows the Message Log Viewer; the Real Time Viewer is identical apart from the title, though its contents may change as it updates – see To enable updates in the Real Time Viewer.

The viewer shows records of the following communications events:

	Message received
	Message transmitted
	Transmission acknowledgment
	Errors/Warnings

You can filter the records in the Message Log Viewer (but not the Real Time Viewer), so that only certain types of record are included – see To Filter the Message Log Viewer below for how to do this and how to apply other filter conditions.

You can choose which columns to display and whether to display in hex or decimal format in either viewer – see To display and hide columns below.

When a viewer window is selected, the menu bar in the main AutoSim window changes to show a different set of menus – in particular, there is a Tools menu and the View menu only contains the Toolbar command, which shows or hides the Viewer toolbar:



In practice, since some tools only apply to the Real Time Viewer and some to the Message Log Viewer, at least one tool will always be disabled in the toolbar.

To close the viewer window:

- Choose File ► Close.

Or

- Click the close button at the top-right hand corner of the window.

When you close the Real Time Viewer, You are asked if you want to save its records in a message log file before it is closed.

To choose to display data in Hexadecimal or Decimal format:

- Choose Tools ► Hexadecimal or Tools ► Decimal.

Or

- Click the HEX or DEC button.

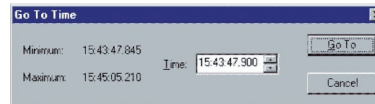
Note: You can change the data format in either viewer at any time.

To jump to a specific time in the Message Log and Real Time Viewers:

Note: You can only jump to a specific time in the Real Time Viewer when updates are disabled; you can jump to a specific time in the Message Log Viewer at any time.

1. Choose Tools ► Go To Time, or click the Go To Time button: .

The Go To Time dialog is displayed:



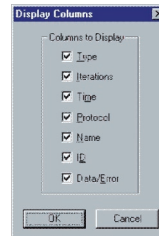
2. Enter the time you want to jump to.
3. Click Go To.

To display and hide columns:

Note: You can only choose which columns to display in the Real Time Viewer when updates are disabled; you can choose which columns to display in the Message Log Viewer at any time.

1. Choose Tools ► Columns, or click the Columns button: .

The Display Columns dialog is displayed:



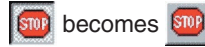
2. Check the boxes for the columns you want to display.

To enable updates in the Real Time Viewer:

- Choose Tools ► Enable Updates.

Or

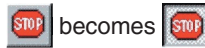
- Click the Stop button to release it:

**To disable updates in the Real Time Viewer:**

- Choose Tools ► Disable Updates.

Or

- Click the Stop button to latch it:



You will need to stop updates if you want to change which columns are displayed in the viewer, to jump to a specific time in the viewer, to view Real Time Viewer statistics or to clear the viewer.

To display Real Time Viewer statistics:

Note: You can only display Real Time Viewer statistics when updates are disabled. You can use the Filter Messages dialog to view the equivalent statistics for a message log – see To Filter the Message Log Viewer.

- Choose Tools ► RTV Statistics.

Or

- Click the Statistics button: .

The Real-time Viewer statistics dialog is displayed, showing a breakdown of the records in the Real Time Viewer.

To clear the Real Time Viewer:

Note: You can only clear the Real Time Viewer when updates are disabled. After you have cleared it, message updating is re-enabled automatically.

- Choose Tools ► Clear Display.

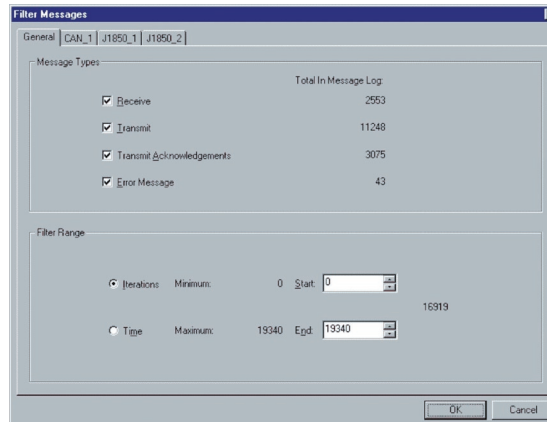
Or

- Click the Clear Display button: .

You are asked if you want to save the records in the Real Time Viewer in a message log file before it is cleared.

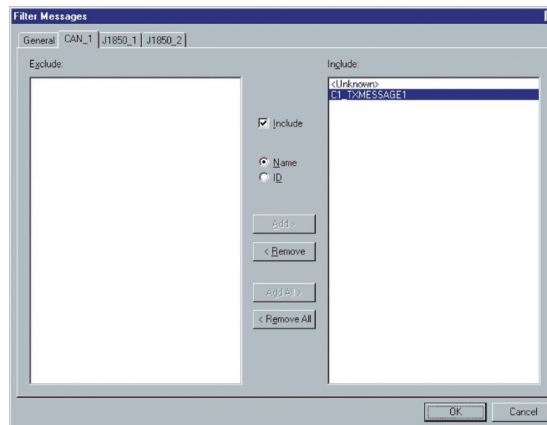
To Filter the Message Log Viewer:

1. Choose Tools ► Filter Messages, or click the Filter button: . The Filter Messages dialog is displayed:



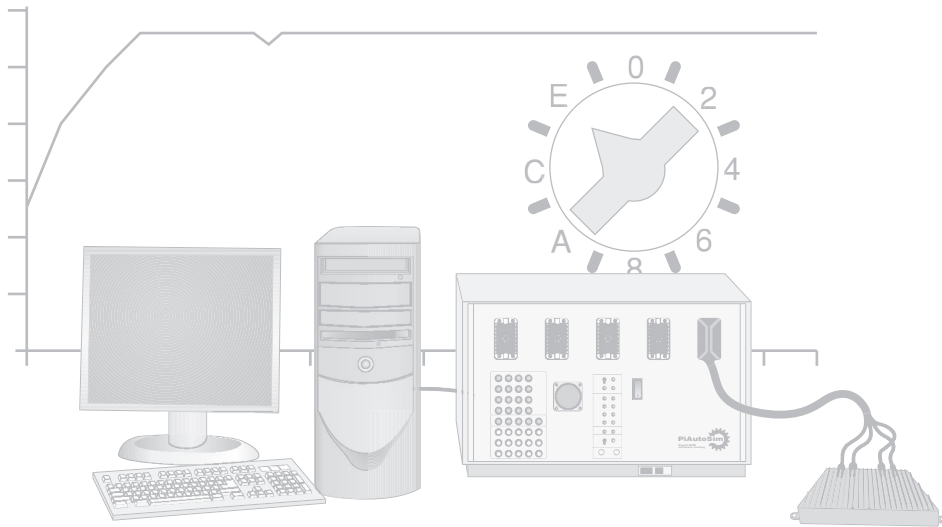
The General page gives a breakdown of the records in the log.

2. Check the Message Types boxes to choose which records to include in the Message Log Viewer display.
3. Use the Filter Range controls to restrict the records included in the Message Log Viewer display to a time range – either defined by iteration number or real time.
4. Click the protocol tabs to set separate filter conditions for the messages sent or received using each protocol:



5. Click Name to show all the names (unknown messages are messages with no name) or ID to list all the IDs (for protocols that assign IDs) of messages included in the message log so that you can set filter conditions on the names and/or IDs. The names/IDs listed in the Exclude box are filtered out of the Message Log Viewer display.
6. To move a name or ID from the Include box to the Exclude box, select it and click < Remove, or just double-click on it.
You can select several names or IDs at the same time in the usual way by **[Shift]**-clicking to extend a selection and **[Ctrl]**-clicking to add to a selection. You can then exclude all the selected names or IDs at the same time by clicking < Remove.
7. To exclude all the names or IDs listed in the tab, click < Remove All.
8. To move a name or ID from the Exclude box to the Include box, select it and click Add >, or just double-click on it.
You can select several names or IDs at the same time in the usual way by **[Shift]**-clicking to extend a selection and **[Ctrl]**-clicking to add to a selection. You can then include all the selected names or IDs at the same time by clicking Add >.
9. To include all the names or IDs listed in a tab, click Add All >.
10. When you have set all the filter conditions you require, click OK.

The Message Log Viewer is reloaded with just those messages satisfying all the filter conditions.



Hardware

I/O System

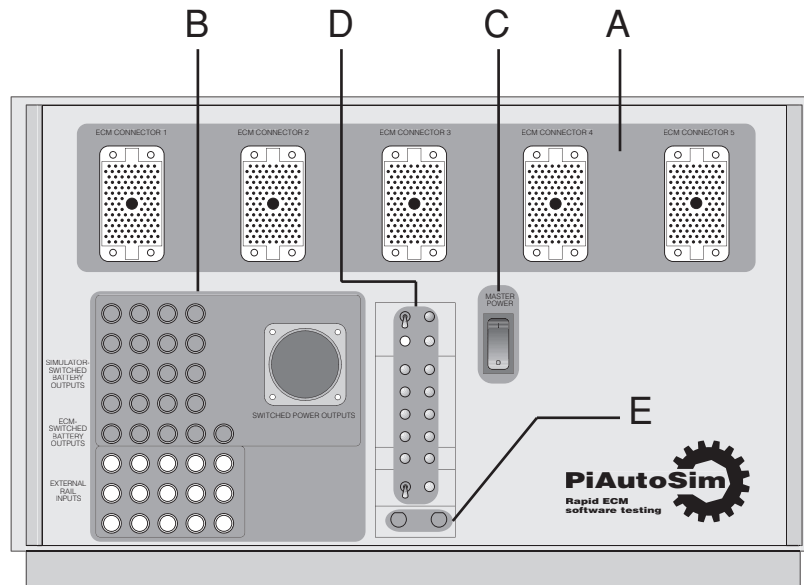
The 19-inch rack in which the I/O System (IOS) is housed has the following main items:

- Front panel – This has the signal and power connectors to connect the IOS to the electronic control module (ECM) under test. It also has switches and indicators for power control.
- ISA motherboard – This is located in the top of the I/O System and contains the main CPU card and other ISA and PCI devices.
- PISA I/O cards – These are the eight types of AutoSim I/O card. They are accessed from the back of the unit. They are controlled by the CPU and connect to the ECM under test via the five front panel connectors. You can install, remove and configure these cards.

Important: You should not attempt to access certain items inside the I/O System, such as the hard disk for the PC, the internal power supply and power distribution.

Note: The maximum load that you can place on top of the I/O System is 25kg; it will support most 17-inch color monitors.

Front panel features



Note: The panel layout may vary slightly depending on the version of AutoSim.

- A** Front panel connectors – These connect the I/O System to the ECM. For further details, see Front panel pin numbering later in this section.
- B** Banana sockets and power output connector – These provide power outputs from the I/O System to the ECM under test. For details, see Front panel power connections in the Power Supplies section later in this chapter .
- C** Master power/circuit breaker – This trips open if more than the maximum current (20A) is drawn from the I/O System. It may be manually reset but should not be prevented from operating, as it protects the internal wiring of the I/O System.
- D** Power indicators and controls – These indicate the status of the various power outputs, and allow some manual control over them. See Front panel power controls in the Power Supplies section later in this chapter .
- E** Trigger outputs – These provide pulses to a scope, logic analyzer or other instrument to allow external measurements to be synchronized with the running of the model. A model script can be made to generate an output pulse on either BNC connector at any time.

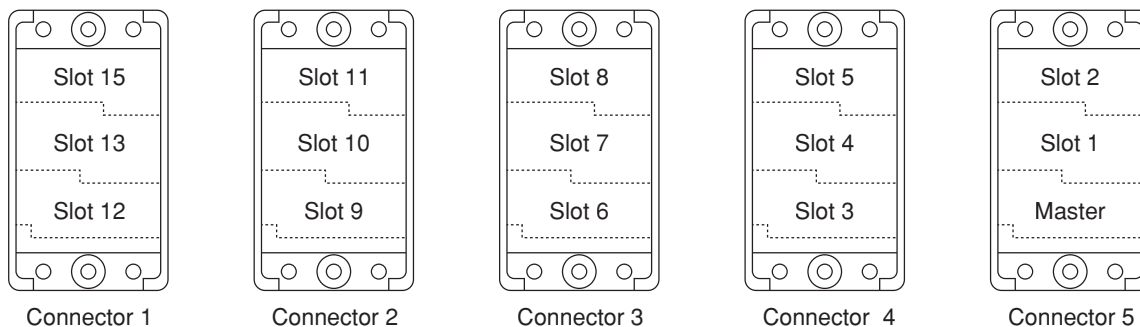
Front panel pin numbering

The PISA I/O cards are housed in slots, numbered from 0 on the left (looking from the back of the I/O System), up to 15 on the right. Slot 0 is reserved for the Master card, which must be present in every system. Slot 14 cannot be used, as it does not connect through to the front panel. The remaining 14 slots can be used for PISA cards in any order.

However, the transformers on the Programmable Waveform Generator card, 25T-060002, mean that you cannot place a card in the next higher numbered slot, so we recommend that slots 13 and 15 are used for this card in the first instance.

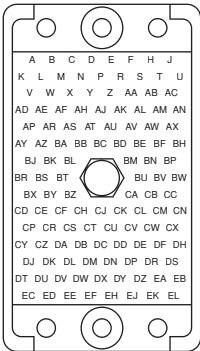
When you fit an I/O card into one of the 14 available I/O slots in the I/O System, each input and output channel is connected to a pin on one of the five large front panel connectors.

The diagram below shows the connections between PISA slots and front panel connectors.



PISA slot to front panel connector mapping

The ECM under test is wired to the front panel connectors, each of which has 127 pins, which are sequentially labeled A to EL (omitting all labels with the letters G, I, O and Q), as shown in the following diagram.



Front panel connector plug (rotated and viewed while facing the front panel)

Mapping channels and pins

For each type of I/O card (Discrete I/O, Pulse Width Modulation input card, etc.) a lookup table is provided that allows you to determine the mapping of input and output channels to pins on a front panel connector, given that the card may be located in any of the available PISA slots of the I/O System.

Each table is in two parts: the top section shows you which front panel connector corresponds to the slot in which you placed the I/O card. Following the column down, the lower section of the table shows you the pin corresponding to each channel of the I/O card. Both input channel and output channel tables are provided, where appropriate.

The tables are easy to use in practice, as the following example demonstrates.

An example...

Suppose you had installed a PWM output card (8 channels) in slot 7 of the I/O System, and you wanted to connect the output on channel 6 to the ECM under test. To find which front panel connector and pins to connect to:

1. Find out which front panel connector to use.

The table for an 8 channel PWM output card is reproduced below. You can see from the upper part of the table that a card located in slot number 7 has its channels mapped to front panel connector number 3 (reading across the row to the left).

Connector	Slot	Slot	Slot
1	15	13	12
2	11	10	9
3	8	7	6
4	5	4	3
5	2	1	0
Output channel	Pin	Pin	Pin
0	J	AU	CH
1	F	AS	CE
2	D	AP	CX
3	B	BF	CV
4	U	BD	CT
5	S	BB	CR
6	P	AZ	DH
7	M	BP	DE

2. Find out which pin on the front panel connector maps to the output channel you want to connect.

Slot 7 is in the second column, so the pins corresponding to a card in slot 7 are in the second column of pin numbers in the lower part of the table.

Channel 6 output therefore connects to (reading across the row to the appropriate column) pin AZ.

Hardware installation

When you receive your I/O system, all the cards installed will already have their memory addresses present, and you do not need to change them. However, you may need to access the hardware, either to change the link settings on a board or install the new hardware.

Note: If you are installing new hardware, it is necessary to determine and configure the base address before installing the card into the IOS.

Tools required

You will need the following tools during hardware installation:

- 4mm flat blade screwdriver (short handle)
- 4mm flat blade screwdriver
- 8mm flat blade screwdriver
- No.1 Pozidrive screwdriver
- Trimming tool to adjust address switches
- 5.5mm nut spinner for M3 nuts.

Precautions for changing any card

When dealing with all mains-powered equipment, the standard precautions should always be observed to avoid electric shock. Always unplug the I/O System from the mains power supply and remove any other external power sources before removing the top or rear panels.

When adding or removing I/O cards, antistatic precautions should also be observed – in particular, you must wear an earthed antistatic band while handling any of the I/O cards.

Steps for installing a new card

1. Determine the base address and set it on the card.
2. Determine and set the hardware links on the card.
3. Open the IOS and fit the I/O card.

Each of these steps is now detailed below.

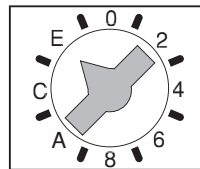
Determining the base address for new hardware:

When you receive your I/O System, all the cards installed will already have their memory addresses preset, and you do not need to change them. However, you will need to set the memory addresses for new cards.

Please refer to the documentation installed with the Pi AutoSim User Interface software for further information.

Setting the base address

Each I/O card in the AutoSim I/O System must have a memory address set up, which is accessed by the software. Memory addresses are 20 bits long, expressed as five hexadecimal characters (e.g. C8140). With boards of rev. 2 and above, the first four digits of the address are set with the hexadecimal switches – the fifth, least significant address digit is implicitly zero. Each switch looks like the one below:



Memory Address Switch

With early boards there is a mixture of methods of setting the addresses detailed in the individual board descriptions, later in this chapter. The amount of memory that a board occupies varies greatly and is also described for the individual boards later in this Hardware section.

Fitting or removing a PISA I/O card

To fit or remove a PISA I/O card:

1. Unplug the mains power supply and any other external power supplies.
2. Wear an earthed antistatic band.
3. Using a 4mm flat-head screwdriver, unscrew the vent panel (for slots 0–12), or the smaller second panel (for the other slots).
4. To remove the I/O card:
 - Pull on the two black levers at the top and bottom edges of the card.
 - Move the card gently up and down to release it and slide it out.

-
5. To fit an I/O card:
 - Ensure that the two black levers are in the unlocked position (i.e. away from the PCB).
 - Slide the PCB into the slot until a slight resistance is met.
 - Push the two black levers in towards the PCB: the card locks into place.

Fitting a PCI card

You must first remove the top panel by following the instructions below:

1. Unplug the mains power supply and any other external power supplies.
2. Wear an earthed antistatic band.
3. Using a No. 1 Pozidrive screwdriver, unscrew the two top lid screws (at the top of the rear panel).
4. Remove the green earthing lead connecting the lid to the I/O system by pulling off the FASTON connector.

Note: Remember to reconnect the lid earthing lead when reassembling the I/O System.

Having removed the EMC top-cover, you now have access to the internal PC module. You can add or remove cards just as you would in a regular PC box.

Removing the internal chassis:

You can slide the internal chassis out of the main I/O System to get access to the fuse boards and other wiring. Follow the steps below:

1. Take off the top panel, as described in the previous section.
2. Remove the EMC top-cover and the ELCO to ISA ground strap.
3. Unscrew the four large flathead screws in the corners of the front panel.
4. Unplug all the ribbon cables, earth leads and power cables.
5. Remove the front panel (this gives you access to the fan fuse board and the relay board).
6. Unscrew the eight large screws at the sides of the rear panel.

You should now be able to slide out the internal chassis to gain access to the wiring and power supply.

Before replacing fuses, investigate the cause of the failure. If in doubt, contact Pi Technology.

Power supplies

External battery power supply

The external battery power supply provides the power for the ECM under test, both directly through the power output connectors on the front of the I/O System, and indirectly through signal lines (such as discrete outputs). It is a separate unit because the requirements on the supply vary from project to project. For small systems, a 12V 1A supply may be sufficient, whereas for driving heavy loads, the I/O System is capable of passing 600W to the ECM under test.

Connecting the power supply

The external power supply connects to the red (positive) and black (negative) 4mm banana sockets labelled 'BATTERY' on the back of the I/O System.

Earthing

The whole I/O System is grounded to the earthing cable in the mains power supply. This is connected to the negative of the external power supply, the ground terminals on the front panel, and internally to the analog and digital grounds. To avoid ground loops, the external power supply should float with respect to earth.

Operating voltage and current

The allowed operating range of the external power supply is 0–30V. Outside this range, relays inside the I/O System will isolate the supply, protecting both the ECM under test and the I/O System against reverse or over voltage.

All current coming from the external supply is routed through the front panel circuit breaker, which is rated at 20A. All the front panel power output connectors can supply this current individually, as well as collectively, so provided that all power to the ECM under test is routed through the I/O System, no damage can occur to the I/O System or its I/O cards. We therefore strongly recommend that no power be applied to the ECM under test other than through the I/O System. In particular, I/O pins can be permanently damaged by the application of voltages outside the 0–30V operating range.

Note: After they have tripped, the previous state of the relays can usually be restored by simply returning the external battery voltage and current back to within the normal operating range. If however a fault causes the relays to rapidly trip on and off for more than two seconds, both the manual override and master power on-of switches will be reset. In this case the the manual override and/or master power switch will need to be pressed on the front panel before power can be restored by the relays.

Controlling the power supply from the I/O System

Two dedicated voltage outputs are available from the I/O System to control the external power supply. These 0–10V DC signals are controlled by the PCI Master card, and are connected to a 9-way D-type connector on the back panel. These signals are controlled from software using the VCTRL and ICTRL variables.

Note: You must apply the correct range and scaling factors to the VCTRL and ICTRL variables. Failure to do so can result in applying up to 36V to your controller, which may damage it irreparably.

Note: When you first test the VCTRL value after setting the scaling, you should set the ICTRL variable to a small value (e.g. 0.1A). You should also disconnect both the controller and the power cables from the back panel of the I/O System.

The scaling for two common power supplies should be set to the following values:

Kepeco ATE 36-30M (using the Pi Harness Part no. 03T-060238)

	Range	Scaling		
		A	B	C
VCTRL	0–36V	0.27778	0	0
ICTRL	0–30A	0.33333	0	0

Xantrax XHR33-33

	Range	Scaling		
		A	B	C
VCTRL	0–33V	0.3	0	0
ICTRL	0–33A	0.3	0	0

You can set a limit on the VCTRL signal in software using the "VCTRL LIMIT" variable in USERVARS (if the model loaded supports it). This sets the maximum output voltage in scaled units (i.e. true battery voltage), and prevents an inexperienced user from setting too high a value. The default limit is 14V.

The connections are as follows:

Function	Pin number
Voltage control	1
Voltage control return	3
Current control	2
Current control return	4

Note: The two returns are connected to ground inside the simulator.

Internal power supply

The primary power for the AutoSim I/O System is delivered by a switched mode power supply, which can cope with any mains input ranging from 115V/230–240V at 50–60Hz. This internal power is used for the PC and logic in AutoSim – it is not available for external equipment. The power supply is located in the rear of the PC Crate.

Front panel power connections

The front panel contains two types of power connector: 4mm banana sockets for power connection, and an optional circular Canon connector. The banana sockets may be divided into three classes by their color:

- Red – power outputs from the simulator
- Black – grounds
- White – external rail power inputs.

Simulator switched supplies 0–7

These eight simulator switched supplies, labeled SSW0 – SSW7, are connected to the external battery power supply via relays, which are under the control of the User Interface PC software. AutoSim can therefore switch these eight outputs on and off individually. The controls for these outputs are on the PCI Master card. When the output is on and power is applied, the associated LED on the front panel is lit.

Hardware on the Master card overrides the software and forces the relays open (outputs off) if either the master power on-off switch on the front panel is off, the external battery supply voltage is outside the 1–30V range, or the current drawn from the external battery supply exceeds 20A.

For ease of use, multiple parallel connectors are provided for the first two supplies (SSW0 and SSW1), labelled SSW0A, SSW0B, etc.

Externally-switched supplies 0–1

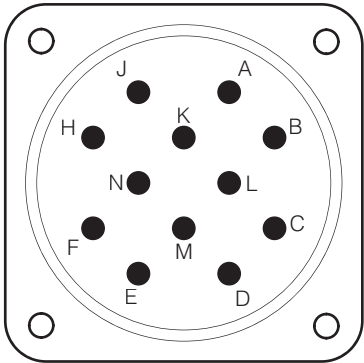
These are identical to the simulator switched supplies, but the control of the relay is from the front panel connector #5, not from the User Interface PC software.

Supply	Control pin (front panel connector #5)
ESW0	DC
ESW1	DB

An input voltage of below 2V, or open circuit, turns the output off. Above 3V, the output is turned on, subject to the master power switch being on and the external power supply being in range (as with all simulator switched supplies).

Optional circular power outlet connector

The simulator and externally-switched power outlets are available on the front panel, both as 4mm banana sockets and as a single power out connector (the pins for the two connectors are in parallel). The circular connector allows the connection of a loom to the I/O System. The pin-out is as follows:



Front panel power out connector pin-out

Signal	Pin	LED
SSW0A/B/C	A	SSW0
SSW1A/B/C	B	SSW1
SSW2	C	SSW2
SSW3	D	SSW3
SSW4	E	SSW4
SSW5	F	SSW5
SSW6	H	SSW6
SSW7	J	SSW7
ESW0A/B/C/D	K	ESW0
ESW1	L	ESW1
Ground	M,N	–

Common switched supply, external rails and high rails

Many of the I/O cards in Pi AutoSim can output power. A simple example is a high side switch – when switched on, it sources power to the external load, which must be earthed.

This power comes from the high rail, which is itself under software control. Typically this high rail is the equivalent of ignition power, switched on by the ignition key. If the software is simulating this ignition key, the high rail must switch on and off with this discrete input.

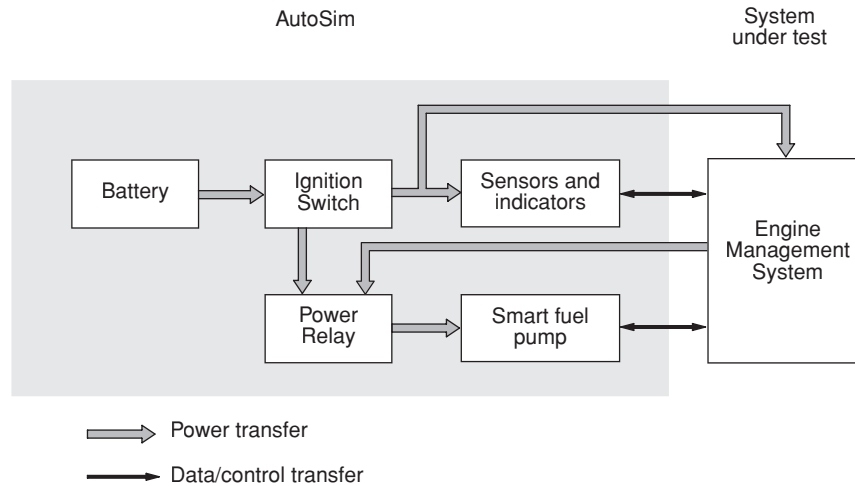
There are three choices of high rail:

- Common switched supply (SSWCOM) – This is a power output much like the simulator-switched supplies SSW0–SSW7. The only difference is that, instead of being brought out on the front panel, it is provided to all I/O cards in the I/O System. It is intended to emulate the most commonly used power supply (usually ignition power).
- Five volts – This is for logic I/O. The 5V is derived from the common switched supply.
- External rail – This is a power input from the front panel to a card – a separate external rail is provided for each slot. The intention is that any of the power outlets SSW0–SSW7 or ESW0–ESW1 can be patched across on the front panel to the relevant external rail input.

On the discrete cards, inputs and outputs can be configured to use any of these three high rails (with the restriction that within any group of four inputs or four outputs, the high rail is the same).

Example use of power supplies

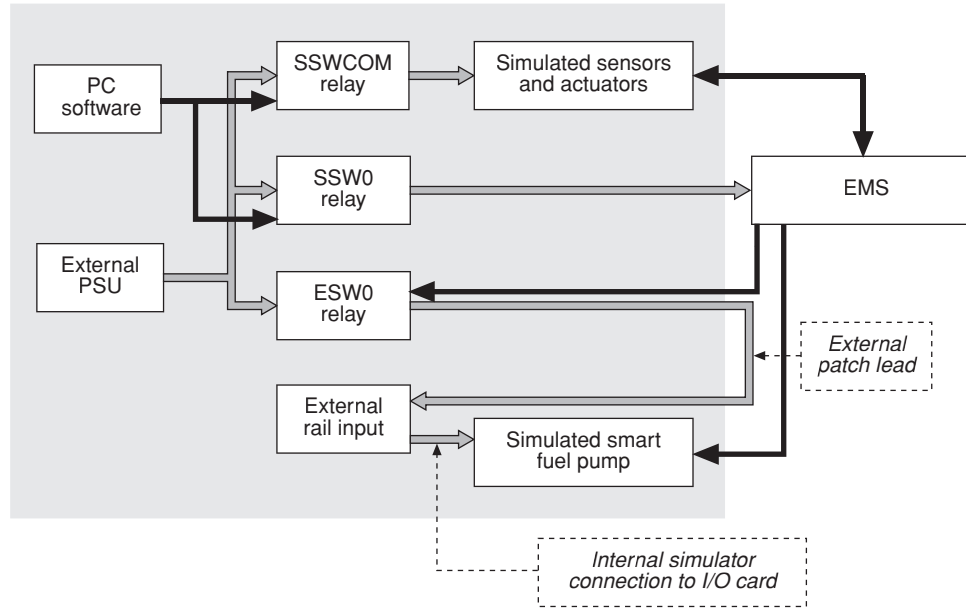
An example application might be the simulation of an engine and smart fuel pump, where the module under test is an Engine Management System. Most of the I/O for the EMS (switches, sensors etc.) are powered by the ignition power rail, but the fuel pump is powered from a relay controlled by the EMS.



System to be simulated

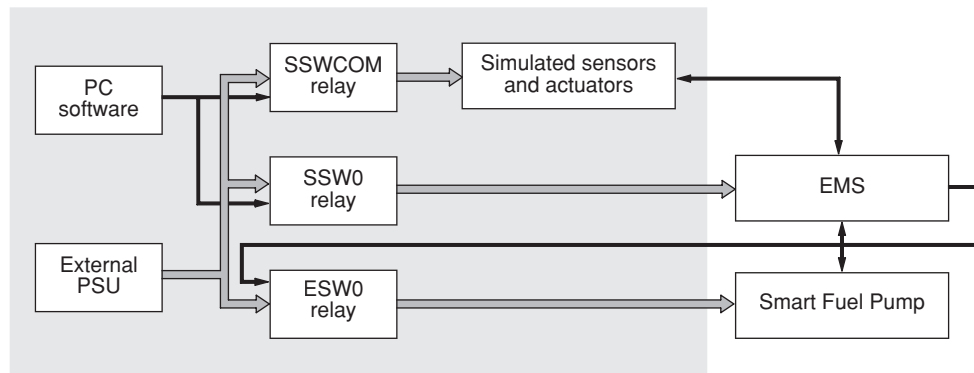
One way of simulating this is as follows:

1. Slave SSW0 and SSWCOM to the simulated ignition switch and power the EMS from SSW0. The sensors and indicators as well as the EMS would then turn on and off with the software-controlled ignition switch.
2. Connect the control signal for the power relay from the EMS to the control input for ESW0 so that the EMS controls this power outlet.
3. Connect the power outlet ESW0 to the external rail for any cards simulating the action of the fuel pump so that these turn on and off when the fuel pump does.



Power and control flows for simulation (simulated fuel pump)

If at a later time you want to use a real fuel pump rather than a simulated one, this can be powered from the ESW0 power outlet.



Power and control flows for simulation (real fuel pump)

There are many other ways to achieve this. For example, the relay control line from the EMS can be imported into the I/O System as a discrete input and the action of the relay modeled in software.

Front panel power controls

Master power (circuit breaker) on/off

After power-up, the On button acts as the switched rail master on-off switch. When off, the master on-off switch turns off all the switched supply rails (unless overridden by manual override). It does not affect the internal power supply.

Manual override

With some applications, it is desirable to have the power maintained on even if the software switches it off – for example if code is held in volatile RAM for development purposes. For a switched supply rail to be in manual override, two things need to be true:

- The manual override for the relevant power output mode must be on.
- The link for the manual override of the relevant power output enable must be present on the Master card (see the section on the Master card later in this chapter).

Order of precedence

The list below summarizes the order of precedence for power outputs.

All power outputs are unconditionally off if any of the following conditions is true:

- The voltage of the external battery supply is out of range.
- The master power switch on the front panel is off.
- The front panel circuit breaker is open.

If none of the above conditions is true, a switched supply rail (SSW0–SSW7 or SSWCOM) is on if either:

- The software has requested it on, the software is running, and the master power switch is on; **or**
- Manual override has been selected and the Master card is link configured to select that channel in this condition.

If none of the above prohibiting conditions is true, an externally switched output (ESW0 to ESW1) is on if either:

- The front panel control pin is above 3V; **or**
- Manual override has been selected and the Master card is link configured to select that channel in this condition.

I/O cards

The following input and output cards are described in this section:

- Master card
- Discrete I/O card
- Analog Input card (including voltage and current daughterboards)
- Analog Output card (including current, thermistor, voltage, thermocouple and LVDT daughterboards)
- Pulse Width Modulation (PWM) Input card
- Pulse Width Modulation (PWM) Output card
- Programmable Waveform Generator (PWG) card
- Event Timer (EVT) card (including injector daughterboard).

For each I/O card listed above, there are details on the following topics:

- Card description – covering the general functionality of the I/O card
- Example application – giving a typical use for the particular I/O card
- Circuit description – giving a schematic diagram to explain the high-level operation of the I/O card (not all cards have this section)
- Card configurations – showing the different ways you can use the I/O card to gain maximum efficiency from it
- I/O channel assignment – giving a table showing how the different channels of the I/O card are connected to the front panel connectors
- Component location – giving a guide to the I/O card's physical layout, showing where each of the main components are located, including jumpers, switches and indicators
- Details of any daughterboards
- Card specifications – showing the operational range of the I/O card.

Master card

The Master card always goes in slot 0 of the I/O System. It controls and synchronizes all the I/O functions within the I/O System. There are Master card connections to one of the connectors on the front of the I/O System. The Master card has 8 built-in Analog Input channels, identical to those on the Analog Input card, and two built-in Analog Output channels, which connect to the power supply control connector on the back panel. There are also eight channels for setting up identification codes for different wiring looms so that you can automatically check that you have the correct loom wired to your ECM.

When you install the Master card using Add New Hardware, you will be given an IRQ setting as well as a base address for the card. Set the IRQ jumper to this IRQ setting.

The Master card also contains user-configurable links to:

- Set up different power supplies on the front panel
- Configure manual override and circuit breaker features
- Set interrupt and master/slave configurations in multiple-rack systems
- Configure two scope trigger outputs that can be written to by the I/O system model.

Card configurations

Input conditioning is exactly the same as that on the Analog Input cards, via the analog input daughterboards.

Board address

The four switches set the base address as described in To set the Base Address in the Hardware installation section, near the beginning of the Hardware chapter.

Watchdog disable

AutoSim has a watchdog that disables power outputs if abnormal I/O System behavior is detected.

The watchdog is controlled by jumper block JP6. There are 11 jumpers in this block, controlling the power outputs described in Front panel power connections in the Power Supplies section, earlier in this Hardware chapter (i.e. SSW0–SSW7, ESW0 and ESW1).

To disable the watchdog timer for a channel, remove the link from the appropriate pair of pins on JP6, located towards the north end of the board.

Manual enable

The Manual Override switch on the front panel allows you to prevent the software taking power from a switched output. In the same way as for the watchdog timer, the manual override can be disabled for specific channels. Then, if the Manual Override switch is thrown and the I/O System's operation is halted, the power supply to the attached equipment is maintained.

To disable the manual override for a channel, remove the link from the appropriate pair of pins on JP5, the bank of 11 jumpers located towards the northeast corner of the board.

If you want to disable manual override for all the channels at once, without having to remove all of the links on JP5, you can simply fit a link onto JP7, located just northwest of the center of the board. This will disable manual override for all channels.

ADC Complete signal

The ADC Complete signal defines whether or not the system waits for the ADC to finish its conversion before starting a new frame.

JP4, located just southwest of the center of the board, controls the ADC Complete signal for both ADC chips. By default, the system waits for both ADCs to complete (i.e. links are fitted to both jumpers 0 and 1 on JP4). We recommend you leave the links fitted.

Master/slave configuration

To daisy-chain more than one I/O System together, one I/O System must be configured as the Master, with the others configured as Slaves.

The following links must be placed to set up Master mode:

- Place a link on JP2, located just southwest of the center of the card, to connect the center pin to the pin marked I (for Internal synchronization).
- Place two links on JP8, located next to JP2, across the northernmost pair and the southernmost pair.

The following links must be configured to set up Slave mode:

- Place a link on JP2, located just southwest of the center of the card, to connect the center pin to the pin marked E (for External synchronization).
- Remove all links on JP8, located next to JP2, unless the I/O System in question is the one furthest from the Master I/O System, in which case a terminating link must be placed across the middle two pins of JP8.

I/O channel assignment

Use the following table to find out which channels on the Master card connect to which pin on the front panel connectors. Mapping channels and pins in the Front Panel section, earlier in this chapter explains how to use these tables.

Channel	Pin	Channel	Pin
Analog In 0	CH	Loom ID 0	CT
Analog In 1	CF	Loom ID 1	CS
Analog In 2	CE	Loom ID 2	CR
Analog In 3	CD	Loom ID 3	CP
Analog In 4	CX	Loom ID 4	DH
Analog In 5	CW	Loom ID 5	DF
Analog In 6	CB	Loom ID 6	DE
Analog In 7	CU	Loom ID 7	DD
		ECM Batt 0	DC
		ECM Batt 1	DB

As this Master card can only be positioned in slot 0, these connections will always be mapped to the bottom section of front panel connector number 5 (extreme right of front panel).

Analog out 1 and Analog out 2 connect to PSU Voltage Control pin 1 and PSU Current Control pin 2, on the Power Supply Control connector on the rear panel, and are used when connecting to the external power supply.

The Loom ID inputs can be used to form an identification code for each cable harness.

The Loom ID inputs are pulled high by the AutoSim, but you can create a unique Loom ID number for each cable harness, which can be read by the system software via the Master card's LOOMID input variable. Unused inputs will be read as a logical 1, while grounded inputs are reported as logical 0. If no pins are grounded, the LOOMID will show a value of 255.

Component location and description

The diagram below shows where the user-adjustable components are found on the Master card.

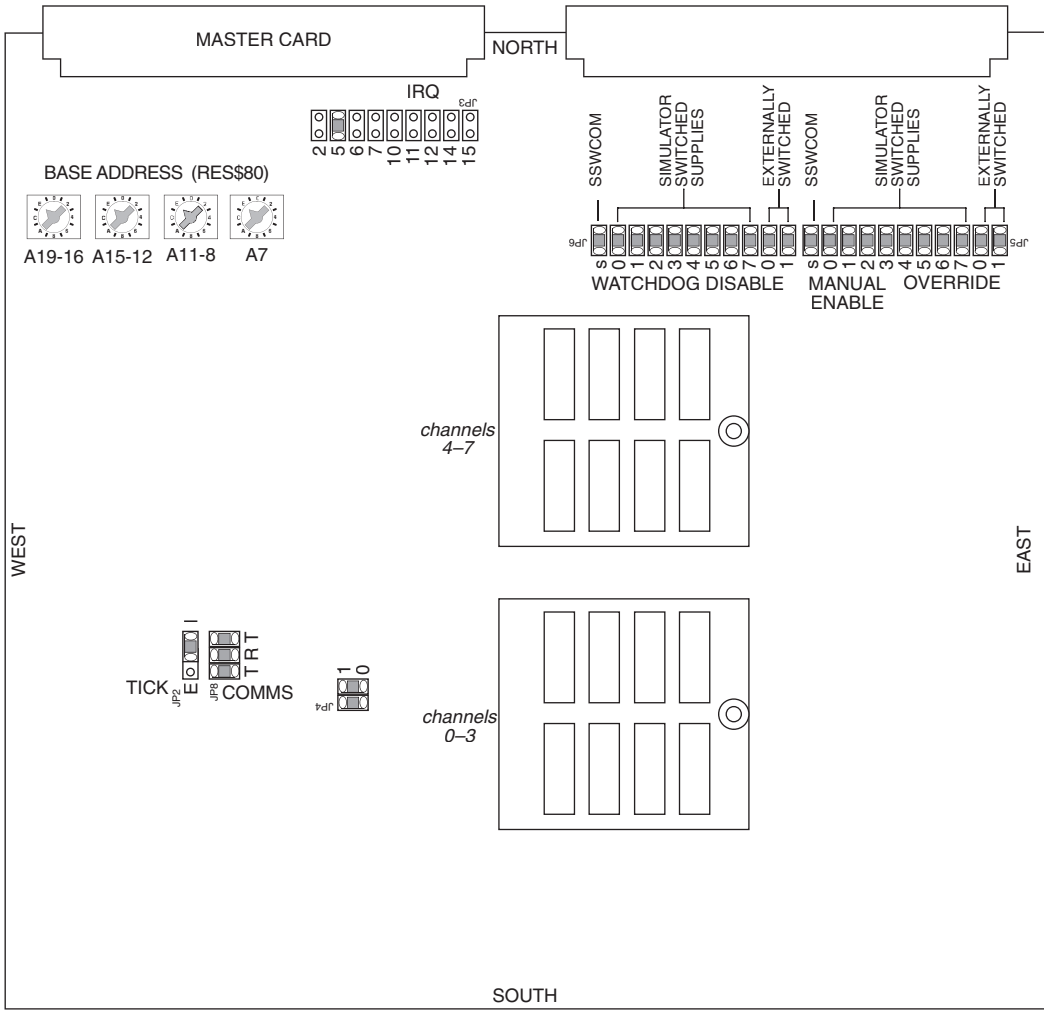


Diagram showing the layout of the MasterCard

The Master card supports a mixture of power control and analog input channels. The analog inputs are split into two groups of four channels, with each group supported by an input daughterboard. The daughterboards are plug-in conditioning cards, which are used to provide voltage or current input. Channel properties are not shown, as all input scaling is dealt with by the standard Variable and Scaling pages.

- Card properties – the standard card property sheet shows the slot number for the card.
- Daughterboard properties – the Daughter property sheet shows the location and type of the daughterboard fitted.
- Variable properties – the standard Variable and Scaling property sheets are provided, as described in Editing model and variable properties in the Software chapter .

There are hardware configuration links on the master card to determine the operation of the power outputs under watchdog and manual override conditions.

Discrete I/O card

This card provides 16 inputs and 16 outputs. These are discrete (battery level) or digitally compatible (logic level) and intended for low speed applications. For each of the 16 input channels, the card provides conversion from external voltages to logic levels, electrical protection and a (software programmable) pull-up or pull-down resistor.

For each of the 16 output channels, the card provides a writable latch, a (software selectable) high side and low side driver, electrical protection and an optional, manually configurable set of bias resistors.

I/O card properties

The properties page for an I/O card displays the slot number for the selected card. You cannot edit this number.

Daughterboard properties

The properties page for a daughterboard displays the type of the selected card. You cannot edit this.

Example application

Example uses of the Discrete I/O card are:

- Outputs – driver switches; low-frequency sensors (e.g. Hall effect)
- Inputs – indicator lamps; on/off fan controls.

Circuit description

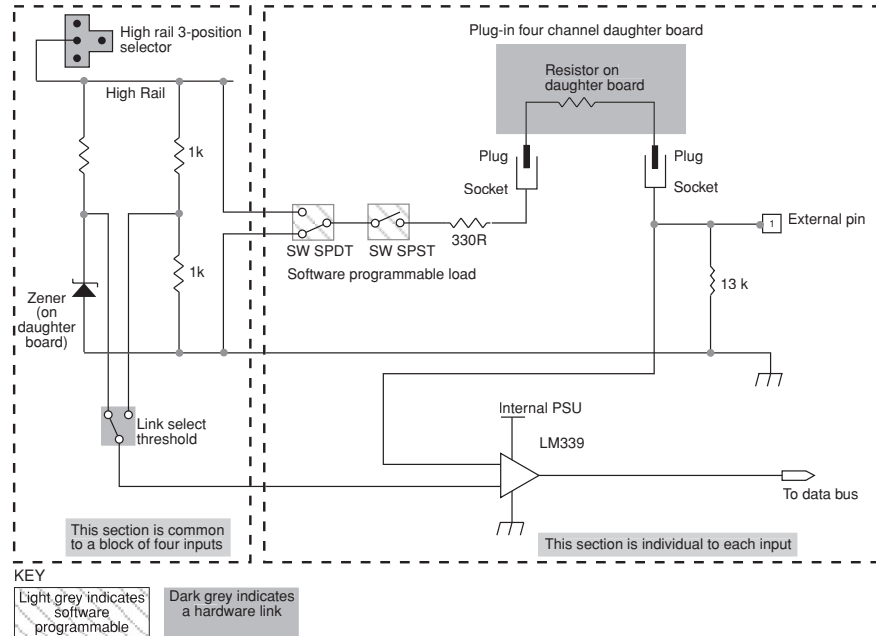
Configuring discrete inputs

Note: In addition to the Discrete I/O card, AutoSim has several types of card (e.g. PWM input and event timers) with discrete inputs that have the same configuration options so that they can be used in many different applications.

The configuration options are:

- High Rail selection
- Threshold
- Load resistors.

The High Rail and Threshold can be set on blocks of four channels. The Load resistors can be set on a per-channel basis. A simplified version of the common circuit used for all types of discrete inputs is shown below:



Common circuit used for discrete inputs

High rail selection

The high rail (VR) for a group of four channels is set by a hardware jumper to one of:

V_{common}	SSWCOM – internally switched output from the external PSU. Known as V_{sensor} on older cards
V_{5V}	Internally derived from V_{common}
V_{Ext}	External rail – an external voltage fed into AutoSim from the front panel connector.

Note: AutoSim has one external rail for each card. Groups of four channels (0–3, 4–7 etc.) can then be configured to use the external rail.

The high rail configuration is used as an input to drive the load resistors and the threshold described in the sections below.

Load resistors

All types of AutoSim discrete input channels have load resistors. These are used to:

- Prevent an ECM incorrectly detecting an open circuit on its outputs
- Enable ECM outputs that only switch to ground or a voltage supply to be monitored by AutoSim.

The normal value of all AutoSim load resistors is 330Ω . This value may be increased using resistor daughterboards, which plug in series with the main 330Ω resistor (standard resistor daughterboards have 0Ω links giving a series total of 330Ω). Each resistor daughterboard has individual load resistors for each of four channels, which can therefore be changed separately for each channel. If either the resistor or the daughterboard, or the entire daughterboard is missing, the input is pulled to ground by approximately $13\text{ k}\Omega$.

The load resistor is connected at one end to the outside connection to the ECM. The other end (330Ω resistor plus the daughterboard in series) is connected to a software-controlled switch that enables the load resistor to be connected to one of:

- Ground
- The high rail VR (described in the previous section)
- High impedance.

The software settings for the load resistors should be programmed as follows, depending on the ECM output type:

ECM Output Type	Load Type
Low Side	High (V_R)
High Side	Low (Ground)
Push Pull	None (High impedance)

Input threshold selection

The input threshold for groups of four channels can be set to one of two options using a jumper:

- $V_R/2$ i.e. half of the high rail voltage – see the previous section for high rail selection
- V_{Zener} determined by a Zener diode on the resistor daughterboard.

The hysteresis on the inputs is fixed at about 300mV.

Configuring discrete outputs

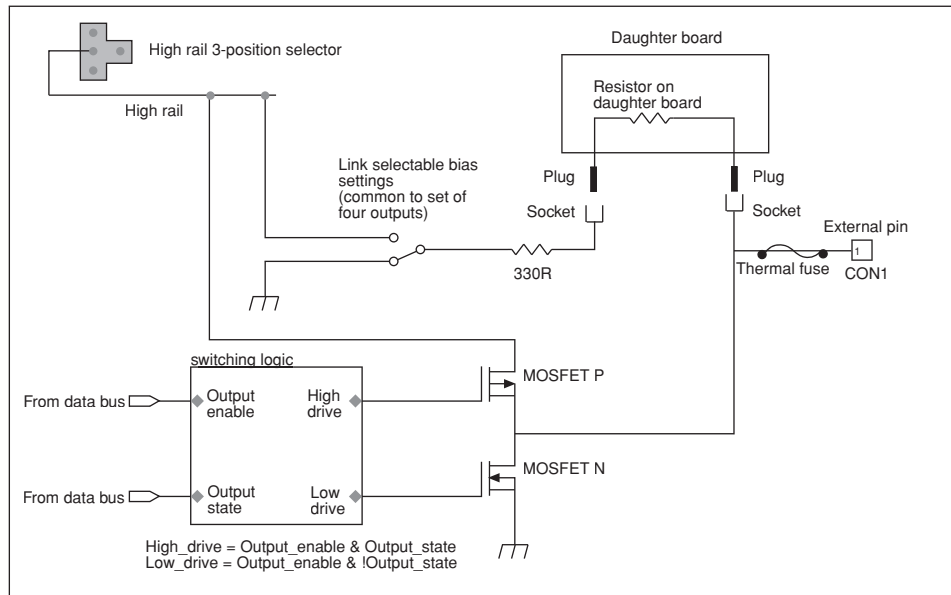
Note: In addition to the Discrete I/O card, AutoSim has several types of card (e.g. PWM output) with discrete outputs that have the same configuration options so that they can be used in many different applications.

AutoSim has different types of discrete outputs (e.g. PWM outputs). Each card has the same configuration options that enable it to be used in many different applications. The configuration options are:

- High side or low side drivers
- High rail selection
- Bias resistance
- Bias state (pull-up or pull-down).

The discrete outputs are configurable, per channel, to be either high side or low side drivers, simulating a switch to supply (high rail V_R) or to ground. The selection is done in software. The high rail can be selected between three options for groups of four channels, as for the discrete inputs.

Bias resistors can be fitted when there is no load resistor on the ECM input. The bias resistors can be connected either to ground or to the high rail. A simplified schematic of a discrete output is shown below:



Simplified schematic of discrete output

High rail selection

The high rail powers the high side drivers and the bias resistors (if the bias jumper is set to HIGH) and is selected in exactly the same way as the discrete input high rail described earlier.

Bias resistors

Bias resistor daughterboards work in the same way as the discrete input daughterboards described earlier. Note that the standard configuration is for no bias resistor daughterboards to be fitted.

Bias jumper

There is one bias jumper for each group of four outputs to determine whether the bias resistors pull the output high or low. The jumper options are:

- If the bias jumper is set to HIGH, the outputs are biased to the high rail. This setting should be used for active low outputs.
- If the bias jumper is set to LOW, the outputs are biased to ground. This setting should be used for active high outputs.

The resistor daughterboard

The resistor daughterboard provides pull-up and pull-down loads for all discrete inputs and outputs in Pi AutoSim. For discrete inputs, the resistor daughterboard may also house a Zener diode to determine the input threshold voltage.

The resistor daughterboard comes in two different varieties:

- 25T-060017 – Standard board with zero ohm links fitted (but no Zener diode). This is used to provide the default 330Ω loads. The zero ohm links may be replaced by the user to increase the load resistance.
- 25T-060017A – A 'virgin' board with no resistors or Zener diode fitted. These are provided so that you can add a different resistor without replacing the zero ohm links to the standard boards.

Note: The serial number on the PCB is the same in both cases (25T-060017).

Load resistors

Each daughterboard provides loads for a group of four consecutive channels (discrete inputs or outputs), whose numbers are written on the main board (on to which this board is mounted). Resistors R1–R4 define the loads in sequential order, thus if the board is for channels 8–11, R1 defines channel 8, R2 defines channel 9, R3 defines channel 10 and R4 defines channel 11.

The motherboard setting decides whether the loads either pull up to a high rail, defined by the motherboard, or down to the ground. The resistance of the load is 330Ω plus the resistor

mounted on the daughterboard. Thus the maximum load (i.e. minimum resistance) of 330Ω is achieved by using zero ohm links on the daughterboard, as provided on the standard board 25T-060017. To increase the resistance and decrease the load, either replace the zero ohm link, or use a virgin board 25T-060017A and mount a resistor 330Ω less than the desired total value.

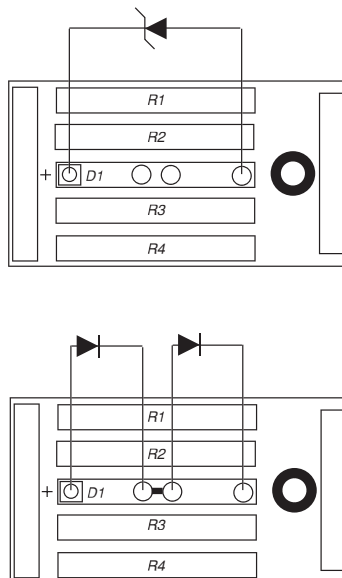
The resistor should be 1W of any type (carbon, metal film etc.). This can be reduced for specific cases as follows. The power dissipated in the resistor of resistance R with the power supply $V_{\max}$ is $V_{\max}^2 * R / (R + 330)^2$. The worst-case resistance value is 330Ω giving $V_{\max}^2 / 1320$. For the highest rated Vmax of 30V, a 750mW (or 1W) resistor will cover all cases.

If the resistor daughterboard is missing, or if there is no resistor mounted on it, there is no bias other than the built-in $13\text{ k}\Omega$ to ground.

Threshold voltage

The concept of threshold voltage does not apply to discrete outputs.

The input threshold voltage is the same for a group of four inputs. This is normally set to half of the high rail but may be set to that of the Zener diode with a link on the motherboard. The ability to mount either a Zener diode, or two series forwards biased diodes (for 1.2V threshold), is provided, as shown in the diagram below.



Diagrams showing the placement of a Zener diode on a resistor daughterboard and the placement of two standard silicon diodes on a resistor daughterboard

The Zener should be at least 400mW. The Zener diode is biased from the high rail, so if insufficient voltage is applied, the threshold will not be at the specified value. Revision 1 daughterboards do not have the facility to mount two series diodes.

Card configurations

This section tells you how to configure the card by setting various jumper links.

See Configuring discrete inputs and Configuring discrete outputs earlier in this section for common configuration information.

Input bias

See the general information in Configuring discrete inputs earlier in this section.

Input threshold voltage

Channels	Jumper	Link for $V_R/2$	Link for V_{Zener}
0–3	JP10	East	West
4–7	JP12	East	West
8–11	JP14	East	West
12–15	JP16	East	West

Input high rail voltage

Channels	Jumper	Link for V_{5V}	Link for V_{common}	Link for V_{Ext}
0–3	JP9	West	North	East
4–7	JP11	West	North	East
8–11	JP13	West	North	East
12–15	JP15	West	North	East

Output load

Outputs can be loaded in the same way as the inputs, by installing daughterboards. Note that without the daughterboard fitted, there is no load.

Output High rail voltage

Channels	Jumper	Link for V _{5V}	Link for V _{sensor}	Link for V _{Ext}
0–3	JP17	West	North	East
4–7	JP19	West	North	East
8–11	JP21	West	North	East
12–15	JP23	West	North	East

Bias switch

Channels	Jumper	Link for High rail	Link for ground
0–3	JP18	West	East
4–7	JP20	West	East
8–11	JP22	West	East
12–15	JP24	West	East

Base address selection

Rev. 2 boards and higher have the standard base address switches described in To set the Base Address in the Hardware installation section near the beginning of this Hardware chapter. Rev. 1 boards have DIL switches as follows:

S4				S3				S2				S1			
1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
a18	a17	a16	a15	a14	a13	a12	a11	a10	a9	a8	a7	a6	a5	–	–

a19 must be high

If a switch is on, the address line must be low for the card selection. Thus for a base address of C8560, the switches must be set as follows:

S4				S3				S2				S1			
1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
a18	a17	a16	a15	a14	a13	a12	a11	a10	a9	a8	a7	a6	a5	–	–
off	on	on	off	on	on	on	on	off	on	off	on	off	off	x	x

I/O channel assignment

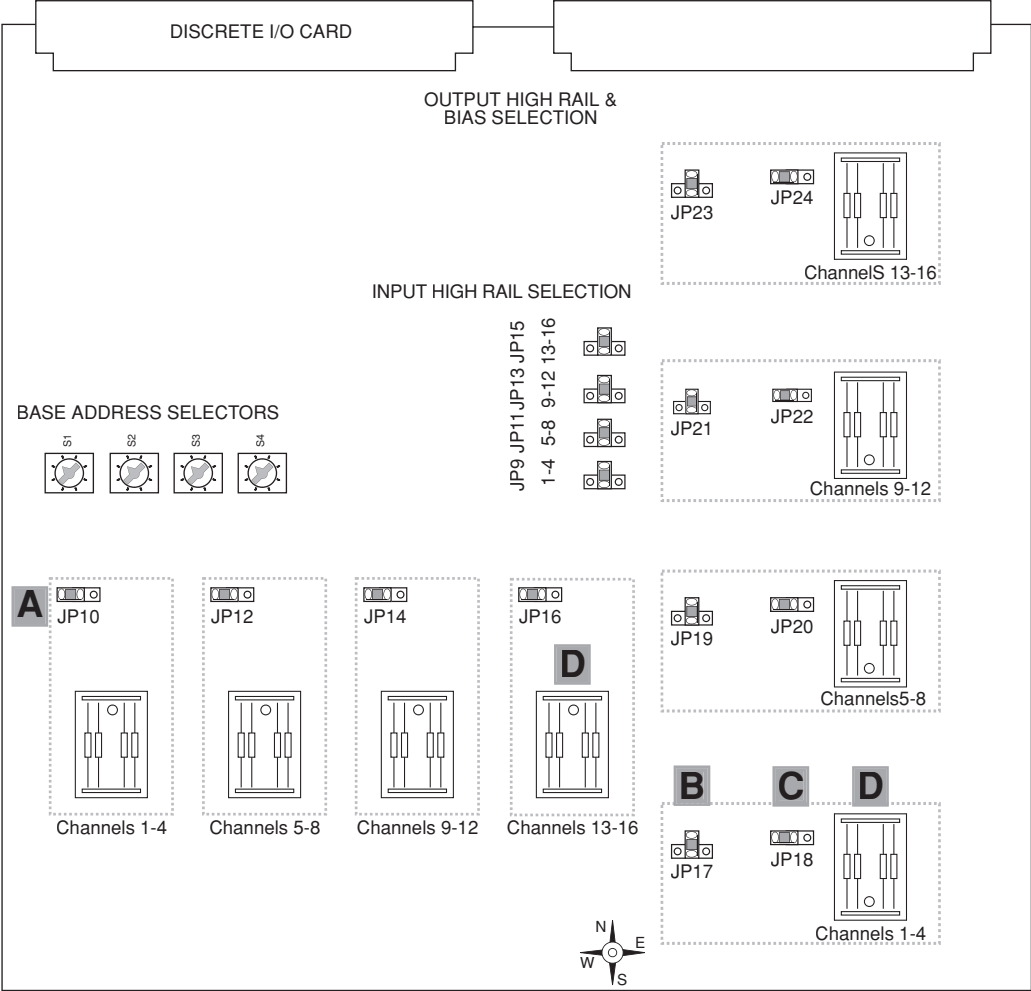
Use the following table to find out which channels on the Discrete I/O card connect to which pin on the front panel connectors. Mapping channels and pins in the Front Panel section of the I/O System earlier in this Hardware chapter explains how to use these tables.

The 16 inputs map to the first 16 I/O channels, and the 16 outputs map to the second 16 channels, as seen in the table below.

Connector	Slot	Slot	Slot	Connector	Slot	Slot	Slot
1	15	13	12	1	15	13	12
2	11	10	9	2	11	10	9
3	8	7	6	3	8	7	6
4	5	4	3	4	5	4	3
5	2	1	0	5	2	1	0
I/O channel	pin	pin	pin	I/O channel	pin	pin	pin
0	J	AU	CH	16	K	BM	DC
1	H	AT	CF	17	AC	BL	DB
2	F	AS	CE	18	AB	BK	DA
3	E	AR	CD	19	AA	BJ	CZ
4	D	AP	CX	20	Z	BW	CY
5	C	BH	CW	21	Y	BV	DS
6	B	BF	CV	22	X	BU	DR
7	A	BE	CU	23	W	BT	DP
8	U	BD	CT	24	V	BS	DN
9	T	BC	CS	25	AN	BR	DM
10	S	BB	CR	26	AM	CC	DL
11	R	BA	CP	27	AL	CB	DK
12	P	AZ	DH	28	AK	CA	DJ
13	N	AY	DF	29	AJ	BZ	EB
14	M	BP	DE	30	AH	BY	EA
15	L	BN	DD	31	AF	BX	DZ

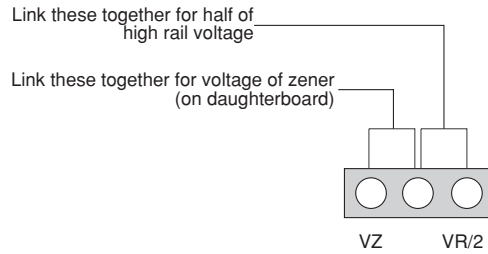
Component location

The diagram below shows the user-adjustable components on the Discrete I/O card. See the key on the next page for detailed component descriptions.

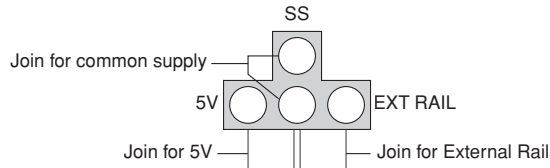


Layout of the Discrete I/O card

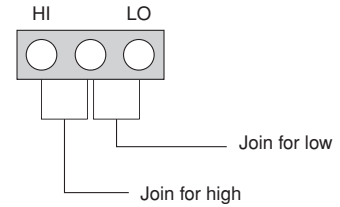
A INPUT THRESHOLD SELECTION LINKS



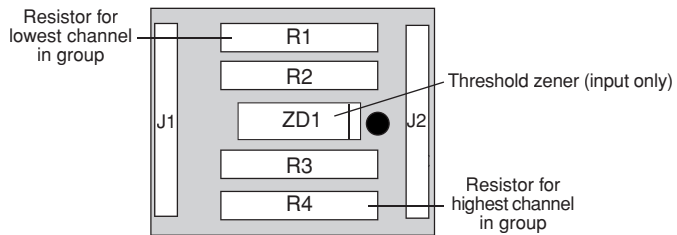
B HIGH RAIL SELECTION LINKS



C SELECT OUTPUT BIAS



D INPUT BIAS DAUGHTERBOARDS



Components of the Discrete I/O card

Note: There are also eight nonadjustable jumpers (JP1 – JP8) just above the High rail voltage selectors (JP9, 11, 13 and 15) – take care not to use these by mistake.

Discrete Inputs	
Number of channels	16
Configuration	Individually software configurable for: pull up to high rail pull down to ground input is floating (assume totem pole output from ECM) High rail is link configurable as either: common supply +5V (derived from battery) external rail, supplied from front panel Restriction: all high rails must be identical within any group of four
Load	Maximum load (i.e. minimum resistance) 330Ω (36mA at 12V), 3W. May be increased in blocks of four by plug-in boards
Input Impedance	$13k\Omega$ maximum
Threshold Voltage	Link selectable ($V_{batt}/2$) $\pm 10\%$, or V_{zener} . Zener diode alterable in blocks of four by plug-in boards
Hysteresis	$>200mV$
Detection time	$<10\mu s$ (hardware), detected at next iteration of software loop, configurable down to $500\mu s$.
Default Configuration	330Ω 3W load Pull up to common supply, or down to ground by software configuration on initialisation Threshold voltage of ($V_{batt}/2$) $\pm 10\%$

Discrete Outputs	
Number of channels	16
Configuration	Individually selectable high or low side switches by software configuration at initialization. For high side switch, the rail voltage is a link-selectable choice of: common supply +5V, derived from battery supply external rail, supplied from the front panel Restriction: within any group of four, all pull-up rail voltages must be the same.
Max external load	100mA
Rail-ground resistance	<3Ω
Internal load	Default is no load, but there are optional pull-up or pull-down resistors (alterable in blocks of four on daughterboards). Maximum load (i.e. minimum resistance) 330Ω (36mA at 12V), 3W.
Set time	<10μs (hardware), set at next iteration of software loop, configurable down to 500μs.
Default Configuration	High or low side switch by software configuration on initialization. High rail is common supply. No load resistors.

Analog Input card

The Analog Input card provides 32 channels, configured as eight daughterboards of four channels each. The Analog Input card provides the ADC inputs (5V, unprotected), with the daughterboards doing the necessary signal conditioning and protection. Note that daughter boards must always be used in pairs for correct operation, although each half of a pair may be of any type.

Example application

Taking outputs from instrumentation in the ECM under test. In an ABS system the input card may be used to check the action of real actuators or read test measurements.

Circuit description

The Analog Input card consists of an array of 12 bit analog-to-digital converters, with a choice of daughterboards that convert a range of input types to 0 – 5V for the motherboard. Each daughterboard has four channels, thus (labeling from zero), daughterboard n conditions channels $4n$ to $4n+3$ for the motherboard.

Base address selection

Rev. 2 cards and upwards have the standard address switches.

Rev. 1 cards, however, have DIL switches – the silkscreen labels for these are incorrect.

Address line	A18	A17	A16	A15	A14	A13	A12	A11	A10	A9	A8
Switch	S3(3)	S3(2)	S3(1)	S2(4)	S2(3)	S2(2)	S2(1)	S1(4)	S1(3)	S1(2)	S1(1)
Silk screen label	17	18		12	13	14	15	8	9	10	11

A switch that is on indicates that the address line must be low for board selection. A19 is assumed to be high.

For example, to set the base address to C8500:

Address line	A18	A17	A16	A15	A14	A13	A12	A11	A10	A9	A8
Switch	S3(3)	S3(2)	S3(1)	S2(4)	S2(3)	S2(2)	S2(1)	S1(4)	S1(3)	S1(2)	S1(1)
Silk screen label	17	18		12	13	14	15	8	9	10	11
State	off	on	on	off	on	on	on	on	off	on	off

Card configurations

Other than the eight daughterboards, there are no configuration options on this board.

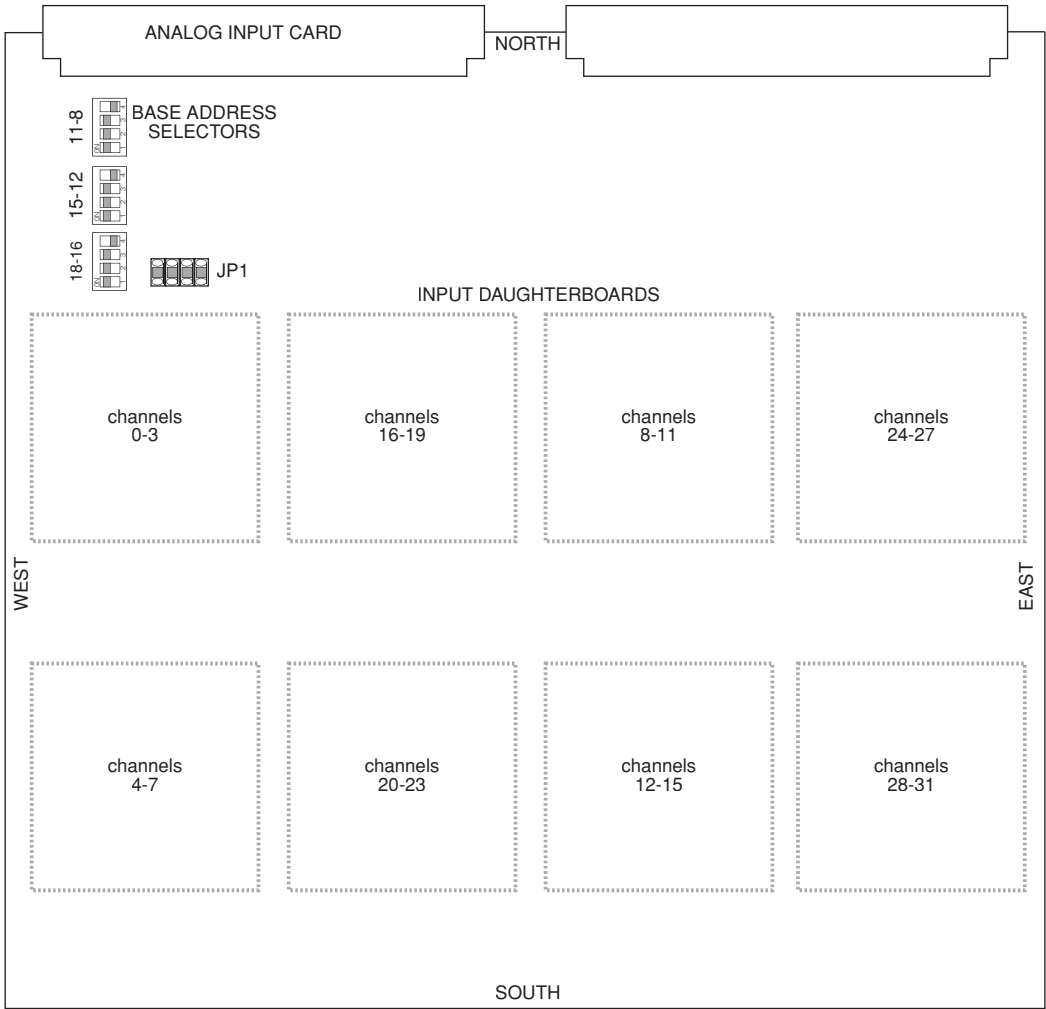
I/O channel assignment

Use the following table to find out which channels on the Analog Input card connect to which pin on the front panel connectors. Mapping channels and pins in the Front Panel section of the I/O System earlier in this Hardware chapter explains how to use these tables.

Connector	Slot	Slot	Slot	Connector	Slot	Slot	Slot
1	15	13	12	1	15	13	12
2	11	10	9	2	11	10	9
3	8	7	6	3	8	7	6
4	5	4	3	4	5	4	3
5	2	1	0	5	2	1	0
Input channel	pin	pin	pin	Input channel	pin	pin	pin
0	J	AU	CH	16	K	BM	DC
1	H	AT	CF	17	AC	BL	DB
2	F	AS	CE	18	AB	BK	DA
3	E	AR	CD	19	AA	BJ	CZ
4	D	AP	CX	20	Z	BW	CY
5	C	BH	CW	21	Y	BV	DS
6	B	BF	CV	22	X	BU	DR
7	A	BE	CU	23	W	BT	DP
8	U	BD	CT	24	V	BS	DN
9	T	BC	CS	25	AN	BR	DM
10	S	BB	CR	26	AM	CC	DL
11	R	BA	CP	27	AL	CB	DK
12	P	AZ	DH	28	AK	CA	DJ
13	N	AY	DF	29	AJ	BZ	EB
14	M	BP	DE	30	AH	BY	EA
15	L	BN	DD	31	AF	BX	DZ

Component location

This diagram shows the user-adjustable components found on the Analog Input card.



Component location on the Analog Input card

Note: For AutoSim applications the links on JP1 should always be fitted.

Daughterboards

Voltage Input daughterboards

This daughterboard accepts four channels of input voltages from 0 – 25V (relative to the simulator ground). It has a DC impedance of 100k Ω down to ground.

Daughterboard channel	Function
0	V0
1	V1
2	V2
3	V3

Channel assignment for the voltage input daughterboard

Current Input daughterboard

This daughterboard accepts two channels of two terminal current inputs. The current range is $\pm 20\text{mA}$, sensed through a resistance of 20 Ω . Current is sensed as positive when passing from the + to the – terminals.

Daughterboard channel	Function
0	I0+
1	I0–
2	I1+
3	I1–

Example application

Taking outputs from instrumentation in the ECM under test. In an ABS system the input card may be used to check the action of real actuators or to read test measurements.

Circuit description

The Analog Input card consists of an array of analog-to-digital converters, with a choice of daughterboards to adapt to a range of input types. Each analog input across a 25V range is converted to a 12-bit digital equivalent at high speed, and with very high accuracy. This digital information is then passed to the data bus where the AutoSim software can read it.

Analog Input card specifications

Analog input card	
Number of channels	32 (8 daughterboards of 4 inputs each)
Input voltage range	0 – 5V
Resolution	12 bits
Absolute accuracy	0.025%
Conversion time	≤100μs (hardware for all channels), detected at next iteration of software loop, configurable down to 500μs

Voltage Input daughterboard	
Number of channels	4 channels
Configuration	Voltage input relative to analog ground
Input impedance	100kΩ to ground
Input voltage range	0 – 25V
Absolute accuracy	0.05%

Current Input daughterboard	
Number of channels	2 channels
Configuration	2 terminal current input. Both voltages must lie between ground and battery
Input impedance	20Ω
Input current range	±20mA
Absolute accuracy	0.05%

Analog Output card

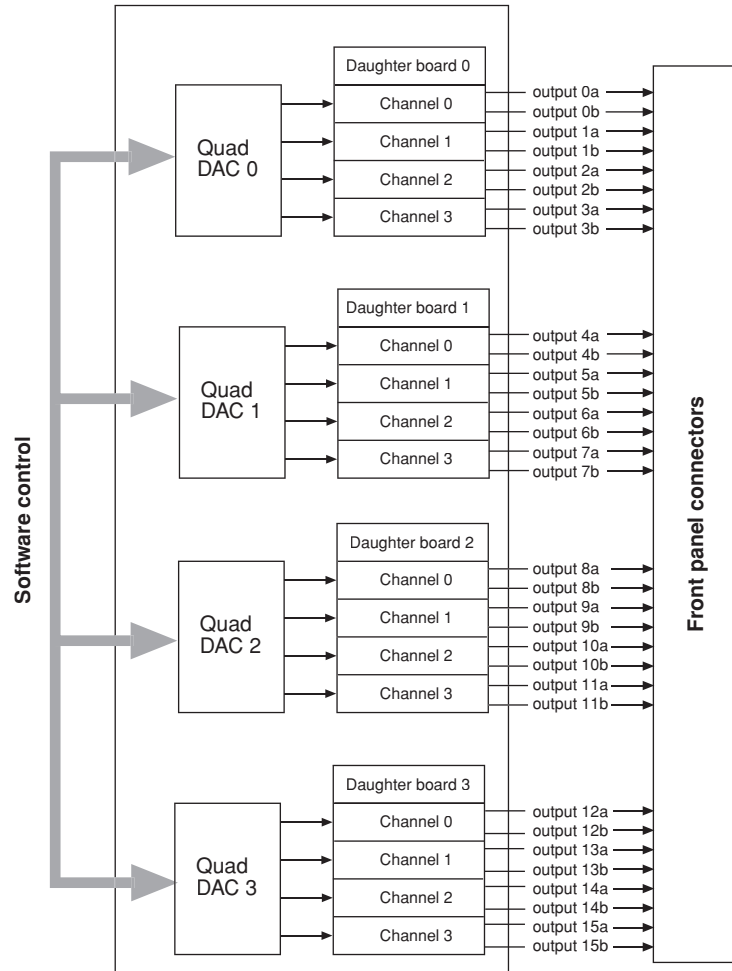
The Analog Output card is an array of digital-to-analog converters, with four daughterboards of varying types providing the analog signal manipulation and conditioning.

Example application

Voltage- and Current-generating sensors, e.g. pressure sensors, throttle pedal position, rotary switches, thermistors and thermocouples.

Circuit description

The Analog Output card provides 16 controllable voltages that are routed through four daughterboards. Each daughterboard has four channels, each of which comprises one input (from the DAC), and two outputs, main (output a) and subsidiary (output b). The four channels 0–3 on the daughterboard map onto the 16 channels on the main analog board as shown in the following diagram.



How the four channels 0–3 on the daughterboard map onto the 16 channels of the main analog board

Card Configurations

Reset and power off

The analog outputs generally emulate sensors that are powered from the common switched power rail. Boards of Rev. 1 therefore zero the output when this power rail is taken below 1V. This is not always appropriate – the simulator may be wanting to control some other instrument (such as another power supply), or a sensor that is not powered from the common rail. Boards of Rev. 3 and above therefore have a link to determine when the output is zeroed. Possible options are:

- COM – output is zeroed when the SSWCOM (common rail) is taken below 1V.
- EXT – output is zeroed when the external rail is taken below 1V.
- NONE – output is always maintained while the software is running.

To select a particular option, link the two pins either side of the label. If no link is installed, the NONE option is installed by default. There are four such sets of links on the board, one for each daughterboard.

Card address

Rev. 2 and above boards have the standard base address switches as described in To set the Base Address in the Hardware installation section near the beginning of the Hardware chapter. Rev. 1 boards have a system of DIL switches that specify the state of the address line should be in for the board to be selected. The silk screen indicates which address line the switch controls.

These are as follows:

S3						S2				S1			
1	2	3	4	5	6	1	2	3	4	1	2	3	4
nc	a18	a14	a15	a16	a17	a10	a11	a12	a13	a6	a7	a8	a9

If a switch is on, the address line must be low for the card selection. Thus for the base address of C8540, the switches must be set as follows:

S3						S2				S1			
1	2	3	4	5	6	1	2	3	4	1	2	3	4
nc	a18	a14	a15	a16	a17	a10	a11	a12	a13	a6	a7	a8	a9
x	off	on	off	on	on	off	on	on	on	off	on	off	on

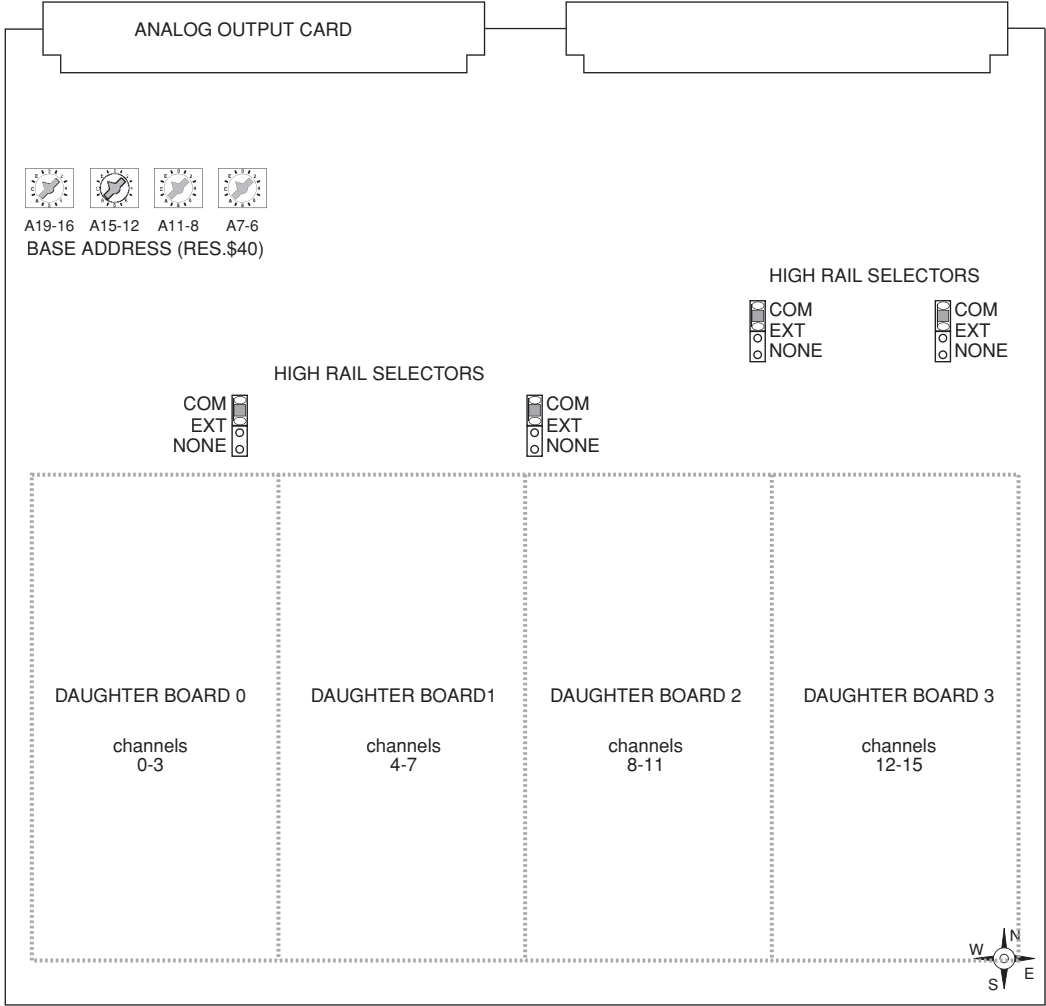
I/O channel assignment

Use the following table to find out which channels on the analog output card connect to which pins on the front panel connector. Mapping channels and pins in the Front Panel section of the I/O System earlier in this Hardware chapter explains how to use these tables.

Connector	Slot	Slot	Slot	Connector	Slot	Slot	Slot
1	15	13	12	1	15	13	12
2	11	10	9	2	11	10	9
3	8	7	6	3	8	7	6
4	5	4	3	4	5	4	3
5	2	1	0	5	2	1	0
Main output	pin	pin	pin	Subsidiary output	pin	pin	pin
0a	J	AU	CH	0b	H	AT	CF
1a	F	AS	CE	1b	E	AR	CD
2a	D	AP	CX	2b	C	BH	CW
3a	B	BF	CV	3b	A	BE	CU
4a	U	BD	CT	4b	T	BC	CS
5a	S	BB	CR	5b	R	BA	CP
6a	P	AZ	DH	6b	N	AY	DF
7a	M	BP	DE	7b	L	BN	DD
8a	K	BM	DC	8b	AC	BL	DB
9a	AB	BK	DA	9b	AA	BJ	CZ
10a	Z	BW	CY	10b	Y	BV	DS
11a	X	BU	DR	11b	W	BT	DP
12a	V	BS	DN	12b	AN	BR	DM
13a	AM	CC	DL	13b	AL	CB	DK
14a	AK	CA	DJ	14b	AJ	BZ	EB
15a	AH	BY	EA	15b	AF	BX	DZ

Component location

The diagram below shows where the user-adjustable components are found on the Analog Output card.



Layout of the Analog Output card

Daughterboard options

Voltage Output daughterboard

This daughterboard provides four channels of 0–20V at a maximum current of 25mA per channel. It also provides protected grounds.

Example application

There are many sensors that output voltages – accelerator pedal position (and other potentiometers), some pressure sensors etc.

Circuit description

Each of the four channels on the daughterboard doubles the 0–10V from the DACs then buffer it and filter it. This forms the main output – the subsidiary output is a ground connection protected by a resettable, thermal fuse of resistance (approx) 2Ω . Each output can source or sink 25mA – above this limit the current is limited by reducing or increasing the output voltage.

Output channel assignment

Channel	Main output (a)	Subsidiary output (b)
0	Voltage output 0	Fused ground
1	Voltage output 1	Fused ground
2	Voltage output 2	Fused ground
3	Voltage output 3	Fused ground

Specifications

Number of channels	4
Voltage range	0–20V
Resolution	12 bits, monotonic
Output current	$\pm 25\text{mA}$
Accuracy	0.05%
Output impedance	$<100\text{m}\Omega$, not including cables
Filter time constant	$100\mu\text{s} \pm 10\%$

Constant Current Source Output daughterboard

The constant current daughterboard sources two channels of 0 – 1.25A at up to 3V into an external load.

Circuit description

Each of the two channels on the card converts the input voltage from the DAC into a current source, deriving the power from the internal 5V supply.

Output channel assignment

Channel	Main output (a)	Subsidiary output (b)
0	Current source 0	Not connected
1	Current source 1	Not connected
2	Not connected	Not connected
3	Not connected	Not connected

Specifications

Number of channels	2
Current range	0–1.25A
Resolution	12 bits, monotonic
Maximum load voltage	3V
Accuracy	0.5%

Thermistor Output daughterboard

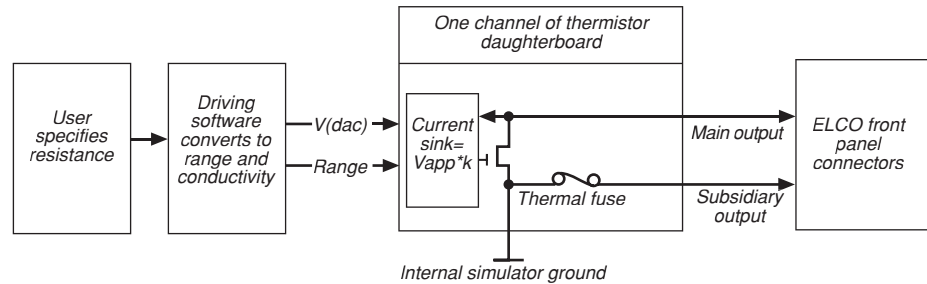
The thermistor output daughterboard provides four channels of resistance. Each resistor has one end connected to the ground; the other is the main output channel. The subsidiary output channel is a ground, protected with a thermally resettable fuse.

Example application

Thermistors are the intended application, but it can be used for other resistive loads to ground.

Circuit description

Because of the wide operating range, each channel on the board works in four different ranges. This is almost transparent to the user because the driving software converts a desired resistance into an appropriate range and conductance and feeds it to the daughterboard. The circuit sinks a current that is equal to this conductance multiplied by the applied voltage on the main output pin. This is shown diagrammatically below:



You should bear the following two points in mind when using the board:

- Protection – the board can only dissipate a limited amount of power, giving it an operating "envelope". If it is taken outside this envelope, the board will increase the resistance to protect itself. This cannot happen at pull-up voltages of 5V or below. See the diagram below for the operating envelope.
- Range changes – when the driving software changes range, there is a short (1ms) glitch on the output.

Output channel assignment

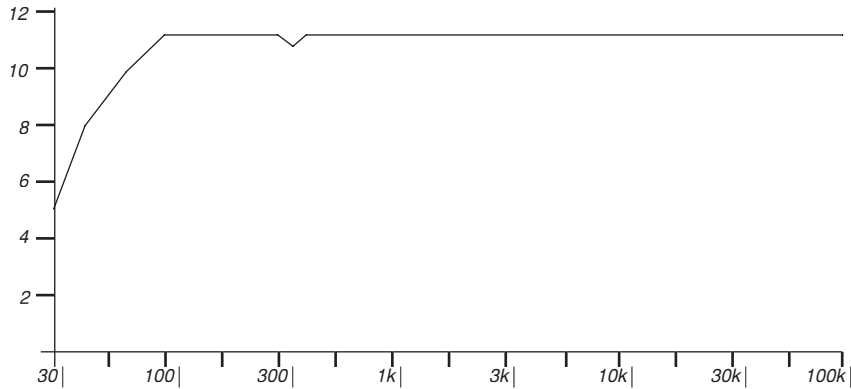
Channel	Main output(a)	Subsidiary output(b)
0	Resistance output 0	Fused ground
1	Resistance output 1	Fused ground
2	Resistance output 2	Fused ground
3	Resistance output 3	Fused ground

Specifications

Number of channels	4
Resistance range	40Ω–100kΩ
Maximum power	500mW per channel
Accuracy	1% (worst case)
Filter time constant	100μs±10%

Maximum voltages

The graph below shows the maximum voltage that may be applied to the main output before the device will increase the resistance to protect itself.



Thermocouple Output daughterboard

The thermocouple output board provides four voltage outputs. The range of each output is 0–50mV, which is the working range of most thermocouples.

Example application

Thermocouples can be used in relatively low temperature applications (e.g. water temperatures) or high temperature ones, notably exhaust gas temperature.

Circuit description

Each of the four channels on the daughterboard reduces the 0–10V voltage input to provide 0–50mV. This forms the main a output – the subsidiary b output is a ground. Both of these outputs are protected by a resettable, thermal fuse of approximately 2m Ω resistance.

Output channel assignment

Channel	Main output	Subsidiary output
0	Voltage output 0	Fused ground
1	Voltage output 1	Fused ground
2	Voltage output 2	Fused ground
3	Voltage output 3	Fused ground

Specifications

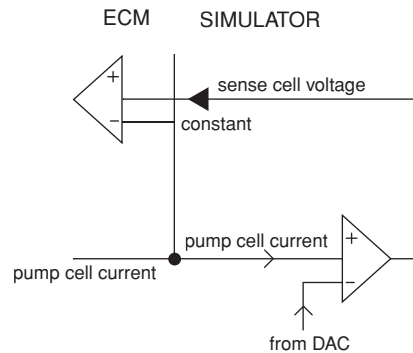
Number of channels	4
Voltage range	0–49.75V
Resolution	12 bits, monotonic
Output impedance	22 Ω (main output) 2 Ω (subsidiary ground)
Accuracy	0.1%

Lambda Oxygen Sensor Output daughterboard

This daughterboard emulates the behavior of a Lambda UEGO sensor, which is used to measure oxygen concentrations in exhaust gases. Each board gives two channels of lambda output (but occupies four channels of the Analog Output card).

Example Application

The board simulates two Lambda sensors consisting of a sense cell and a pump cell where the pump cell current is varied by the ECM to keep the sense cell voltage constant.



Circuit description for one channel

The type of Lambda sensor simulated by this board consists of two parts – a sense cell and a pump cell. The sense cell produces an output voltage, which changes abruptly at $\lambda=1$. The pump cell simulates adding or removing oxygen from the sense cell at a rate controlled by the current passing through it – this has the effect of shifting the switching point of the sense cell. By adjusting the pump cell current to keep the sense cell voltage just at its switching point, the ECM can determine the lambda voltage value (from the current).

The simulator works by trying to make the ECM produce a particular pump cell current. It adjusts the sense cell voltage up or down according to whether the pump cell current is too high or too low. The pump cell current is related to the DAC voltage as follows:

$$\text{Pump cell current} = 1/250 ((\text{DAC voltage} - 5) + (\text{sense cell voltage}/4.7))$$

Output Channel Assignment

Channel	Main	Subsidiary
0	Pump0+	Pump0–
1	Sense0+	Sense0–
2	Pump1+	Pump1–
3	Sense1+	Sense1–

Specifications

Number of channels	2
Pump cell current (when sense cell voltage = 0V)	–20mA to +20mA
Sense cell voltage range	0 ... 5V
Current accuracy	±1%

Differential Output daughterboard

Sensors such as strain gages and pressure sensors produce a small variable differential voltage (up to 100 mV) superimposed on a larger fixed common mode voltage (up to 5V). This daughterboard simulates four such sensors.

Example Application

Any sensors that produce a small differential voltage superimposed on a larger common mode voltage – pressure sensors, strain gauges etc.

Circuit description

Each pair of outputs is driven with the same common mode voltage of 5V. The individual pins are then offset slightly from this value, so one increases as the other decreases. The maximum offset is when one pin is at 5V + 50mV, and the other is at 5V – 50mV, giving a differential voltage of 100mV.

Output Channel Assignment

Channel	Main	Subsidiary
0	Output0+	Output0-
1	Output1+	Output1-
2	Output2+	Output2-
3	Output3+	Output3-

Specifications

Number of channels	4
Common Mode Voltage	0...5V (set by resistor)
Common Mode Voltage Accuracy	±3%
Differential Voltage	−100mV ... ±100mV
Differential Voltage Accuracy	±0.5%
Output Impedance	20Ω (each pin)
Max output current	1mA

LVDT Sensor daughterboard

The LVDT sensor daughterboard simulates either two or four LVDT (Linear Variable Differential Transformer) sensors. These consist of a primary coil and a center-tapped secondary coil. The primary coil is energized by an AC voltage produced by the conditioning circuit. Moving a metal core changes the coupling between the primary and the secondary windings, which changes the voltage induced. The secondary voltage is measured by the conditioning circuit and processed to produce a voltage proportional to the core position.

Example Application

LVDT sensors are used wherever position has to be measured with as little friction as possible – throttles, suspension measurements etc.

Specifications

Number of channels	2 or 4 output channels depending on mode
Configuration	LVDT emulator
Frequency range	1 ... 10kHz
Excitation Voltage Range	0 ... 10V peak (0...7Vrms)
Output Voltage	–100% to +100% of excitation voltage
Accuracy	1%

Circuit Description

Each LVDT board contains two ‘circuits’. Each circuit can simulate either one or two LVDTs. Thus each LVDT daughterboard can simulate either two or four LVDTs.

Each circuit works by multiplying the primary voltage by a value depending on the DAC setting and the mode selected. Each circuit has three main operating modes. The modes and the output polarity can be changed in software (by changing the daughterboard properties).

Mode	Secondary Output +	Secondary Output -
0	$\text{primary} * (\text{dac0} / 10)$	$\text{primary} * (1 - \text{dac0} / 10)$
1	$\text{primary} * (\text{dac0} / 5 - 1)$	$\text{primary} * (\text{dac0} / 5 - 1)$
2	$\text{primary} * (\text{dac0} / 5 - 1)$	$\text{primary} * (\text{dac1} / 5 - 1)$

1 LVDT 5 wire mode (default)

Both the outputs are controlled from a single DAC channel. The outputs do not change polarity but vary from 0 to 100% and from 100 to 0% as the DAC voltage changes from 0 to 10V. This simulates an LVDT with five wires connected. The center tap of the secondary is earthed or at a fixed DC voltage – the secondary voltage of each half of the secondary varies between 0 and its maximum value.

1 LVDT 4 wire mode

This mode simulates an LVDT in 4-wire mode, with the two secondaries in series. The output voltage in this case varies from -100 to $+100\%$ as the DAC voltage changes from 0 to 10V. Both outputs are controlled from DAC0, so a 'push-pull' output can be made by setting one output to be inverted.

2 LVDTs mode

This allows two four-wire LVDTs to be simulated if they have the same primary voltage. The output voltages vary as for mode 1, but the secondary output is controlled by a separate DAC channel. This means the two secondaries can be altered independently.

The primary input only has one pin assigned to it, so if the driver for the LVDT is a 'push-pull' type (rather than one end of the primary coil being connected to earth, the two ends are driven in anti-phase), then the output voltages will only be half of the expected value.

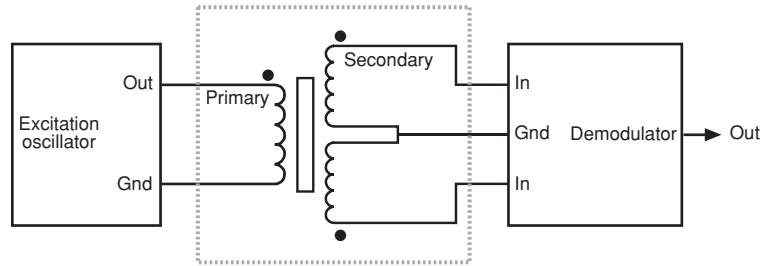
Output Channel Assignment

Channel	Main Output	Subsidiary Output
0	Primary Input 0	Secondary Center Tap 0
1	Secondary Output 0+	Secondary Output 0-
2	Primary Input 1	Secondary Center Tap 1
3	Secondary Output 1+	Secondary Output 1-

Using the AutoSim LVDT daughterboard

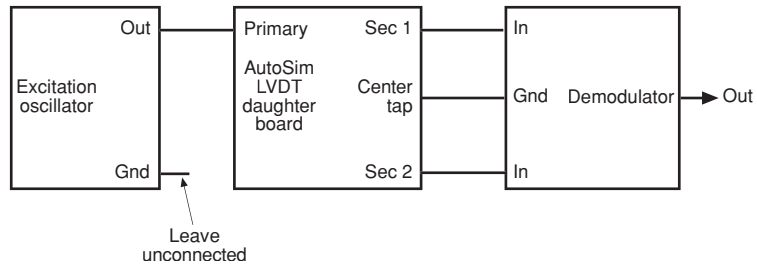
You can use the LVDT daughterboard in several ways, which are shown in the following diagrams.

Five-wire mode



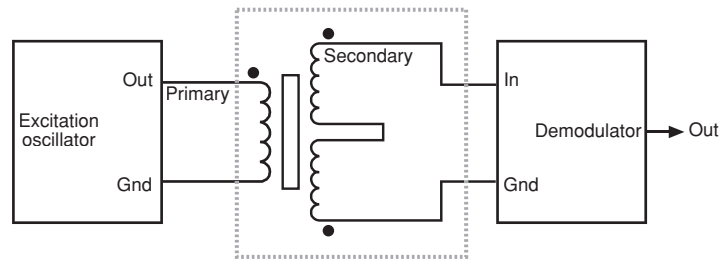
Five-wire mode connections for a real LVDT

For this configuration set the AutoSim LVDT daughterboard to 1 LVDT five-wire mode, with both secondary polarities set to 'non inverting'.



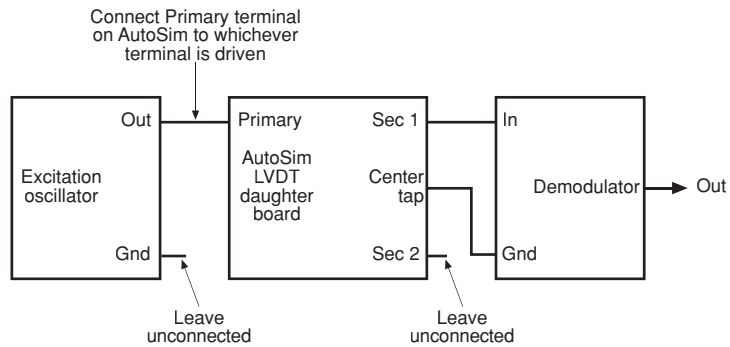
LVDT daughterboard five wire mode connections for AutoSim

Four-wire mode



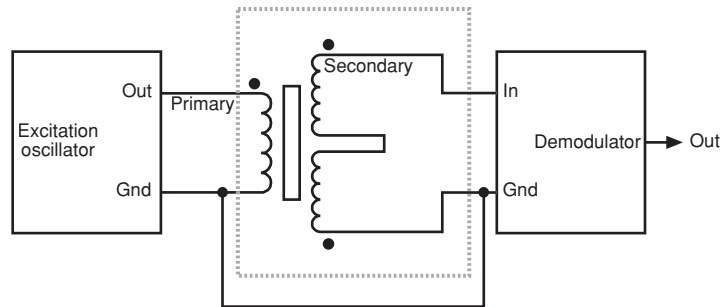
Four wire mode connections for a real LVDT

For this configuration set the AutoSim LVDT daughterboard to 1 LVDT four-wire mode, and set both secondary polarities to 'non inverting'.



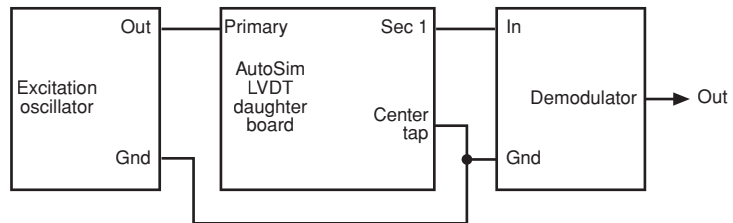
LVDT daughterboard four-wire mode connections for AutoSim

Three-wire mode



Three wire mode connections for a real LVDT

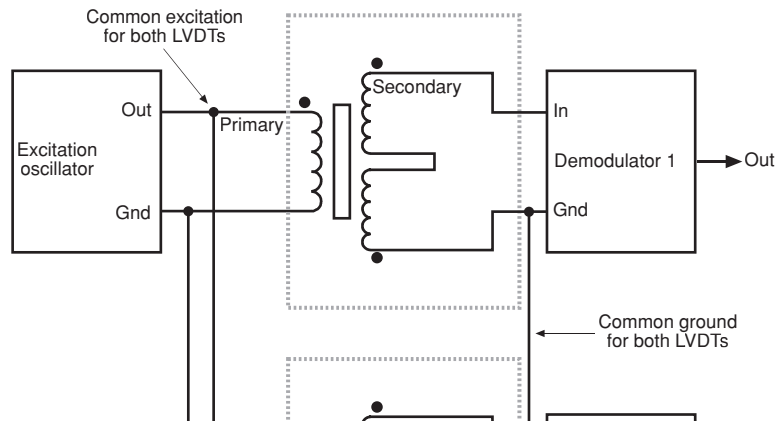
For this configuration set the AutoSim LVDT daughterboard to 1 LVDT four-wire mode, with both secondary polarities set to 'non inverting'.



LVDT daughterboard three wire mode connections for AutoSim

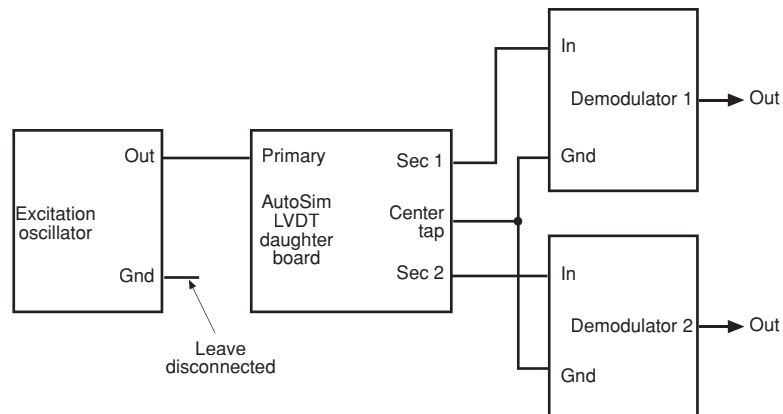
Simulating two LVDTs with one AutoSim circuit

If two LVDTs are wired in 3- or 4-wire mode, and have the same excitation source, they can both be simulated with one AutoSim LVDT circuit (i.e. one daughterboard can simulate four LVDTs).



Connections for two real LVDTs

For this configuration set the AutoSim LVDT daughterboard to 2 LVDTs mode, with both secondary polarities set to 'non inverting'.



AutoSim connections when using one LVDT daughterboard to simulate 2 LVDTs

Analog Output card specifications

Analog Output card	
Number of channels	4 daughterboards of 4 channels each
Configuration	Voltage source
Output impedance	<10Ω
Basic voltage range	0–10V
Resolution	12 bits
Absolute accuracy	0.025%, guaranteed monotonic
Set time	<10μs (hardware), set at next iteration of software loop, configurable down to 500μs

Voltage Output daughterboard	
Number of channels	4
Configuration	Voltage source
Basic voltage range	0–20V
Resolution	12 bits
Absolute accuracy	0.05%
Output impedance	<100mΩ
Maximum load	25mA (some load sharing possible)
Output response	100μs

Constant Current Output daughterboard	
Number of channels	2
Configuration	Constant current source
Range	0–1.25A
Maximum output voltage	3V
Absolute accuracy	0.5%

Thermistor Simulator Analog Output daughterboard

Number of channels	4
Configuration	Thermistor emulator
Basic range	40 Ω –100k Ω
Resolution	0.6% (worst case)
Absolute accuracy	1%

Thermocouple Emulator Output daughterboard

Number of channels	4
Configuration	Thermocouple emulator
Absolute accuracy	0.1%

Lambda Oxygen Sensor Output daughterboard

Number of channels	2
Pump cell current	–20mA to +20mA
Sense cell voltage range	0...5V
Current Accuracy	$\pm 1\%$

Differential Output daughterboard

Number of channels	4
Common Mode Voltage	0...5V (set by resistor)
Common Mode Voltage	Accuracy $\pm 3\%$
Differential Voltage	–100mV $\pm$ 100mV
Differential Voltage Accuracy	$\pm 0.5\%$
Output Impedance	20 Ω (each pin)
Max output current	1mA

LVDT Output daughterboard	
Number of channels	2 or 4 output channels depending on mode
Configuration	LVDT emulator
Frequency range	1...10kHz
Excitation Voltage Range	0...10V peak (0...7Vrms)
Output Voltage	−100% to +100% of excitation voltage
Accuracy	1%

Pulse Width Modulation Input card

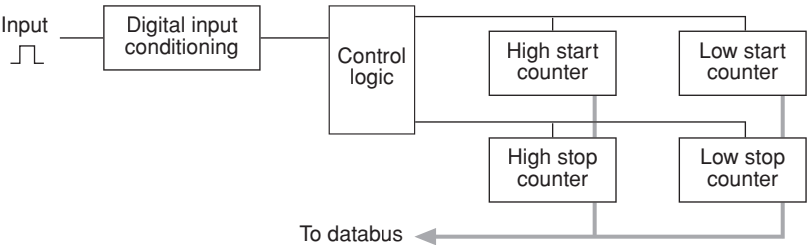
The Pulse Width Modulation (PWM) Input card directly measures the high and low periods and frequencies of 16 separate discrete PWM signals using sets of 24-bit counters. The interface circuitry is identical to the Discrete Input circuitry and has the same hardware and software options, so it can also measure the state of the signal.

Example application

Frequently, an ECM has to provide a continuously variable voltage or current output to drive a solenoid or motor. If the ECM used a linearly controlled output, it would have to be able to dissipate as much power as was delivered to the load. Using PWM outputs on the ECM greatly reduces the power dissipation. The PWM input card enables the output waveform to be measured to determine the duty cycle and frequency.

Circuit description

The PWM Input card measures the low and high times of its input signals using 16 individual clock sets, one of which is shown below.



Low and high times of the PWM input signal

In each of the sixteen channels, the input is first put through the standard discrete input circuits. The control logic waits for rising and falling edges in the signal, and sends start and stop signals to the two sets of counters – one to measure the high time, and the other to

measure the low-time. At any time, the counters can be read across the data bus and will always contain the latest high and low counts, which the software can use to calculate the original analog signal level.

Card configurations

This section tells you how to configure the card by setting various jumper links.

See Configuring discrete inputs in the Discrete I/O card section earlier in this chapter for common configuration information. See the component diagram at the end of this section for jumper positions.

Input load

See the general information in Configuring discrete inputs in the Discrete I/O card section.

Input threshold voltage

Channels	Jumper	Link for $V_R/2$	Link for V_{Zener}
0–3	JP1	East	West
4–7	JP2	East	West
8–11	JP3	East	West
12–15	JP4	East	West

Input High rail voltage

Channels	Jumper	Link for V_{5V}	Link for V_{common}	Link for V_{Ext}
0-3	JP5	North	East	South
4-7	JP6	North	East	South
8-11	JP7	North	East	South
12-15	JP8	North	East	South

Base address selection

The four switches set the base address described in 'To set the Base Address' in the Hardware installation section near the beginning of the Hardware chapter.

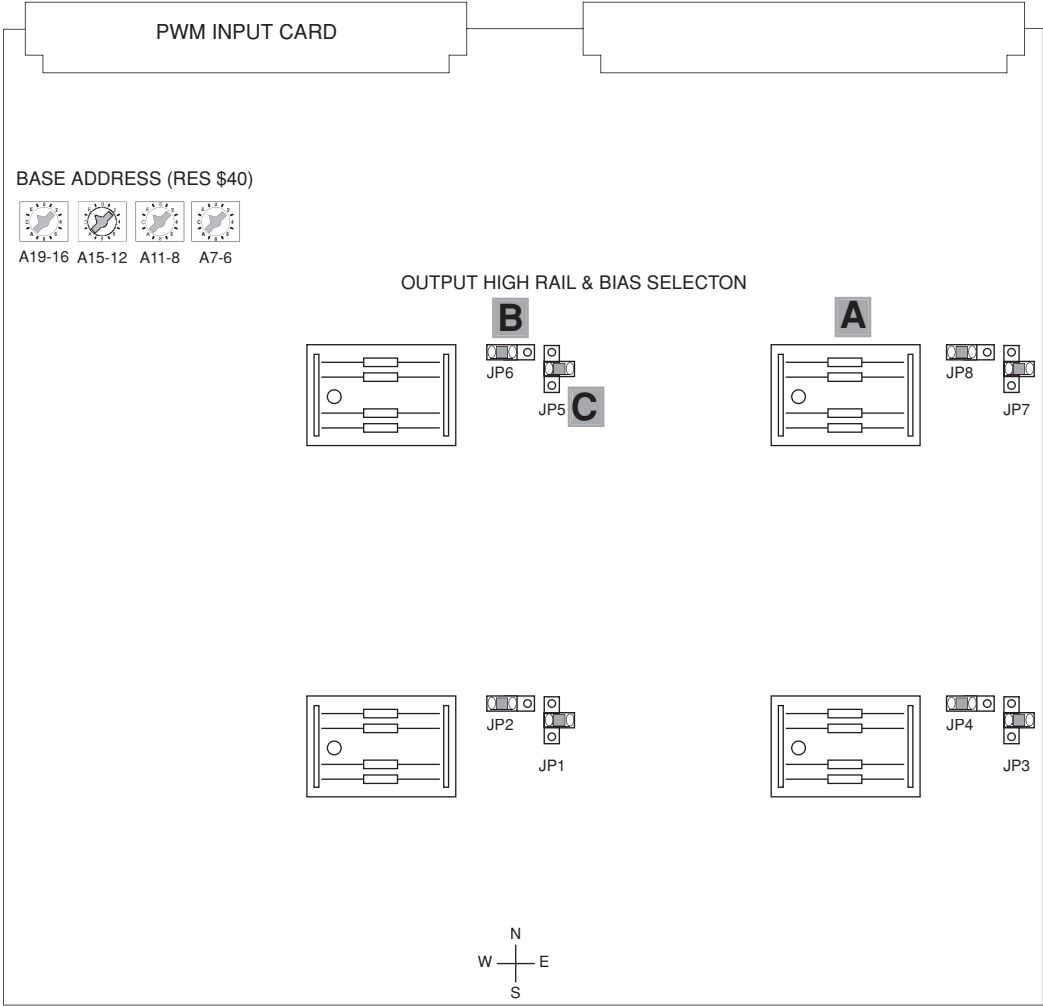
I/O channel assignment

Use the following table to find out which channels on the PWM Input card connect to which pin on the front panel connectors. Mapping channels and pins in the Front Panel section of the I/O System earlier in this Hardware chapter explains how to use these tables.

Connector	Slot	Slot	Slot
1	15	13	12
2	11	10	9
3	8	7	6
4	5	4	3
5	2	1	0
Main output	pin	pin	pin
0	J	AU	CH
1	F	AS	CE
2	D	AP	CX
3	B	BF	CV
4	U	BD	CT
5	S	BB	CR
6	P	AZ	DH
7	M	BP	DE
8	K	BM	DC
9	AB	BK	DA
10	Z	BW	CY
11	X	BU	DR
12	V	BS	DN
13	AM	CC	DL
14	AK	CA	DJ
15	AH	BY	EA

Component location

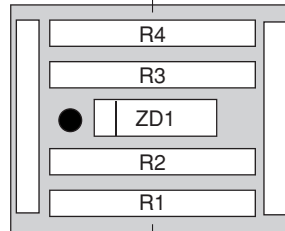
The diagram below shows where the user-adjustable components are found on the PWM Input card. See the key on the next page for detailed component descriptions.



Layout of the PWM Input card

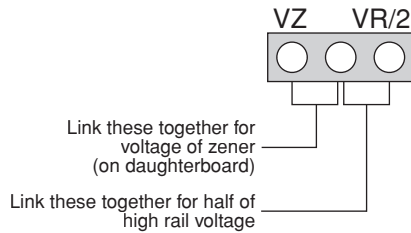
A OUTPUT BIAS DAUGHTERBOARD

Resistor for highest channel in group

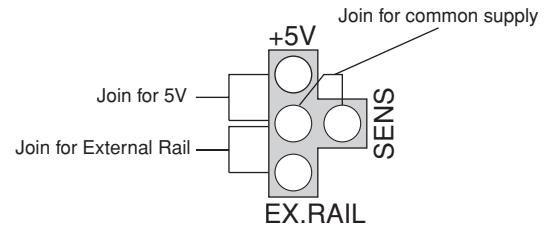


Resistor for lowest channel in group

B INPUT THRESHOLD SELECTION LINKS



C HIGH RAIL SELECTION LINKS



Components of the PWM Input card

PWM Input card specifications

PWM Input card	
Number of channels	16
Configuration	Individually software configurable for: ■ pull up to high rail; ■ pull down to ground; ■ input is floating (totem pole output assumed from ECM). High rail is link configurable as one of: ■ sensor supply; ■ +5V (derived from battery); ■ external rail, supplied from front panel. Restriction: within any group of four, all high rails must be identical.
Load	Maximum load (i.e. minimum resistance) 330Ω (36mA at 12V), 3W. Alterable in blocks of four by plug-in boards.
Threshold voltage	Link selectable $(VR/2) \pm 5\%$, or V_{Zener} . Zener diode alterable in blocks of four by plug-in boards.
Hysteresis	>200mV
Detection time	Next edge (hardware), detected at next iteration of software loop, configurable down to 500 μ s.
Frequency range	0.12Hz – 20kHz and DC (Higher with degraded resolution).
Frequency resolution	500ns on each edge (e.g. 0.05% at 1kHz, 1% at 20kHz).
Frequency accuracy	0.1%
Duty cycle range	0% – 100%
Duty cycle resolution	500ns on each edge (e.g. 0.05% at 1kHz, 1% at 20kHz)
Duty cycle accuracy	Limited by resolution; ± 250 ns on each edge

The PWM input card supports 16 channels, where each channel supports Frequency, Duty Cycle, State and On-Time variables. Inputs are arranged in groups of four, with common resistor daughterboards and common high rail selection, as for discrete inputs.

Resistor daughterboards are not shown in the property pages.

- Card properties – The standard card property sheet shows the slot number for the card.
- Channel properties (inputs) – PWM input channels show the property page:
 - Channel Bias sets the input bias for the channel, which may be set to pull high (to the channel group's high rail), pull low (to ground), or left floating. If you change this property, you will need to stop and start the model before the active change to the channel takes effect.
 - Active State sets the logic level for the channel. When set to Active High, the channel will report to a value of 1 (mark) when the input is pulled to the high rail.
 - When set to Active Low, the channel reports to a value of 1 (mark) when the input is pulled to ground. Active State takes effect when the property is saved using Apply of Low, and the model is stopped and restarted.
 - Low frequency cut-off sets the lowest frequency that the channel variables reports before the input is assumed to be in a steady state. When in a steady state the channel variables report a frequency of 0Hz and a duty cycle of 0 or 100% according to the channel state. The model should be stopped and restarted for the change to take effect.
 - Variable properties – The standard Variable and Scaling property sheets are provided – see 'Editing model and variable properties' in the Software chapter.

There are hardware configuration links on the card to select a high rail and sense voltage for groups of inputs, and to set the high rail for groups of outputs.

Pulse Width Modulation Output card

The Pulse Width Modulation (PWM) Output card offers eight discrete PWM outputs with signal conditioning and an output frequency of up to 100kHz. The frequency resolution is constant over the entire range. This card works in exactly the opposite way to the PWM Input card, generating PWM signals from discrete high time and low time values. The interface circuitry is identical to the Discrete output circuitry and has the same hardware and software options.

Analog signals are generally affected by noise, and so cannot be totally relied upon to carry data signals with a high enough degree of accuracy. PWM is a method of fitting analog data into a discrete signal, thereby increasing its signal-to-noise ratio.

The analog signal is represented as a high-low discrete cycle, where the ratio of the high time to the period represents the analog voltage level. For example, a high analog measurement, say 80% of the highest voltage level, is translated into a discrete cycle where the signal is high for the first 80% of the total cycle, and then turns low for the last 20%.

Example application

Applications for PWM output cards are rare, as ECMs do not often accept PWM signals, but there are signals such as those from antitheft devices and other on-board computer signals that can be generated by this card. The signals 'requested torque' and 'delivered torque', which are sent between an ECM and ABS controller, are examples of PWM signals.

Circuit description

The PWM Output card generates the appropriate high- and low-times that make up the PWM signals using 16 individual clock sets. The software places the appropriate high time and low time counts on the data bus, and the high time counter then generates a high pulse for the appropriate amount of time, followed by the low-time counter pulling the output low for the remaining cycle time.

Card configurations

This section tells you how to configure the card by setting various jumper links.

See Configuring discrete inputs in the Discrete I/O card section earlier in this chapter for common configuration information. See the component location diagram at the end of this section for jumper positions.

Output bias

Outputs can be biased by installing up to two resistor daughterboards. Note that there can be no bias unless the daughterboard is fitted.

Output High rail voltage

Channels	Jumper	Link for V _{5V}	Link for V _{common}	Link for V _{Ext}
0–3	JP1	North	East	South
4–7	JP3	North	East	South

Bias switch

Channels	Jumper	Link for bias to V _R (high)	Link for bias to ground (low)
0–3	JP2	East	West
4–7	JP4	East	West

Note: The Output channel properties are software configurable to one of active-high or active-low. See the Hardware branches section in the Software chapter for instructions on how to set this. The default setting is active-high.

Board address

The four switches set the base address as described in To set the Base Address in the Hardware installation section near the beginning of the Hardware chapter.

I/O channel assignment

Use the following table to find out which channels on the PWM Output card connect to which pin on the front panel connectors. Mapping channels and pins in the Front Panel section of the I/O System earlier in this Hardware chapter explains how to use these tables.

Connector	Slot	Slot	Slot
1	15	13	12
2	11	10	9
3	8	7	6
4	5	4	3
5	2	1	0
Main output	pin	pin	pin
0	J	AU	CH
1	F	AS	CE
2	D	AP	CX
3	B	BF	CV
4	U	BD	CT
5	S	BB	CR
6	P	AZ	DH
7	M	BP	DE

Component location and description

The diagram below shows where the user-adjustable components are found on the PWM Output card. See the key on the following page for detailed component descriptions.

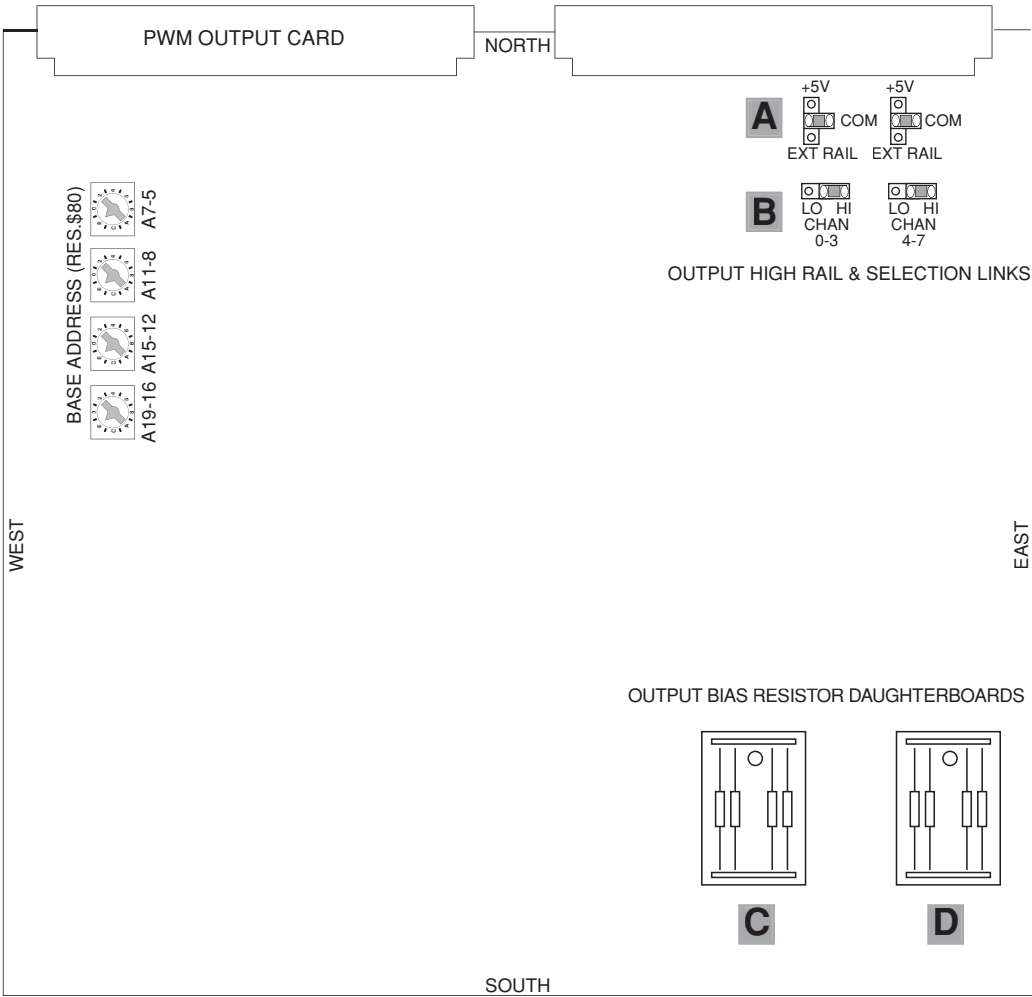
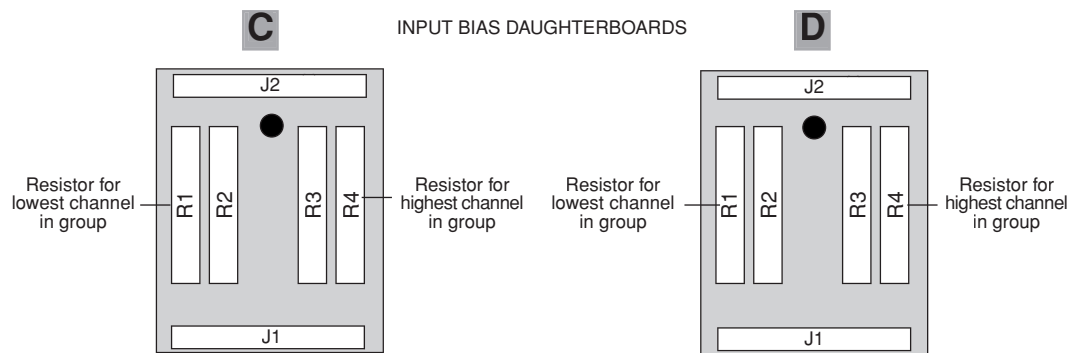
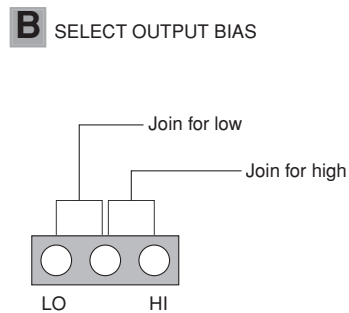
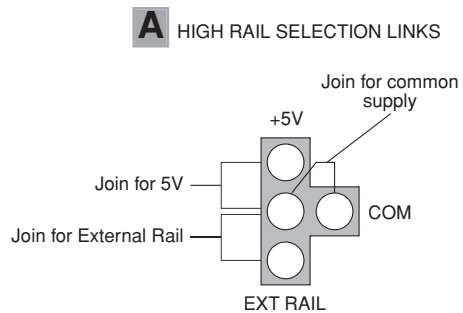


Diagram showing the layout of the PWM Output card



Components on the PWM Output card

**PWM Output card
specifications**

PWM Output card	
Number of channels	8
Configuration	Individually high-side or low-side to ground by software configuration at initialization. Pull-up rail voltage is a link selectable choice of: ■ common supply; ■ +5V; ■ external rail, supplied from the front panel. Restriction: within any group of four, all pull-up rail voltages must be the same.
Max. load	100mA
Internal bias	Default is no bias. Optional bias with pull-up or down resistors (alterable in blocks of four by plug-in boards).
Maximum bias	(i.e. minimum resistance) 330Ω (36mA at 12V), 3W.
Set time	<10μs (hardware), set at next iteration of software loop, configurable down to 500μs
Frequency range	0.12Hz–100 kHz
Frequency resolution	0.12Hz
Frequency accuracy	0.1%
Duty cycle range	0–100%
Duty cycle resolution	0.025% plus edge jitter
Duty cycle accuracy	100ns worst case when measured at $V_R/2$, not including edge jitter
Edge jitter	±250ns

The PWM output card supports eight channels, where each channel supports a Duty Cycle and Frequency variable. Outputs are arranged in groups of four, with common resistor daughterboards and common output biasing, as for discrete outputs. Resistor daughterboards are not shown in the property pages.

- Card properties – The standard card property sheet shows the slot number for the card.
- Channel properties – PWM output channels show the standard channel property page.
 - Channel bias reflects the output bias for the channel, which is set by hardware links to pull high (to the channel group's high rail), pull low (to ground), or left floating. This property page does not control the bias, it merely records the current state and must be manually set when changes are made to the hardware settings. It will be used in the future to control a 'virtual rail' facility, but is currently unused.
 - Active State sets the logic level for the channel. When set to Active High, the channel will drive the channel high for the mark part of the duty cycle. When set to Active Low, the channel will drive the channel low (to ground), for the mark part of the cycle. If you change the Active State for an output channel you will need to stop and start the model before the change will take effect.
- Variable properties – The standard Variable and Scaling property sheets are provided – see Editing model and variable properties in the Software chapter.

There are hardware configuration links on the card to select the high rail for groups of outputs.

Programmable Waveform Generator card

The Programmable Waveform Generator (PWG) card is a specialist I/O card that simulates sensors such as crank, cam and wheel speed sensors by generating the appropriate signals for input to the ECM under test. The PWG card simulates either variable reluctance sensors (VRS), where a sinusoidal wave is produced by the sensor each time a crank or cam tooth passes, or Hall effect sensors, where an open collector discrete signal is produced.

The PWG card has 8 output channels, each of which can represent a particular crank, cam or wheel speed signal. Each channel has a transformer-isolated analog output for VRS simulation along with two discrete outputs. One discrete output is open-collector and the other has a pull-up resistor. The analog outputs can provide up to 64V peak-to-peak and the discrete outputs can sink up to 100mA.

The board can generate arbitrarily shaped waveforms, stored as a series of 7-bit samples in a memory on the card. The card has two banks of memory, one for channels 0 to 3 and the other for channels 4 to 7. Each bank holds 128K samples, which can hold 16 patterns up to 8K long, or fewer, longer patterns up to a maximum length of 64K. The patterns in the memory cannot be altered while the card is generating waveforms, but a channel can switch to any of the patterns in the memory bank while the model is running. This allows simulation of faults such as a timing wheel losing a tooth while the engine is running.

Any contiguous group of channels can be locked together in frequency and phase, so that, for example, the cam sensor signals can be set to follow the frequency and phase of the crankshaft. The lowest numbered channel in the group acts as master, and the slave channels can have their phase changed independently to any value over the range of a complete engine cycle. The phase can be altered while the model is running, so that conditions such as the camshaft drive slipping can be simulated.

The Event Timer card can measure the timing of events in relation to a PWG waveform. To do this the PWG generates an angle clock signal, which is fed along the back-plane to the Event Timer card. There are switches on the PWG card to select which channels are used to provide angle clock signals.

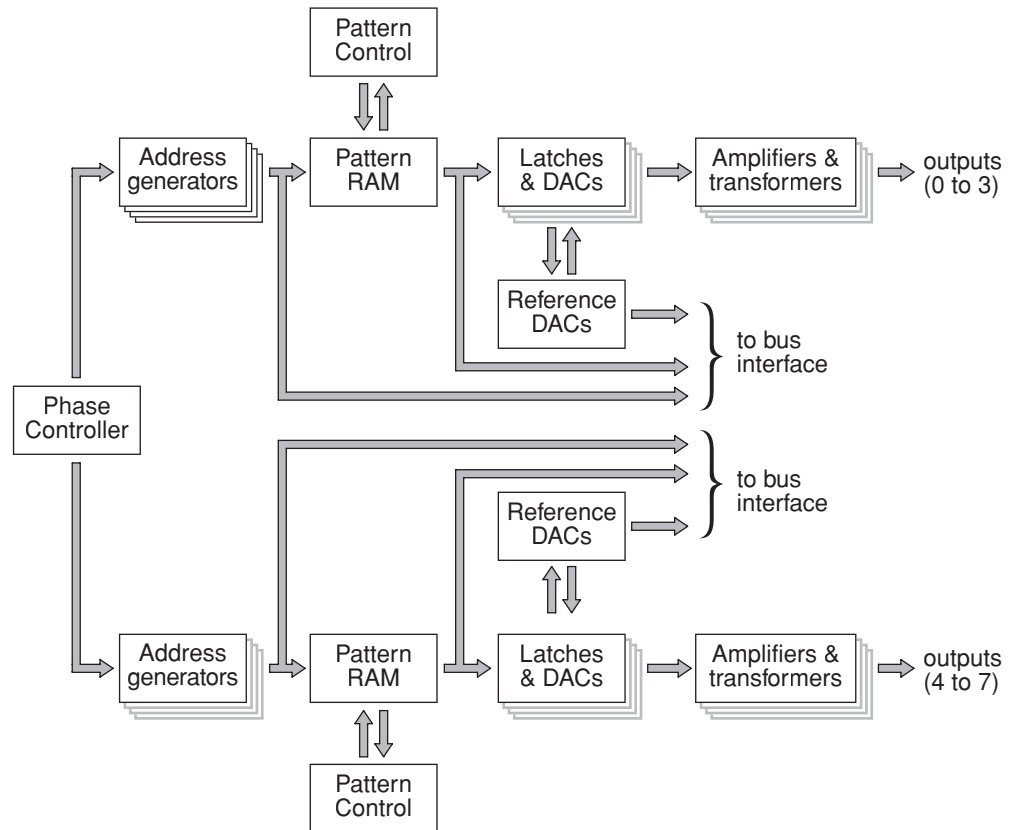
Example applications

Example applications for the PWG card are:

- Crank and cam position sensors
- Wheelspeed sensors.

Circuit description

The schematic diagram below shows the high-level structure of the PWG card. There are eight output channels altogether, organized as two banks of four.



The high-level structure of the PWG card

Address generators

The eight address generators are a combination of a digitally controlled variable frequency oscillator and a counter for the RAM address.

Pattern RAM and control

The waveforms to be produced are defined as a series of 7-bit samples stored in a waveform RAM. The waveform RAM has 8 bits, of which the 7 least significant bits are used for the output samples and the other bit is used for the end of pattern flag. There are two waveform RAMs, each with 128K bytes capacity, one for each group of four channels. Each pattern file is typically 8K samples in length, though it can be up to 64K samples long (obviously not all four pattern files in a group can be 64K).

There can be up to 16 pattern files in each RAM and each channel within the group may be switched to use any of the 16 files. It is acceptable for more than one channel in the group to use the same pattern file.

The pattern files can only be updated when the card is not generating waveforms. While the card is generating waveforms, each channel may be switched between a number of alternative patterns, but the waveform RAM cannot be accessed.

The patterns may be any number of samples in length up to the maximum length of 65536 samples. The end of the pattern is marked by a 1 in the most significant bit of the last sample of the pattern (the end of pattern flag).

Latches and output DACs

The output DACs convert the discrete patterns into the analog output voltages.

Reference DACs

The outputs of the reference DACs are fed to the reference inputs of the main DACs in order to control the overall amplitude of the output signal from each channel.

Output Amplifiers

The output amplifiers scale the output voltages from the DACs to achieve the full 0 – 64V range. The output voltage range is 0.25V – 64V peak-to-peak with a 20k Ω output impedance, and are transformer isolated.

Outputs

The board has eight output channels (numbered 0 to 7) with independent frequency and amplitude control, and can produce either a simulated VRS signal (in fact any arbitrary waveform) or an open collector discrete output (with or without pull-up resistor).

Phase controller

Outputs can be locked together at the same frequency with a controlled phase relationship. This is necessary if the outputs are used to simulate crank and cam signals from an engine.

The phase control is achieved by generating a signal that is true whenever an address counter is at zero. This signal is delayed by a channel of an 8253 counter timer chip, and the output of the timer chip is used to reset the slave channel at the appropriate time.

Each of the channels may be set to operate in master or slave mode. If a channel is in master mode, its frequency and phase are independent of all the other channels.

If a channel is in slave mode, it is phase locked to the previous channel. For example, if channel 2 were in slave mode, it would be locked in phase to channel 1. If channel 1 was in slave mode as well, then both channel 1 and channel 2 would be locked to channel 0.

Two or more cards can be connected together in a daisy chain style so that the output of one card is connected to the input of the next. This enables channels to be synchronized between cards. If channel 0 of a card is set to slave mode, its phase is locked to the next card along. In this way, channels may be synchronized across multiple cards. The 4MHz clock signals are not synchronized between cards, so there can be up to 250ns time difference in the outputs of two adjacent cards.

Each channel that is in slave mode can have its phase changed relative to its master channel, which is the first channel in master mode, counting downward from the slave channel. If channel 0 is in slave mode, then continue counting on the board that is driving the cascade input. For example, if channels 0, 1, 2 and 3 are in slave mode, the master channel is on the board driving the cascade input.

The phases of slave mode channels are controlled independently, i.e. if the phase of channel 1 is varied and channel 2 is in slave mode as well, the phase of channel 2 will not be affected. When a channel is in slave mode, the frequency is determined by the master channel and the slave channel frequency setting is ignored.

Synchronizing a Pi Event Timer card for angle measurement

The I/O connector of the PISA bus system contains four angle clock signals and four TDC signals. The event timer card can measure events by engine angle as well as time if its angle counter is synchronized to these signals. The event timer board has jumpers so that any one of the four pairs of angle signals can drive its angle counter.

The PWG card has jumpers so that any two of the four angle clock signals (and their corresponding TDC signals) can be synchronized.

Card configurations

This section tells you how to configure the card by setting various jumper links. See Configuring discrete inputs and Configuring discrete outputs in the Discrete I/O card section earlier in this chapter for common configuration information. See the component location diagram at the end of this section for jumper positions.

Card address

There are two hex switches in the northwest corner of the card to set the card's base address. These two switches should be set to the first two digits (in hex) of the base address.

Interrupt

The interrupt facility is not used in AutoSim, so do not fit a jumper to JP1.

Input High rail voltage

The pull-up resistors on the discrete outputs can be connected to one of three different high rails. All eight of the outputs have the same high rail.

Channels	Jumper	Link for V _{5V}	Link for V _{sensor}	Link for V _{Ext}
0-7	JP10	North	East	South

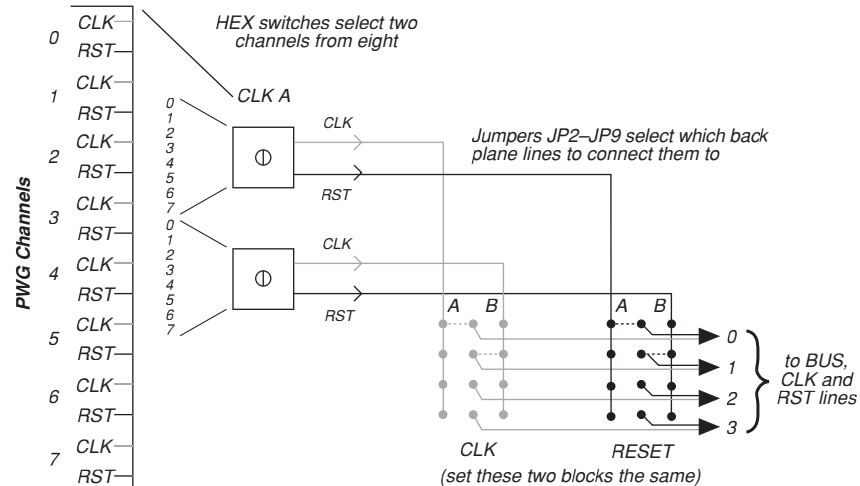
Output bias resistance daughterboards

These determine the pull-up resistance for the discrete outputs.

Angle clock channel selectors

If the Event Timer card is to measure the angle of events, it needs an angle clock and an angle reset signal. These two signals make the angle counter in the Event Timer card keep in step with the sample counter on the PWG. There are 4 pairs of lines on the backplane to carry the signals between the PWG and the Event Timer card. The Event Timer card can get its clock and reset signals from any of one of the four backplane signals, and the PWG card can drive any two pairs of the backplane signals. The two pairs can be operated by only two PWG channels.

To set the PWG card up, select which channel(s) need to control the Event Timer card. Usually this would be the channel that produces the crank position sensor waveform. Select these channels on the CLK A and CLK B switches. Next decide which pair of backplane lines (0-3) to connect to. Unless there are several Event Timer cards in the system, line 0 should be used. Fit the jumpers JP2 to 9 to connect to the required backplane lines (see diagram below).



Jumpers connected to the required backplane lines

Angle clock bus selectors

In the NE corner of the card there are jumpers (JP2 – JP9) to select which bus lines are driven. There are two groups of 12 pins.

There can be angle clocks on the bus for up to four Event Timer cards. The default position of these jumpers, if this is the first or only PWG card in the system, is that the jumpers should be in Column A, Row 0 and Column B, Row 1, with both blocks of jumpers looking the same. This generates angle clocks for the first two Event Timer cards (0 and 1). If this is the second PWG card in a system, move the jumpers in column A to Row 2 and in Column B to Row 3. This makes the card generate angle clocks for Event Timer cards 2 and 3 (if fitted). If there are more than 2 PWG cards, leave these jumpers off all cards apart from the first two.

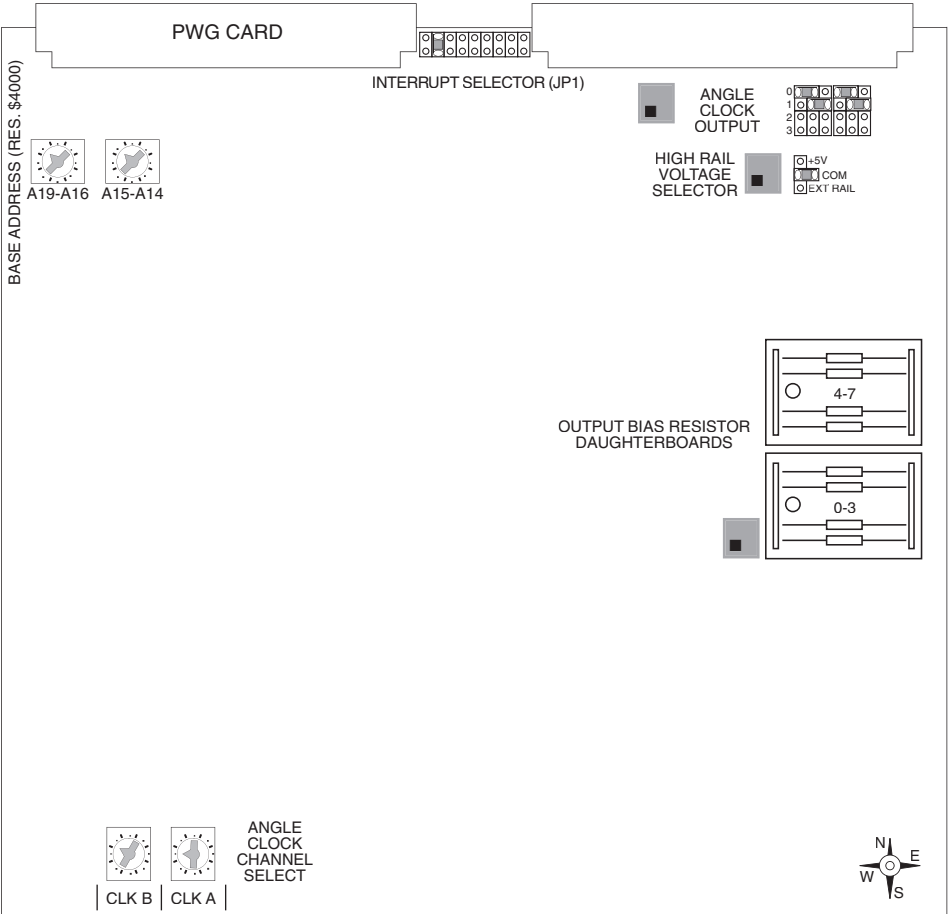
I/O channel assignment

Use the following table to find out which channels on the PWG card connect to which pin on the front panel connectors. Mapping channels and pins in the Front Panel section of the I/O System earlier in this Hardware chapter explains how to use these tables.

Connector	Slot	Slot	Slot	Connector	Slot	Slot	Slot
1	15	13	12	1	15	13	12
2	11	10	9	2	11	10	9
3	8	7	6	3	8	7	6
4	5	4	3	4	5	4	3
5	2	1	0	5	2	1	0
Input channel	pin	pin	pin	Input channel	pin	pin	pin
0+	J	AU	CH	0 o/c	K	BM	DC
0–	H	AT	CF	1 o/c	AC	BL	DB
1+	F	AS	CE	2 o/c	AB	BK	DA
1–	E	AR	CD	3 o/c	AA	BJ	CZ
2+	D	AP	CX	4 o/c	Z	BW	CY
2–	C	BH	CW	5 o/c	Y	BV	DS
3+	B	BF	CV	6 o/c	X	BU	DR
3–	A	BE	CU	7 o/c	W	BT	DP
4+	U	BD	CT	0 pull-up	V	BS	DN
4–	T	BC	CS	1 pull-up	AN	BR	DM
5+	S	BB	CR	2 pull-up	AM	CC	DL
5–	R	BA	CP	3 pull-up	AL	CB	DK
6+	P	AZ	DH	4 pull-up	AK	CA	DJ
6–	N	AY	DF	5 pull-up	AJ	BZ	EB
7+	M	BP	DE	6 pull-up	AH	BY	EA
7–	L	BN	DD	7 pull-up	AF	BX	DZ

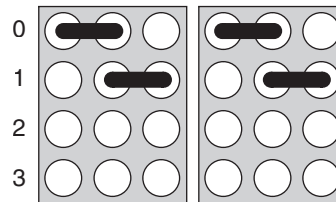
Component location and description

The diagram below shows where the user-adjustable components are found on the PWG card. See the key on the following page for detailed component descriptions.

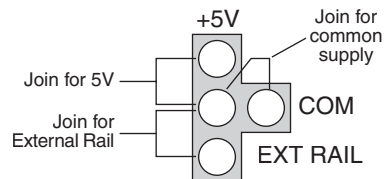


Layout of the PWG card

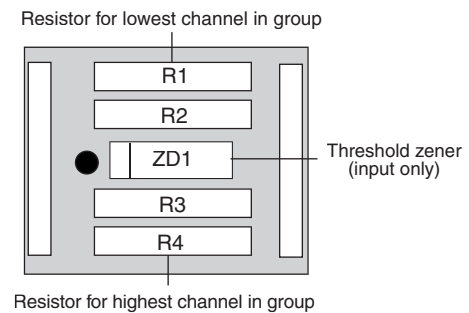
A ANGLE CLOCK OUTPUT



B HIGH RAIL SELECTION LINKS



C OUTPUT BIAS RESISTOR DAUGHTERBOARDS



Components of the PWG card

Card specifications

PWG card	
Number of outputs	8
Configuration	Harness selectable choice of: ■ VRS emulation with variable amplitude, transformer isolated ■ open collector discrete output ■ discrete output with pull-up to high rail. Resistor value 330W min.
VRS O/P amplitude range	0.25V – 64V peak-to-peak
VRS O/P impedance	20kΩ
Max. o/c sink current	100mA
Set time	Takes effect at start of next cycle after an iteration of software loop alters it.
Frequency range	0 to 30,000rpm
Frequency resolution	0.51rpm
Frequency accuracy	0.1%
Jitter	±125ns
Fault simulation	Sixteen tooth patterns available for all four outputs, software switchable.
Relative output phasing	Any phase in steps of 0.1°, dynamically programmable

Creating pattern files

This section describes how to create and load pattern data to be used by the PWG card.

The PWG card stores pattern data in two banks – the first bank serves channels 0 to 3, the second serves channels 4 to 7. Both banks store data in the same format, as described below. You will need to generate your own pattern data to synthesize cam and crank patterns, or the output from wheelspeed or other sensors.

Pattern data consists of a large array of 8-bit values, logically split into 16 patterns, each containing up to 8192 bytes of data. When a PWG channel replays a pattern, it replays the sequence of bytes in one pattern at a rate determined by the model (or the User Interface if it is an open-loop model). Typical patterns will include 7200 bytes of pattern data, with the remaining bytes (up to 8192) being unused. This allows for a 0.1 degree angle resolution when generating cam and crank signals.

Within each byte of data, the lower 7 bits (0 to 6) define the amplitude of the pattern. They can therefore assume a value between 0 and 127, with 64 representing the mid-range value, or 0V for analog outputs. Values of 0 and 127 then represent the most negative or positive values respectively.

The pattern should be centered around 64 to ensure that there is no DC component fed to the output transformers. In a correct pattern, the mean value of all the samples is 64.

For discrete outputs only, the most significant bit (6) is used, so the pattern data should be set to 0 for off and 127 for on. This will also ensure that the pattern is available via the analog output as well as the discrete output, which can be useful when debugging the system for the first time.

For both analog and discrete patterns, the top bit (7) of the last byte of the pattern data is set to signify the end of the pattern. This causes the PWG channel to skip back to the beginning of the pattern and repeat it. Typically the top bit is set in byte 7200 of the pattern to generate a 720 degree pattern with a 0.1 degree resolution. You can, however, set it to any position, with 3600 being a common alternative.

Using Excel to generate pattern data

You can use the spreadsheet application, Microsoft Excel™ to generate pattern data, and convert that Excel data into a binary pattern file that AutoSim can load:

1. Create an Excel spreadsheet with 16 columns and 8192 rows (note that any extra rows and columns will prevent the conversion working properly). Each column represents a single pattern.
2. In each column, insert the values you need for your pattern. The values can be in the range of 0 to 127. You can use other sheets in the spreadsheet to perform calculations (such as sine calculations) to generate patterns.
3. In each column, ensure there is an end of pattern marker (the top bit set in the value field) typically at row 3600 or 7200. We strongly recommend that you set the values in row 8192 to 255 to ensure there is a 'fail safe' end marker for each pattern.
4. Save a copy of the spreadsheet as a CSV file (e.g. pattern.csv). You will probably want to retain your original spreadsheet as the master copy if it contains other formulae and notes.
5. Load the pattern into Pi AutoSim via the PWG's card property sheet (see Loading Patterns under Programmable Waveform Generator card (PWG) in the Using Open Loop Models section of the Software chapter of this manual).

Event Timer card

The Event Timer card records the time and angle of events (i.e. changes in the state of its inputs) to a high resolution. The card takes four daughterboards, each of which provides eight input channels.

Example application

The Event Timer card is used to detect ignition and injector timings from an electronic control module. In a typical application you would have a PWG card producing crank and cam sensor waveforms, which would be connected to the ECM under test. The injection and spark signals produced by the ECM would then be monitored by the Event Timer card. The Event Timer card can be synchronized to the PWG card to record the angle (crank) of the event as well as the absolute time.

Circuit description

The standard discrete input circuit is used – See Configuring discrete inputs in the Discrete I/O card section earlier in this chapter for common configuration information.

These TTL signals then pass through the input daughterboards, which perform further processing before the signals reach the event timer itself. The input daughterboards can be changed for different applications. A typical function would be to pass the injector coil drive signals through a re-triggerable monostable to remove unwanted transitions caused by a PWM drive of the coil. This would produce a clean pulse showing when the injector coil was active. The default function of the daughterboard is to pass the signals through unmodified.

Whenever any of the inputs changes state, the Event Timer card records the time as the value of a 24-bit counter, which is clocked at 4MHz. The time is therefore recorded to a resolution of 250ns and the longest gap that can be recorded is just over 4s.

The angle is recorded as the count of a 16-bit counter, which is clocked by the sample clock of a PWG channel. This means that the angle counter records the sample number reached by the PWG when the event occurred. If the angle counter is controlled by the channel generating the crank sensor waveform, this gives the engine angle at which the event occurred.

Channels 29 and 31 of the event timer are reserved for timing signals, leaving 30 channels for event capture. Channel 29 is the Angle Reset signal, which carries a pulse every time the angle is zero. This enables the I/O System model to determine when a new engine cycle has started. Channel 31 is the frame tick signal, which enables the model to determine the time and angle of the start of each frame. These two channels cannot be used as standard inputs.

Card configurations

This section tells you how to configure the card by setting various jumper links.

See Configuring discrete inputs at the beginning of this chapter for common configuration information. See the component location diagram at the end of this section for jumper positions.

Input threshold voltage

Channels	Jumper	Link for $V_R/2$	Link for V_{Zener}
0–3	JP22	East	West
4–7	JP7	East	West
8–11	JP19	East	West
12–15	JP4	East	West
16–19	JP21	East	West
20–23	JP6	East	West
24–27	JP20	East	West
28–31	JP5	East	West

Input High rail voltage

Channels	Jumper	Link for V _{5V}	Link for V _{common}	Link for V _{Ext}
0–3	JP27	North	East	South
4–7	JP11	North	East	South
8–11	JP24	North	East	South
12–15	JP8	North	East	South
16–19	JP26	North	East	South
20–23	JP10	North	East	South
24–27	JP25	North	East	South
28–31	JP9	North	East	South

Injector daughterboard

The Injector daughterboard treats the eight signals as four pairs. Each pair can be passed through unmodified, or gated together to make one signal (e.g. when an injector coil is controlled by both high and low side switches). Both the high and low sides of the injector coil outputs from the ECM could be connected to the Event Timer card, and the Injector daughterboard could be configured to make its output high only when one input was high and the other low.

The Injector daughterboard can also add a 500µs monostable to the gated output, so that any unwanted edges caused by the PWM drive to the injector coil are removed.

The selection of the operating mode is under software control by changing the properties of the input channels in the Navigator. Assuming the two inputs in the pair are called In0 and In1, the following modes are available:

Mode	Out0	t1	Comments
0	In0	In1	Straight through
1	In0 and not In1	In1	
2	(In0 and not In1) extended 500µs	In1	Uses re-triggerable monostable
3			Reserved

Out1 follows In1 but can be disabled (held at 0) if required. The default mode is 0 with Out1 enabled, so the daughterboard passes all the signals through unmodified.

Card address

The four switches set the base address as described in the section Setting the base address at the beginning of this chapter.

Angle clock selection switch

The angle clock selection switch (SW5) determines which of the four angle clock signals on the bus are used to control the angle counter on the Event Timer card. This switch has to be set to the same number as the PWG card that is providing the angle clock signal. Move the switch according to the markings on the PCB rather than those on the switch itself. The switch will most commonly be set at zero.

If you are using more than one Event Timer card in a single rack, you can either set them all to the same angle clock to use a single PWG clock source or set a different angle clock on each card corresponding to a multiple PWG clock source – see Programmable Waveform Generator card earlier in this Hardware chapter for details of how to generate multiple PWG clock sources (see also Master/Slave jumper below for further information about setting up multiple Event Timer cards).

IRQ jumpers

These jumpers (JP3) select which IRQ line is used by the card. This facility is not used in AutoSim, so do not fit a jumper to this block.

Master/Slave jumper

In order to use multiple Event Timer cards you must:

- use Event Timer cards that are REV1 MOD4 or above,
- install each Event Timer card with its own unique address,
- remove the Master/Slave jumpers (JP2) from all of the cards.

See Angle clock selection switch above for how to set the angle clock used by each Event Timer card.

Channel IRQ

This facility is not used in AutoSim – leave the jumper off (JP1).

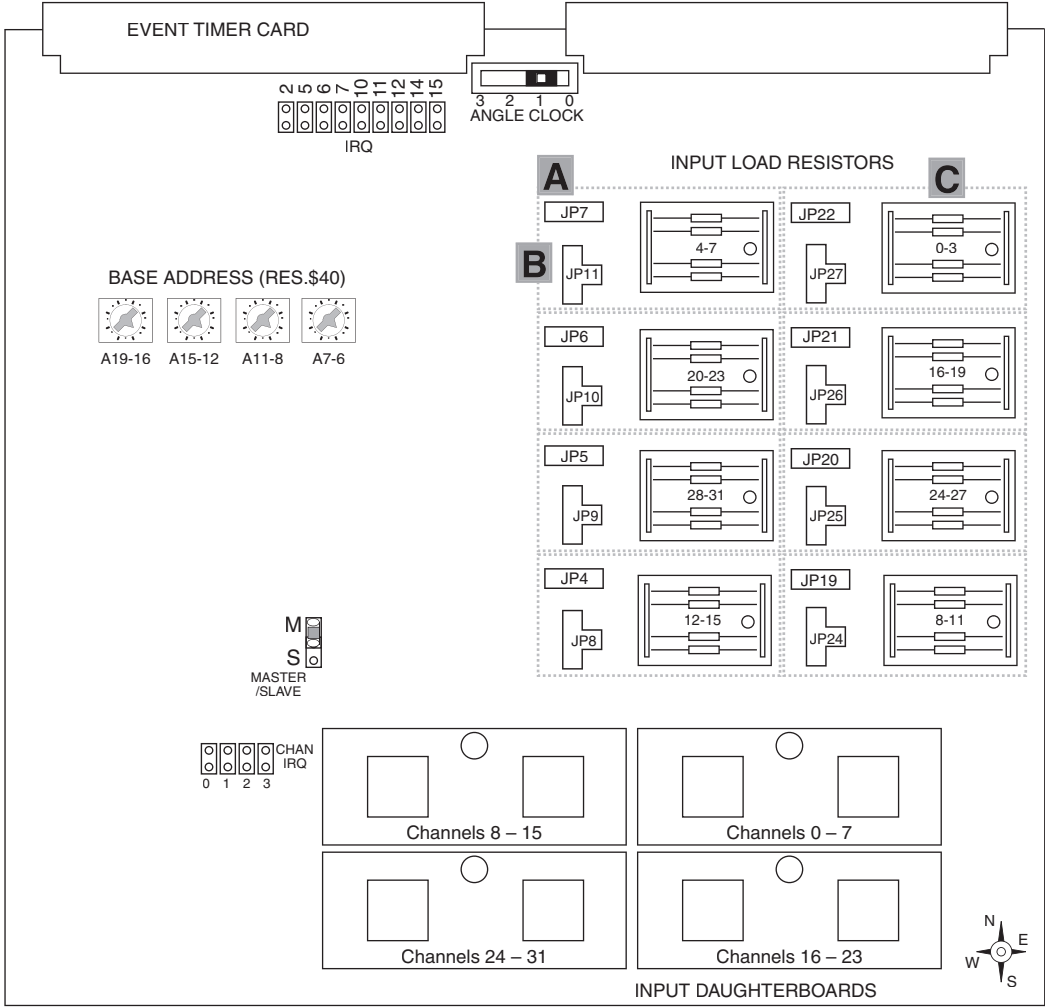
I/O channel assignment

Use the following table to find out which channels on the Event Timer card connect to which pin on the front panel connectors. Mapping channels and pins in the Front Panel section of the I/O System earlier in this Hardware chapter explains how to use these tables.

Connector	Slot	Slot	Slot	Connector	Slot	Slot	Slot
1	15	13	12	1	15	13	12
2	11	10	9	2	11	10	9
3	8	7	6	3	8	7	6
4	5	4	3	4	5	4	3
5	2	1	0	5	2	1	0
Input channel	pin	pin	pin	Input channel	pin	pin	pin
0	J	AU	CH	16	K	BM	DC
1	H	AT	CF	17	AC	BL	DB
2	F	AS	CE	18	AB	BK	DA
3	E	AR	CD	19	AA	BJ	CZ
4	D	AP	CX	20	Z	BW	CY
5	C	BH	CW	21	Y	BV	DS
6	B	BF	CV	22	X	BU	DR
7	A	BE	CU	23	W	BT	DP
8	U	BD	CT	24	V	BS	DN
9	T	BC	CS	25	AN	BR	DM
10	S	BB	CR	26	AM	CC	DL
11	R	BA	CP	27	AL	CB	DK
12	P	AZ	DH	28	AK	CA	DJ
13	N	AY	DF	30	AH	BY	EA
14	M	BP	DE				
15	L	BN	DD				

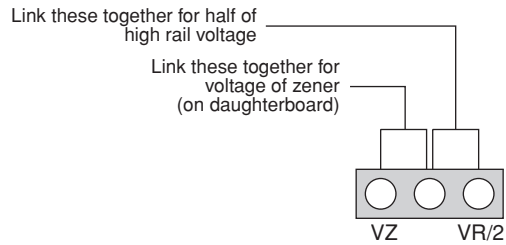
Component location

The diagram below shows where the user-adjustable components are found on the Event Timer card. See the key on the next page for detailed component descriptions.

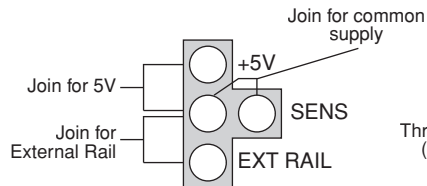


Component location on the Event Timer card

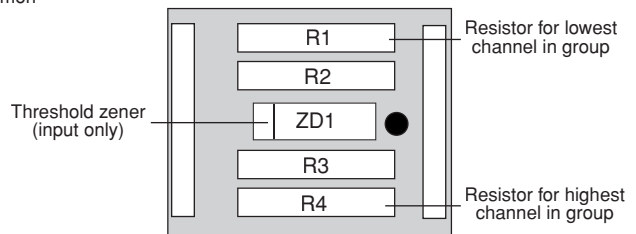
A INPUT THRESHOLD SELECTION LINKS



B HIGH RAIL SELECTION LINKS



C INPUT BIAS DAUGHTERBOARDS



Components of the Event Timer card

Event Timer card specifications

Event Timer card	
Number of channels	32 channels, provided by four daughterboards with eight channels each (two of these channels are reserved)
Angle resolution	Typical 0.1° (slaved to any PWG angle clock)
Time resolution	250ns

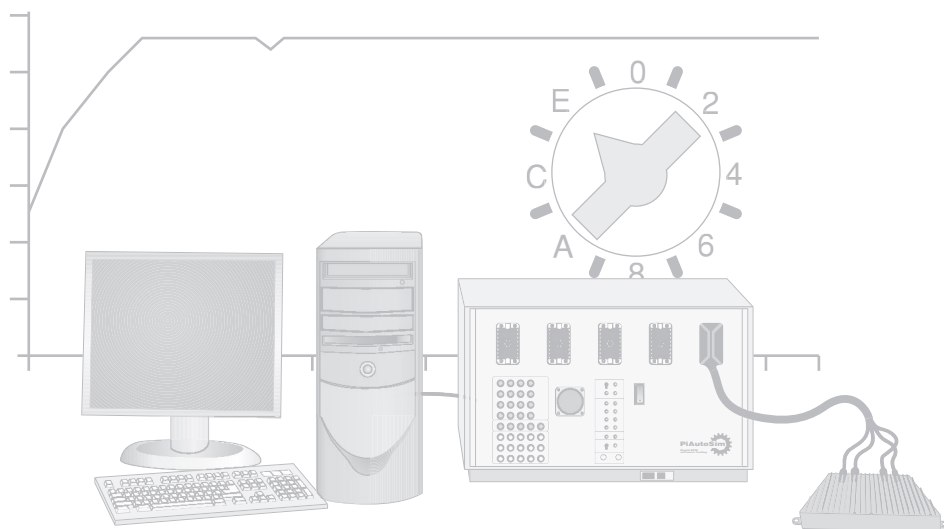
Injector daughterboard		
	With chopped mode filtering	Without
Minimum pulse width	100µs	1µs
Minimum pulse separation	1µs	1µs
Angle resolution	0.1°	0.1°
Pulse width accuracy	±1µs	±250ns
Pulse start accuracy	±250ns	±250ns
Min. chopping frequency	2kHz	N/A

The Event Timer card supports 32 channels of discrete input, which can be used for the measurement of high speeds in both time and angle domains. It is typically used in conjunction with the PWG card to measure spark and injection events in engine control applications, with measurements taken relative to crank angle.

The 32 channels are arranged in four groups of eight, with a daughterboard for each group. The daughterboards can support either single-ended or double-ended drive circuits, and each channel supports stop and start event, event duration (on-time), current state and number of events per engine cycle.

- Card properties – The standard card property sheet shows the slot number for the card.
- Daughterboard properties – The Daughter property sheet shows the location and type of the daughterboard fitted.
- Channel properties (Inputs) – Shows the property page:
 - Channel bias sets the input bias for the channel, which may be set to pull high (to the channel group's high rail), pull low (to ground), or left floating. If you change this property, you will need to stop and start the model before the change to the channel bias takes effect.
 - Active State sets the logic level for the channel. When set to Active High, the channel will report a start event when the input is pulled to the high rail. When set to Active Low, the channel will report a start event when the input is pulled to ground. Stop events are reported on the opposite edge, and duration is measured as the time between start and stop events. Active State will take immediate effect when the property is saved using Apply or OK.
 - TDC is the Top Dead Center value for the input, which can be used in engine applications to apply an offset from TDC cylinder 1 for each cylinder event. When set, this value is subtracted from the absolute crank angle measured for the event, to give cylinder relative value.
- Variable properties – The standard Variable and Scaling property sheets are provided: see Editing model and variable properties in the Software chapter.

There are hardware configuration links on the card to select the angle clock source, and set the high rail for groups of input.



Index

Index

A

- Active State (Digital Inputs) 74
- add I/O system 35
- AIN (Analog Input Card) 73
- Amplitude variable (PWG) 77
- Analog Input card (AIN) 73, 205
- Analog meter 97
- Analog Output card (AOT) 72, 211
- Angle clock channel selectors 252
- angle measurement 251
- AOT card
 - Card address 213
 - software 72
- AutoSim SSC
 - description 17
 - typical setup 17
- Auto command 119

B

- bar control 98
- base address
 - for new hardware 171
 - setting 171
- Base address selection
 - AIN card 205
 - DIO card 199
 - PIN card 234

C

- CE compliance 14
- changing cards, precautions 170
- Channel Bias (Digital Inputs) 74
- Channel properties (EVT) 80
- Chart Recorder 102
- Check command 120
- closed loop model 18, 59
- COMMS Navigator branch 140
- Compliance 9

- connecting an ECM to the I/O System 53
- connecting to the I/O system 35, 51–71
- Cosmetic branches (Navigator) 83
- Count variable (EVT) 80
- Crank and cam position sensors 248
- Creating and manipulating panels 87–161
- creating a control panel 39–48

D

- Data logger 130–161
- daughterboard
 - Lambda Oxygen Sensor Output 221
 - load resistors 195
- Daughterboard options
 - AIN card 209
 - AOT card 216
- Differential Output daughterboard
 - AOT card 222
- DIO (Discrete Input/Output card) 74
- Discrete Input/Output card (DIO) 74, 189
- discrete inputs
 - configuring 190
- discrete outputs
 - configuring 193
- Duration variable (EVT) 80

E

- Earthing
 - power supply 173
- editing model and variable properties 60
- Edit control 101
- EndBlock command 121
- Event Capture card 260
- Event Timer Card (EVT) 79
- EVT (Event Timer Card) 79
- Excel
 - generating pattern data 259
- exporting XML properties 66
- External power supply 173

F

- FCC compliance 13
- Front panel
 - diagram 20
 - power connections 176
 - power controls 181
- Front panel features 166
- Function command 118

G

- GUI Macros 122–161
- GUI Macro Commands 127–161
- GUI Macro Editor 123–161

H

- Hardware branches (Navigator) 68
- Hardware installation 170
- Health and Safety Information 2
- Help system 44–48

I

- I/O cards 18, 183
- I/O channel assignment
 - AIN card 207
 - AOT card 214
 - DIO card 200
 - EVT card 264
 - PIN card 235
 - POT card 242
 - PWG card 254
- I/O System 19–48, 165
- importing XML Properties 66
- Indicator control 100
- Input bias
 - DIO card 198
- Installation 23–48
 - User Interface software 23
- Instruments
 - adding, deleting & manipulating 89
 - and controls 97–112
 - common properties 93
 - editing properties 92

Internal power supply 175
Introduction to Pi AutoSim 17–48
IOS scripts 109–112
 commands 117
 summary 115
IP Address
 setting 29

K

Kepeco power supply 174

L

Lambda Oxygen Sensor Output daughter-
 board 221
LCD panel and buttons 170
linear scaling 63
loading a model 36, 51
Loading Patterns (PWG) 78
logged data
 viewing 136
Logging data 131
log files 131
LVDT Sensor daughterboard
 AOT card 223

M

macros 18
managing I/O Systems 54
Mapping channels and pins 168
Master card 189
Master Card properties 189
Memory Address Switch 171
Message command 121
Message Log
 setting up 151
 using 156
Message variables 144
MFX branch 68
Model Framework eXtension (MFX) 68
model variable properties 61
MonitorStop command 121

Monitor command 120
MST (Master card) 69

N

Navigator
 add an I/O system 56
 dock and undock 55
 hardware branches 68
 using the 54
 workspace 57
Networking
 setup 28
non-linear scaling 63

O

on-line help system 44–48
open loop model 18, 58
Overlaid modes 106

P

Pan mode 107
pattern files
 creating 258
Pattern variable (PWG) 77
PC
 User Interface 19
Phase controller 250
Phase variable (PWG) 78
PIN (Pulse Width Modulation Input card) 75
PISA I/O card
 fitting or removing 171
power outputs
 order of precedence 181
power supplies 173
 connecting 173
 controlling from the I/O System 174
Programmable Waveform Generator card
 77, 247
Pulse Width Modulation Input card (PIN)
 75, 233
Pulse Width Modulation Output card (POT)
 76, 240

Push-button control 99
PWG (Programmable Waveform Generator
 card) 77

R

Ramp command 118
Real Time Viewer 154
rear panel diagram 22
Recorder toolbar 111
reference variables (Navigator) 83
resistor daughterboard 195
Rpm variable (PWG) 77
running a model 43

S

scaling properties tab 63
scripts 18
Script Editor 112
 checking syntax 114
 tool bar 113
Script Recorder toolbar 110
Serial comms
 configuring hardware 141
 statistics 143
Serial Communications 140
Serial Communications Branch (Navi-
 tor) 83
Session Logs 138
Set command 117
Simulator switched supplies 176
Slave variable (PWG) 78
Slider control 99
SSWCOM (common switched supply) 178
starting the I/O system 31
Start variable (EVT) 79
State variable (EVT) 79
Static Text control 101
Statistics branch (Navigator) 82
stopping a model 43
Stop variable (EVT) 79
system file problems 28

T

- TCP/IP 28
- Thermistor Output daughterboard
 - AOT card 218
- Thermocouple Output daughterboard
 - AOT card 220
- Thermometer 100
- Tiled modes 106
- timestamp
 - for a command 126
- Time Series Graph 103
- toolbar options 33, 85
- Tools required for installation 170
- Trace tab (Chart Recorder) 105
- TRIGGERn 70
- Trigger variables 148
- troubleshooting 28
 - network connection 30
- Tutorial 31

U

- User Interface software 23
- User Variables Branch (Navigator) 83
- using open loop models 68

V

- variable properties tab 62

W

- Wheelspeed sensors 248
- working with models 58
- Working with panels 85
- workspace (Navigator snapshot) 57

X

- Xantrax power supply 174
- XML properties 66

Z

- Zener diode 197
- Zoom mode 107

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